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Discovery

Discover the world of microcontrollers through [Rust!](#)

This book is an introductory course on microcontroller-based embedded systems that uses Rust as the teaching language rather than the usual C/C++.

Scope

The following topics will be covered (eventually, I hope):

- How to write, build, flash and debug an “embedded” (Rust) program.
- Functionality (“peripherals”) commonly found in microcontrollers: Digital input and output, Pulse Width Modulation (PWM), Analog to Digital Converters (ADC), common communication protocols like Serial, I2C and SPI, etc.
- Multitasking concepts: cooperative vs preemptive multitasking, interrupts, schedulers, etc.
- Control systems concepts: sensors, calibration, digital filters, actuators, open loop control, closed loop control, etc.

Approach

- Beginner friendly. No previous experience with microcontrollers or embedded systems is required.
- Hands on. Plenty of exercises to put the theory into practice. *You* will be doing most of the work here.
- Tool centered. We’ll make plenty use of tooling to ease development. “Real” debugging, with GDB, and logging will be introduced early on. Using LEDs as a debugging mechanism has no place here.

Non-goals

What’s out of scope for this book:

- Teaching Rust. There’s plenty of material on that topic already. We’ll focus on microcontrollers and embedded systems.

- Being a comprehensive text about electric circuit theory or electronics. We'll just cover the minimum required to understand how some devices work.
- Covering details such as linker scripts and the boot process. For example, we'll use existing tools to help get your code onto your board, but not go into detail about how those tools work.

Also I don't intend to port this material to other development boards; this book will make exclusive use of the STM32F3DISCOVERY development board.

Reporting problems

The source of this book is in [this repository](#). If you encounter any typo or problem with the code report it on the [issue tracker](#).

Other embedded Rust resources

This Discovery book is just one of several embedded Rust resources provided by the [Embedded Working Group](#). The full selection can be found at [The Embedded Rust Bookshelf](#). This includes the list of [Frequently Asked Questions](#).

Sponsored by



Many thanks to [integer 32](#) for sponsoring me to work on this book! Please give them lots of work (they do Rust consulting!) so they'll have no choice but to hire more Rustaceans <3.

Background

What's a microcontroller?

A microcontroller is a *system* on a chip. Whereas your computer is made up of several discrete components: a processor, RAM sticks, a hard drive, an ethernet port, etc.; a microcontroller has all those components built into a single “chip” or package. This makes it possible to build systems with a minimal part count.

What can you do with a microcontroller?

Lots of things! Microcontrollers are the central part of systems known as *embedded* systems. These systems are everywhere but you don't usually notice them. These systems control the brakes of your car, wash your clothes, print your documents, keep you warm, keep you cool, optimize the fuel consumption of your car, etc.

The main trait of these systems is that they operate without user intervention even if they expose a user interface as a washing machine does; most of their operation is done on their own.

The other common trait of these systems is that they *control* a process. And for that these systems usually have one or more sensors and one or more actuators. For example, an HVAC system has several sensors, thermometers, and humidity sensors spread across some areas, and several actuators as well, heating elements and fans connected to ducts.

When should I use a microcontroller?

All these applications I've mentioned, you can probably implement with a Raspberry Pi, a computer that runs Linux. Why should I bother with a microcontroller that operates without an OS? Sounds like it would be harder to develop a program.

The main reason is cost. A microcontroller is much cheaper than a general-purpose computer. Not only the microcontroller is cheaper; it also requires many fewer external electrical components to operate. This makes Printed Circuit Boards (PCBs) smaller and cheaper to design and manufacture.

The other big reason is power consumption. A microcontroller consumes orders of magnitude less power than a full-blown processor. If your application will run on batteries that makes a huge difference.

And last but not least: (hard) *real-time* constraints. Some processes require their controllers to respond to some events within some time interval (e.g. a quadcopter/drone hit by a wind gust). If this *deadline* is not met, the process could end in catastrophic failure (e.g. the drone crashes to the

ground). A general-purpose computer running a general-purpose OS has many services running in the background. This makes it hard to guarantee the execution of a program within tight time constraints. 10

When should I *not* use a microcontroller?

Where heavy computations are involved. To keep their power consumption low, microcontrollers have very limited computational resources available to them. For example, some microcontrollers don't even have hardware support for floating-point operations. On those devices, performing a simple addition of single-precision numbers can take hundreds of CPU cycles.

Why use Rust and not C?

Hopefully, I don't need to convince you here as you are probably familiar with the language differences between Rust and C. One point I do want to bring up is package management. C lacks an official, widely accepted package management solution whereas Rust has Cargo. This makes development *much* easier. And, IMO, easy package management encourages code reuse because libraries can be easily integrated into an application which is also a good thing as libraries get more "battle testing".

Why should I not use Rust?

Or why should I prefer C over Rust?

The C ecosystem is way more mature. Off the shelf solution for several problems already exist. If you need to control a time-sensitive process, you can grab one of the existing commercial Real-Time Operating Systems (RTOS) out there and solve your problem. There are no commercial, production-grade RTOSes in Rust yet so you would have to either create one yourself or try one of the ones that are in development.

Hardware/knowledge requirements

The primary knowledge requirement to read this book is to know *some* Rust. It's hard for me to quantify *some* but at least I can tell you that you don't need to fully grok generics but you do need to know how to *use* closures. You also need to be familiar with the idioms of the [2018 edition](#), in particular with the fact that `extern crate` is not necessary in the 2018 edition.

Due to the nature of embedded programming, it will also be extremely helpful to understand how binary and hexadecimal representations of values work, as well as the use of some bitwise operators. For example, it would be useful to understand how the following program produces its output.

```
fn main() {
    let a = 0x4000_0000 + 0xa2;

    // Use of the bit shift "<<" operation.
    let b = 1 << 5;

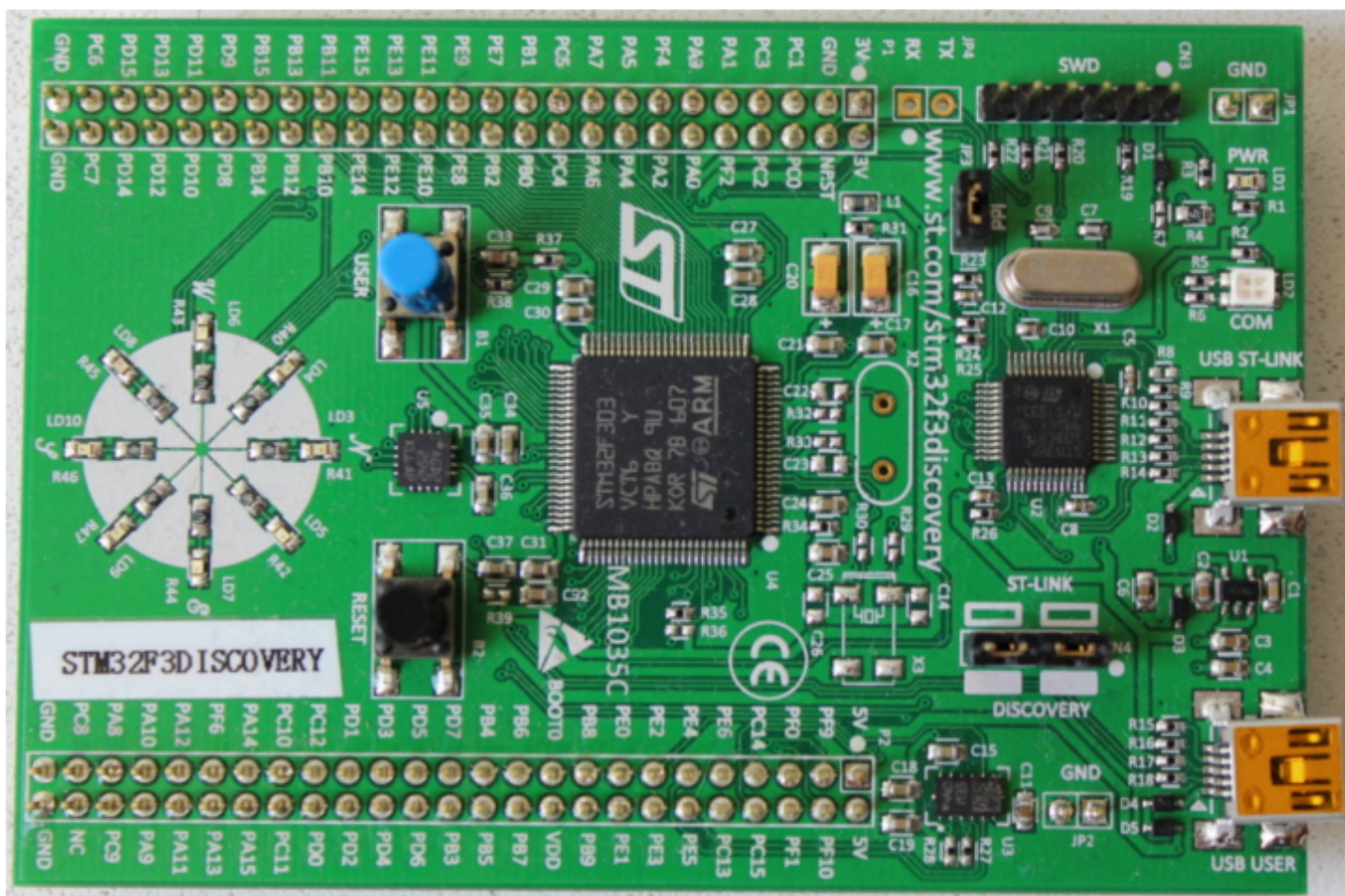
    // {:X} will format values as hexadecimal
    println!("{:X}: {:X}", a, b);
}
```

Also, to follow this material you'll need the following hardware:

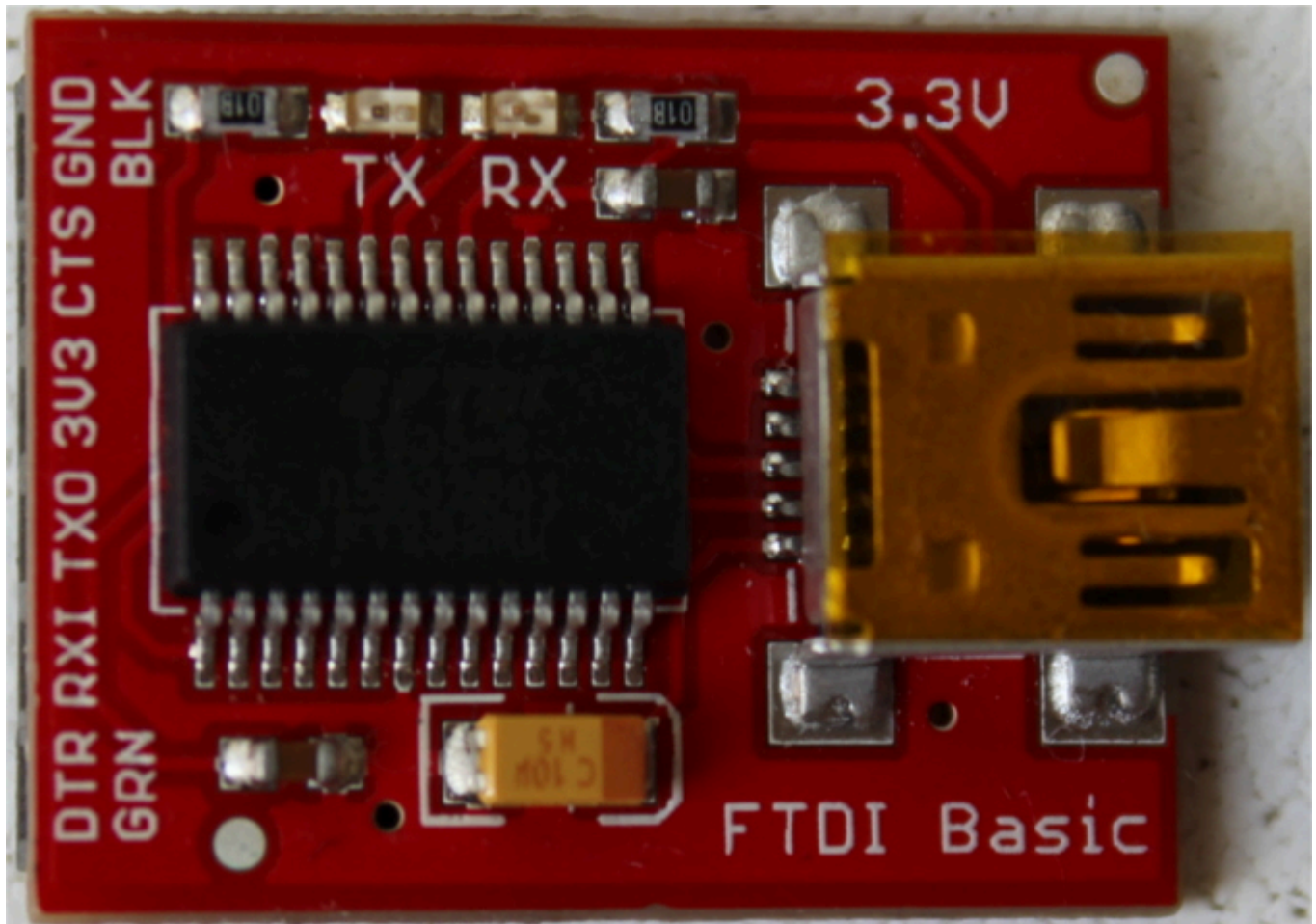
(Some components are optional but recommended)

- A [STM32F3DISCOVERY](#) board.

(You can purchase this board from “big” [electronics suppliers](#) or from [e-commerce sites](#))

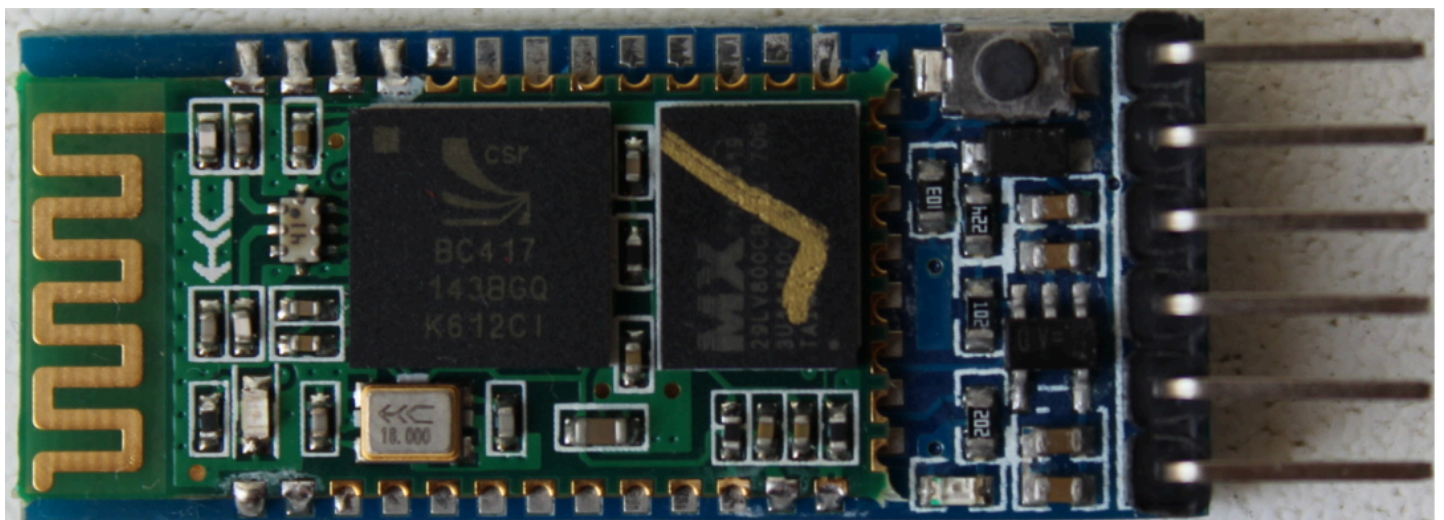


- OPTIONAL. A **3.3V** USB <-> Serial module. To elaborate: if you have one of the latest revisions of the discovery board (which is usually the case given the first revision was released years ago) then you do *not* need this module because the board includes this functionality on-board. If you have an older revision of the board then you'll need this module for chapters 10 and 11. For completeness, we'll include instructions for using a Serial module. The book will use [this particular model](#) but you can use any other model as long as it operates at 3.3V. The CH340G module, which you can buy from [e-commerce](#) sites works too and it's probably cheaper for you to get.



- OPTIONAL. A HC-05 Bluetooth module (with headers!). A HC-06 would work too.

(As with other Chinese parts, you pretty much can only find these on [e-commerce sites](#). (US) Electronics suppliers don't usually stock these for some reason)



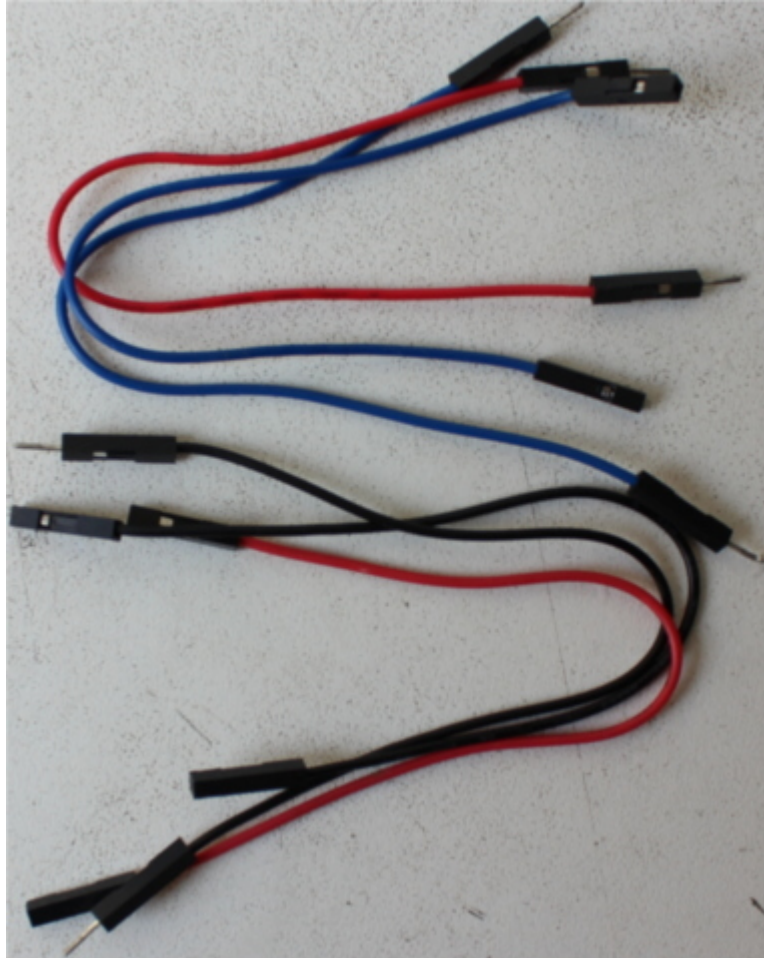
- Two mini-B USB cables. One is required to make the STM32F3DISCOVERY board work. The other is only required if you have the Serial <-> USB module. Make sure that the cables both support data transfer as some cables only support charging devices.



NOTE These are **not** the USB cables that ship with pretty much every Android phone; those are *micro* USB cables. Make sure you have the right thing!

- MOSTLY OPTIONAL. 5 female to female, 4 male to female and 1 Male to Male *jumper* (AKA Dupont) wires. You'll *very likely* need one female to female to get ITM working. The other wires are only needed if you'll be using the USB <-> Serial and Bluetooth modules.

(You can get these from electronics [suppliers](#) or from [e-commerce sites](#))



FAQ: Wait, why do I need this specific hardware?

It makes my life and yours much easier.

The material is much, much more approachable if we don't have to worry about hardware differences. Trust me on this one.

FAQ: Can I follow this material with a different development board?

Maybe? It depends mainly on two things: your previous experience with microcontrollers and/or whether there already exists a high level crate, like the [f3](#), for your development board somewhere.

With a different development board, this text would lose most if not all its beginner friendliness and “easy to follow”-ness, IMO.

If you have a different development board and you don't consider yourself a total beginner, you are better off starting with the [quickstart](#) project template.

Setting up a development environment

Dealing with microcontrollers involves several tools as we'll be dealing with an architecture different than your computer's and we'll have to run and debug programs on a "remote" device.

Documentation

Tooling is not everything though. Without documentation it is pretty much impossible to work with microcontrollers.

We'll be referring to all these documents throughout this book:

HEADS UP All these links point to PDF files and some of them are hundreds of pages long and several MBs in size.

- [STM32F3DISCOVERY User Manual](#)
- [STM32F303VC Datasheet](#)
- [STM32F303VC Reference Manual](#)
- [LSM303DLHC *](#)
- [L3GD20 *](#)

* **NOTE:** Newer (from around 2020/09) Discovery boards may have a different e-compass and gyroscope (see the user manual). As such, much in chapters 14-16 will not work as is. Checkout the github issues like [this](#).

Tools

We'll use all the tools listed below. Where a minimum version is not specified, any recent version should work but we have listed the version we have tested.

- Rust 1.31 or a newer toolchain. Chapter [USART](#) requires 1.51 or newer.
- `itmdump` `>=0.3.1` (`cargo install itm`). Tested versions: 0.3.1.
- OpenOCD `>=0.8`. Tested versions: v0.9.0 and v0.10.0
- `arm-none-eabi-gdb` . Version 7.12 or newer highly recommended. Tested versions: 7.10, 7.11, 7.12 and 8.1
- `cargo-binutils` . Version 0.1.4 or newer.
- `minicom` on Linux and macOS. Tested version: 2.7. Readers report that `picocom` also works but we'll use `minicom` in this text.
- `PuTTY` on Windows.

If your computer has Bluetooth functionality and you have the Bluetooth module, you can additionally install these tools to play with the Bluetooth module. All these are optional:

- Linux, only if you don't have a Bluetooth manager application like Blueman.
 - `bluez`
 - `hcitool`
 - `rfcomm`
 - `rfkill`

macOS / OSX / Windows users only need the default bluetooth manager that ships with their OS.

Next, follow OS-agnostic installation instructions for a few of the tools:

rustc & Cargo

Install rustup by following the instructions at <https://rustup.rs>.

If you already have rustup installed double check that you are on the stable channel and your stable toolchain is up to date. `rustc -V` should return a date newer than the one shown below:

```
$ rustc -V
rustc 1.31.0 (abe02cefd 2018-12-04)
```

itmdump

```
cargo install itm
```

Verify the version is $\geq 0.3.1$

```
$ itmdump -V
itmdump 0.3.1
```

cargo-binutils

Install `llvm-tools`

```
rustup component add llvm-tools
```

Install `cargo-binutils`

```
cargo install cargo-binutils
```

Verify tools are installed

Run the following commands at your terminal

```
cargo new test-size
```

```
cd test-size
```

```
cargo run
```

```
cargo size -- --version
```

The results should be something like:

```
~
$ cargo new test-size
  Created binary (application) `test-size` package

~
$ cd test-size

~/test-size (main)
$ cargo run
  Compiling test-size v0.1.0 (~/test-size)
  Finished dev [unoptimized + debuginfo] target(s) in 0.26s
  Running `target/debug/test-size`
Hello, world!

~/test-size (main)
$ cargo size -- --version
  Finished dev [unoptimized + debuginfo] target(s) in 0.00s
LLVM (http://llvm.org/):
  LLVM version 11.0.0-rust-1.50.0-stable
  Optimized build.
  Default target: x86_64-unknown-linux-gnu
  Host CPU: znver2
```

OS specific instructions

Now follow the instructions specific to the OS you are using:

- [Linux](#)
- [Windows](#)
- [macOS](#)

Linux

Here are the installation commands for a few Linux distributions.

REQUIRED packages

Ubuntu 18.04 or newer / Debian stretch or newer

NOTE `gdb-multiarch` is the GDB command you'll use to debug your ARM Cortex-M programs

```
sudo apt-get install \  
  gdb-multiarch \  
  minicom \  
  openocd
```

Ubuntu 14.04 and 16.04

NOTE `arm-none-eabi-gdb` is the GDB command you'll use to debug your ARM Cortex-M programs

```
sudo apt-get install \  
  gdb-arm-none-eabi \  
  minicom \  
  openocd
```

Fedora 23 or newer

```
sudo dnf install \  
  minicom \  
  openocd \  
  gdb
```

Arch Linux

NOTE `arm-none-eabi-gdb` is the GDB command you'll use to debug your ARM Cortex-M programs

```
sudo pacman -S \
  arm-none-eabi-gdb \
  minicom \
  openocd
```

Other distros

NOTE `arm-none-eabi-gdb` is the GDB command you'll use to debug your ARM Cortex-M programs

For distros that don't have packages for [ARM's pre-built toolchain](#), download the "Linux 64-bit" file and put its `bin` directory on your path. Here's one way to do it:

```
mkdir -p ~/local && cd ~/local
```

```
tar xjf /path/to/downloaded/file/gcc-arm-none-eabi-10-2020-q4-major-x86_64-
linux.tar.bz2
```

Then, use your editor of choice to append to your `PATH` in the appropriate shell init file (e.g. `~/.zshrc` or `~/.bashrc`):

```
PATH=$PATH:$HOME/local/gcc-arm-none-eabi-10-2020-q4-major-x86_64-linux/bin
```

Optional packages

Ubuntu / Debian

```
sudo apt-get install \
  bluez \
  rfkill
```

Fedora

```
sudo dnf install \
  bluez \
  rfkill
```

```
sudo pacman -S \  
bluez \  
bluez-utils \  
rfkill
```

udev rules

These rules let you use USB devices like the F3 and the Serial module without root privilege, i.e. `sudo`.

Create `99-openocd.rules` in `/etc/udev/rules.d` using the `idVendor` and `idProduct` from the `lsusb` output.

For example, connect the STM32F3DISCOVERY to your computer using a USB cable. Be sure to connect the cable to the “USB ST-LINK” port, the USB port in the center of the edge of the board.

Execute `lsusb`:

```
lsusb | grep ST-LINK
```

It should result in something like:

```
$ lsusb | grep ST-LINK  
Bus 003 Device 003: ID 0483:374b STMicroelectronics ST-LINK/V2.1
```

So the `idVendor` is `0483` and `idProduct` is `374b`.

Create `/etc/udev/rules.d/99-openocd.rules`:

```
sudo vi /etc/udev/rules.d/99-openocd.rules
```

With the contents:

```
# STM32F3DISCOVERY - ST-LINK/V2.1  
ATTRS{idVendor}=="0483", ATTRS{idProduct}=="374b", MODE:="0666"
```

For older devices with OPTIONAL USB <-> FT232 based Serial Module

Create `/etc/udev/rules.d/99-ftdi.rules`:

```
sudo vi /etc/udev/rules.d/99-openocd.rules
```

With the contents:

```
# FT232 - USB <-> Serial Converter
ATTRS{idVendor}=="0403", ATTRS{idProduct}=="6001", MODE:="0666"
```

Reload the udev rules with:

```
sudo udevadm control --reload-rules
```

If you had any board plugged to your computer, unplug them and then plug them in again.

Now, go to the [next section](#).

Windows

arm-none-eabi-gdb

ARM provides `.exe` installers for Windows. Grab one from [here](#), and follow the instructions. Just before the installation process finishes tick/select the “Add path to environment variable” option. Then verify that the tools are in your `%PATH%`:

Verify gcc is installed:

```
arm-none-eabi-gcc -v
```

The results should be something like:

```
(..)  
$ arm-none-eabi-gcc -v  
gcc version 5.4.1 20160919 (release) (..)
```

OpenOCD

There's no official binary release of OpenOCD for Windows but there are unofficial releases available [here](#). Grab the 0.10.x zipfile and extract it somewhere in your drive (I recommend `C:\OpenOCD` but with the drive letter that makes sense to you) then update your `%PATH%` environment variable to include the following path: `C:\OpenOCD\bin` (or the path that you used before).

Verify OpenOCD is installed and in your `%PATH%` with:

```
openocd -v
```

The results should be something like:

```
$ openocd -v  
Open On-Chip Debugger 0.10.0  
(..)
```

PuTTY

Download the latest `putty.exe` from [this site](#) and place it somewhere in your `%PATH%`.

You'll also need to install [this USB driver](#) or OpenOCD won't work. Follow the installer instructions and make sure you install the right (32-bit or 64-bit) version of the driver.

That's all! Go to the [next section](#).

macOS

All the tools can be installed using [Homebrew](#):

```
$ # ARM GCC debugger  
$ brew install arm-none-eabi-gdb  
  
$ # Minicom and OpenOCD  
$ brew install minicom openocd
```

That's all! Go to the [next section](#).

Verify the installation

Let's verify that all the tools were installed correctly.

Linux only

Verify permissions

Connect the STM32F3DISCOVERY to your computer using an USB cable. Be sure to connect the cable to the "USB ST-LINK" port, the USB port in the center of the edge of the board.

The STM32F3DISCOVERY should now appear as a USB device (file) in `/dev/bus/usb`. Let's find out how it got enumerated:

```
lsusb | grep -i stm
```

This should result in:

```
$ lsusb | grep -i stm
Bus 003 Device 004: ID 0483:374b STMicroelectronics ST-LINK/V2.1
$ #   ^^^           ^^^
```

In my case, the STM32F3DISCOVERY got connected to the bus #3 and got enumerated as the device #4. This means the file `/dev/bus/usb/003/004` is the STM32F3DISCOVERY. Let's check its permissions:

```
$ ls -la /dev/bus/usb/003/004
crw-rw-rw-+ 1 root root 189, 259 Feb 28 13:32 /dev/bus/usb/003/00
```

The permissions should be `crw-rw-rw-`. If it's not ... then check your [udev rules](#) and try re-loading them with:

```
sudo udevadm control --reload-rules
```

For older devices with OPTIONAL USB <-> FT232 based Serial Module

Unplug the STM32F3DISCOVERY and plug the Serial module. Now, figure out what's its associated file:

```
$ lsusb | grep -i ft232
Bus 003 Device 005: ID 0403:6001 Future Technology Devices International, Ltd FT232
Serial (UART) IC
```

In my case, it's the `/dev/bus/usb/003/005`. Now, check its permissions:

```
$ ls -l /dev/bus/usb/003/005
crw-rw-rw- 1 root root 189, 21 Sep 13 00:00 /dev/bus/usb/003/005
```

As before, the permissions should be `crw-rw-rw-`.

Verify OpenOCD connection

Connect the STM32F3DISCOVERY using the USB cable to the USB port in the center of edge of the board, the one that's labeled "USB ST-LINK".

Two *red* LEDs should turn on right after connecting the USB cable to the board.

IMPORTANT There is more than one hardware revision of the STM32F3DISCOVERY board. For older revisions, you'll need to change the "interface" argument to `-f interface/stlink-v2.cfg` (note: no `-1` at the end). Alternatively, older revisions can use `-f board/stm32f3discovery.cfg` instead of `-f interface/stlink-v2-1.cfg -f target/stm32f3x.cfg`.

NOTE OpenOCD v0.11.0 has deprecated `interface/stlink-v2.cfg` in favor of `interface/stlink.cfg` which supports ST-LINK/V1, ST-LINK/V2, ST-LINK/V2-1, and ST-LINK/V3.

*Nix

FYI: The `interface` directory is typically located in `/usr/share/openocd/scripts/`, which is the default location OpenOCD expects these files. If you've installed them somewhere else use the `-s /path/to/scripts/` option to specify your install directory.

```
openocd -f interface/stlink-v2-1.cfg -f target/stm32f3x.cfg
```

or

```
openocd -f interface/stlink.cfg -f target/stm32f3x.cfg
```

Windows

Below the references to `C:\OpenOCD` is the directory where OpenOCD is installed.

```
openocd -s C:\OpenOCD\share\scripts -f interface/stlink-v2-1.cfg -f
target/stm32f3x.cfg
```

NOTE cygwin users have reported problems with the `-s` flag. If you run into that problem you can add `C:\OpenOCD\share\scripts\` directory to the parameters.

cygwin users:

```
openocd -f C:\OpenOCD\share\scripts\interface\stlink-v2-1.cfg -f
C:\OpenOCD\share\scripts\target\stm32f3x.cfg
```

All

OpenOCD is a service which forwards debug information from the ITM channel to a file, `itm.txt`, as such it runs forever and does **not** return to the terminal prompt.

The initial output of OpenOCD is something like:

```
Open On-Chip Debugger 0.10.0
Licensed under GNU GPL v2
For bug reports, read
    http://openocd.org/doc/doxygen/bugs.html
Info : auto-selecting first available session transport "hla_swd". To override use
'transport select <transport>'.
adapter speed: 1000 kHz
adapter_nsrst_delay: 100
Info : The selected transport took over low-level target control. The results might
differ compared to plain JTAG/SWD
none separate
Info : Unable to match requested speed 1000 kHz, using 950 kHz
Info : Unable to match requested speed 1000 kHz, using 950 kHz
Info : clock speed 950 kHz
Info : STLINK v2 JTAG v27 API v2 SWIM v15 VID 0x0483 PID 0x374B
Info : using stlink api v2
Info : Target voltage: 2.915608
Info : stm32f3x.cpu: hardware has 6 breakpoints, 4 watchpoints
```

(If you don't ... then check the [general troubleshooting](#) instructions.)

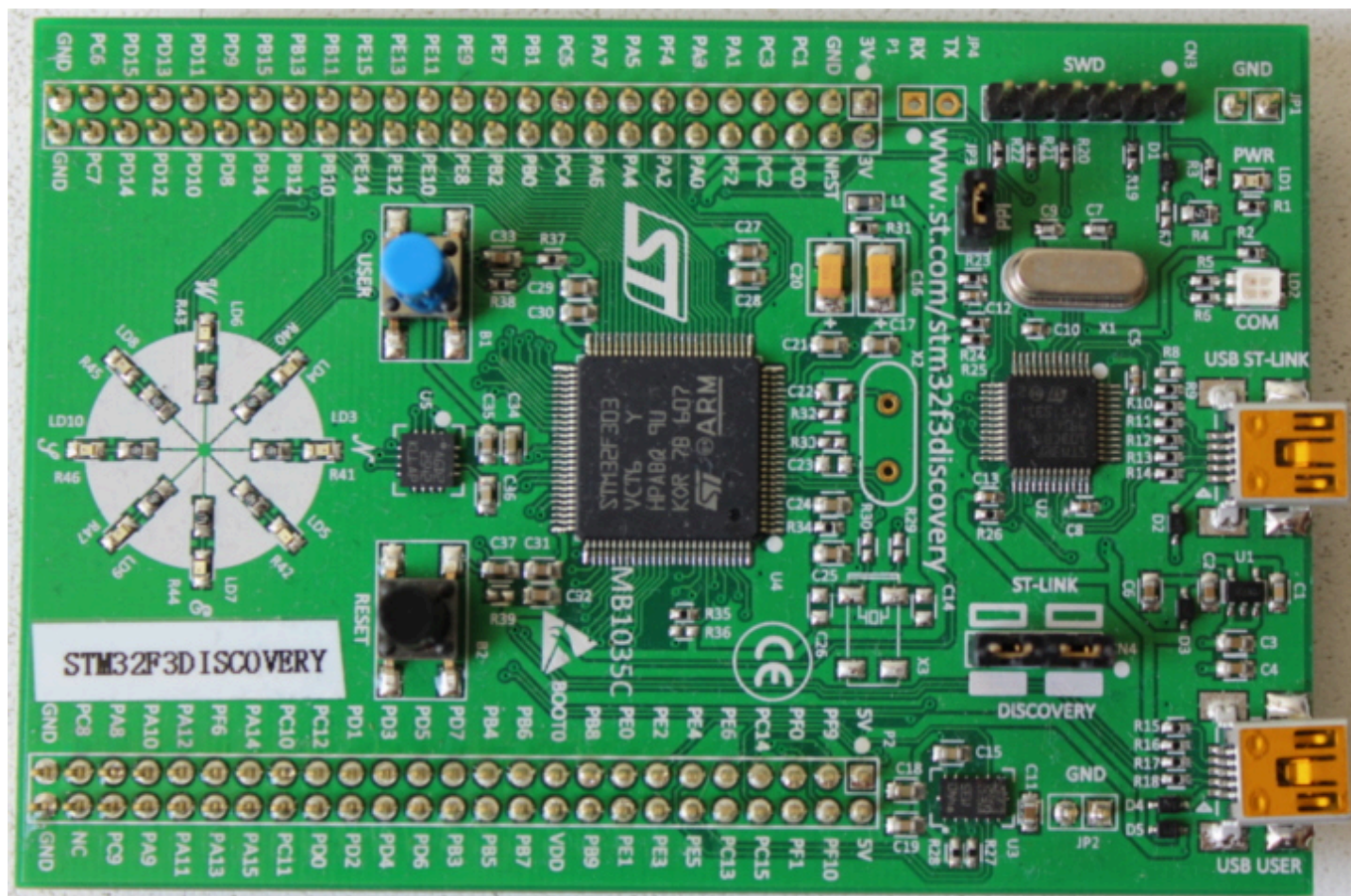
Also, one of the red LEDs, the one closest to the USB port, should start oscillating between red light and green light.

That's it! It works. You can now use `Ctrl-C` to stop OpenOCD or close/kill the terminal.

Meet your hardware

Let's get familiar with the hardware we'll be working with.

STM32F3DISCOVERY (the “F3”)



We'll refer to this board as “F3” throughout this book. Here are some of the many components on the board:

- A [microcontroller](#).
- A number of LEDs, including the eight aligned in a “compass” formation.
- Two buttons.
- Two USB ports.
- An [accelerometer](#).
- A [magnetometer](#).
- A [gyroscope](#).

Of these components, the most important is the microcontroller (sometimes shortened to “MCU” for “microcontroller unit”), which is the large black square sitting in the center of your board. The MCU is what runs your code. You might sometimes read about “programming a board”, when in reality what we are doing is programming the MCU that is installed on the board.

Since the MCU is so important, let's take a closer look at the one sitting on our board.

Our MCU is surrounded by 100 tiny metal **pins**. These pins are connected to **traces**, the little “roads” that act as the wires connecting components together on the board. The MCU can dynamically alter the electrical properties of the pins. This works similar to a light switch altering how electrical current flows through a circuit. By enabling or disabling electrical current to flow through a specific pin, an LED attached to that pin (via the traces) can be turned on and off.

Each manufacturer uses a different part numbering scheme, but many will allow you to determine information about a component simply by looking at the part number. Looking at our MCU's part number (**STM32F303VCT6**), the **ST** at the front hints to us that this is a part manufactured by **ST Microelectronics**. Searching through **ST's marketing materials** we can also learn the following:

- The **M32** represents that this is an Arm®-based 32-bit microcontroller.
- The **F3** represents that the MCU is from ST's “STM32F3” series. This is a series of MCUs based on the Cortex®-M4 processor design.
- The remainder of the part number goes into more details about things like extra features and RAM size, which at this point we're less concerned about.

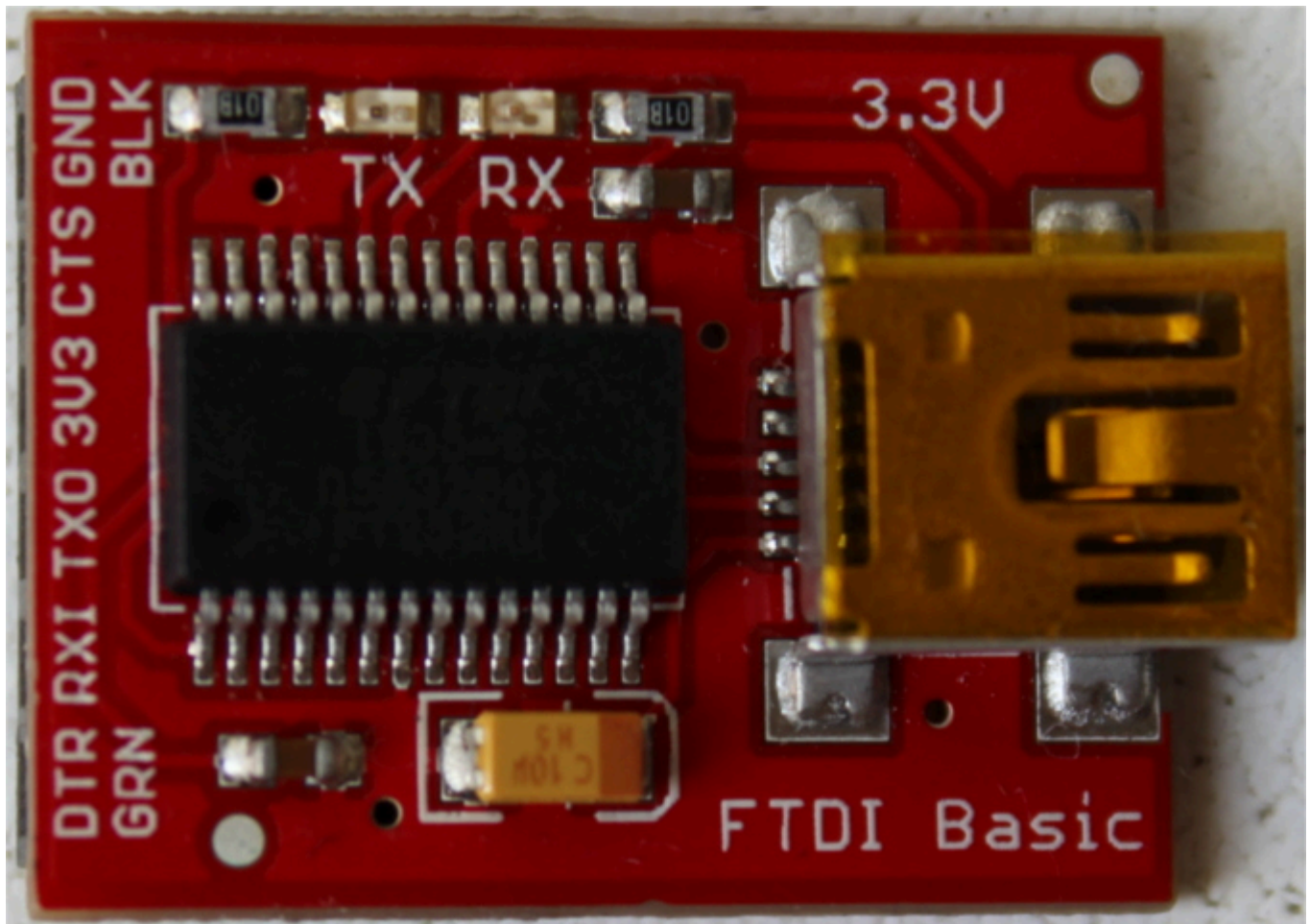
Arm? Cortex-M4?

If our chip is manufactured by ST, then who is Arm? And if our chip is the STM32F3, what is the Cortex-M4?

You might be surprised to hear that while “Arm-based” chips are quite popular, the company behind the “Arm” trademark (**Arm Holdings**) doesn't actually manufacture chips for purchase. Instead, their primary business model is to just *design* parts of chips. They will then license those designs to manufacturers, who will in turn implement the designs (perhaps with some of their own tweaks) in the form of physical hardware that can then be sold. Arm's strategy here is different from companies like Intel, which both designs *and* manufactures their chips.

Arm licenses a bunch of different designs. Their “Cortex-M” family of designs are mainly used as the core in microcontrollers. For example, the Cortex-M0 is designed for low cost and low power usage. The Cortex-M7 is higher cost, but with more features and performance. The core of our STM32F3 is based on the Cortex-M4, which is in the middle: more features and performance than the Cortex-M0, but less expensive than the Cortex-M7.

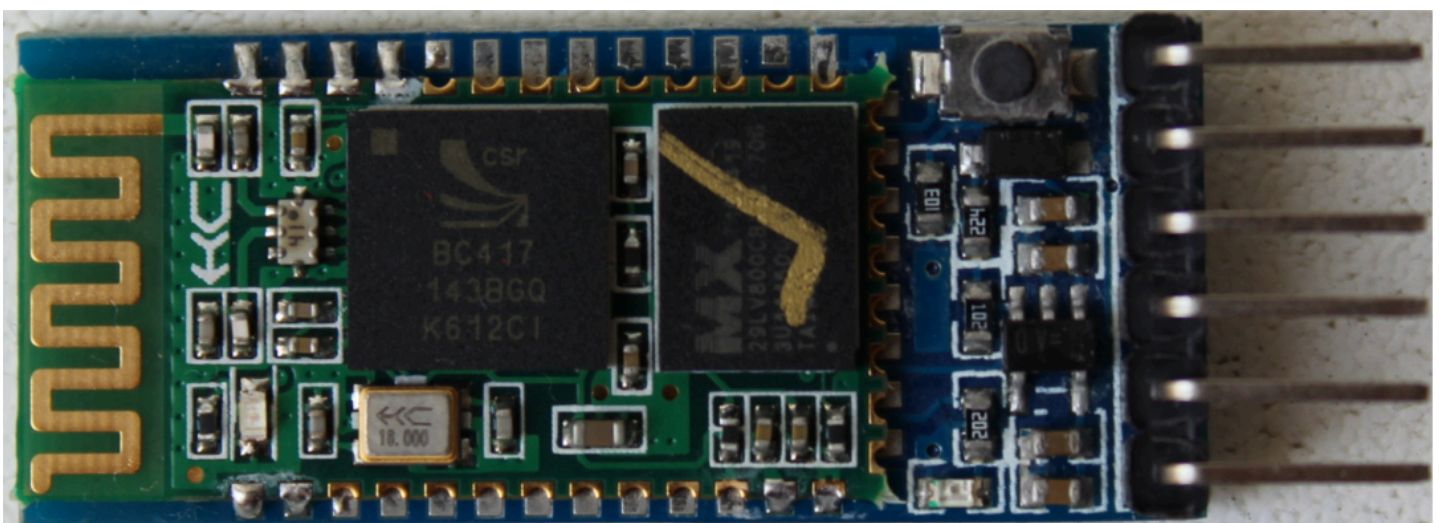
Luckily, you don't need to know too much about different types of processors or Cortex designs for the sake of this book. However, you are hopefully now a bit more knowledgeable about the terminology of your device. While you are working specifically with an STM32F3, you might find yourself reading documentation and using tools for Cortex-M-based chips, as the STM32F3 is based on a Cortex-M design.



If you have an older revision of the discovery board, you can use this module to exchange data between the microcontroller in the F3 and your computer. This module will be connected to your computer using an USB cable. I won't say more at this point.

If you have a newer release of the board then you don't need this module. The ST-LINK will double as a USB<->serial converter connected to the microcontroller USART1 at pins PC4 and PC5.

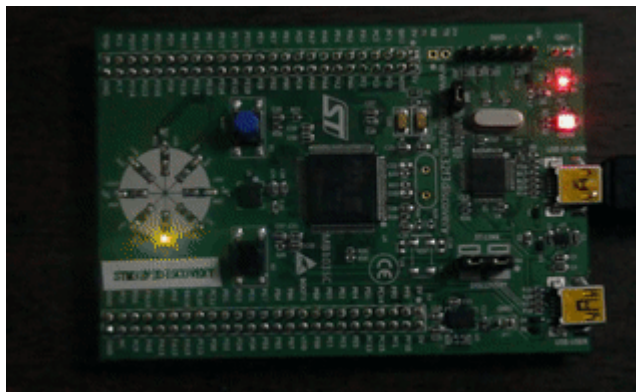
The Bluetooth module



This module has the exact same purpose as the serial module but it sends the data over Bluetooth instead of over USB.

LED roulette

Alright, let's start by building the following application:



I'm going to give you a high level API to implement this app but don't worry we'll do low level stuff later on. The main goal of this chapter is to get familiar with the *flashing* and debugging process.

Throughout this text we'll be using the starter code that's in the [discovery](#) repository. Make sure you always have the latest version of the master branch because this website tracks that branch.

The starter code is in the `src` directory of that repository. Inside that directory there are more directories named after each chapter of this book. Most of those directories are starter Cargo projects.

Now, jump into the `src/05-led-roulette` directory. Check the `src/main.rs` file:

```
#![deny(unsafe_code)]
#![no_main]
#![no_std]

use aux5::entry;

#[entry]
fn main() -> ! {
    let _y;
    let x = 42;
    _y = x;

    // infinite loop; just so we don't leave this stack frame
    loop {}
}
```

Microcontroller programs are different from standard programs in two aspects: `#![no_std]` and `#![no_main]`.

The `no_std` attribute says that this program won't use the `std` crate, which assumes an underlying OS; the program will instead use the `core` crate, a subset of `std` that can run on bare metal systems (i.e., systems without OS abstractions like files and sockets).

The `no_main` attribute says that this program won't use the standard `main` interface, which is tailored for command line applications that receive arguments. Instead of the standard `main` we'll use the `entry` attribute from the `cortex-m-rt` crate to define a custom entry point. In this program we have named the entry point "main", but any other name could have been used. The entry point function

must have the signature `fn() -> !`; this type indicates that the function can't return – this means that the program never terminates.

If you are a careful observer, you'll also notice there is a `.cargo` directory in the Cargo project as well. This directory contains a Cargo configuration file (`.cargo/config`) that tweaks the linking process to tailor the memory layout of the program to the requirements of the target device. This modified linking process is a requirement of the `cortex-m-rt` crate. You'll also be making further tweaks to `.cargo/config` in future sections to make building and debugging easier.

Alright, let's start by building this program.

Build it

The first step is to build our “binary” crate. Because the microcontroller has a different architecture than your computer we’ll have to cross compile. Cross compiling in Rust land is as simple as passing an extra `--target` flag to `rustc` or Cargo. The complicated part is figuring out the argument of that flag: the *name* of the target.

The microcontroller in the F3 has a Cortex-M4F processor in it. `rustc` knows how to cross compile to the Cortex-M architecture and provides 4 different targets that cover the different processor families within that architecture:

- `thumbv6m-none-eabi` , for the Cortex-M0 and Cortex-M1 processors
- `thumbv7m-none-eabi` , for the Cortex-M3 processor
- `thumbv7em-none-eabi` , for the Cortex-M4 and Cortex-M7 processors
- `thumbv7em-none-eabihf` , for the Cortex-M4F and Cortex-M7F processors

For the F3, we’ll use the `thumbv7em-none-eabihf` target. Before cross compiling you have to download a pre-compiled version of the standard library (a reduced version of it actually) for your target. That’s done using `rustup` :

```
rustup target add thumbv7em-none-eabihf
```

You only need to do the above step once; `rustup` will re-install a new standard library (`rust-std` component) whenever you update your toolchain.

With the `rust-std` component in place you can now cross compile the program using Cargo.

NOTE Make sure you are in the `src/05-led-roulette` directory and run `cargo build` command below to create the executable:

```
cargo build --target thumbv7em-none-eabihf
```

On your console you should see something like:

```
$ cargo build --target thumbv7em-none-eabihf
Compiling typenum v1.12.0
Compiling semver-parser v0.7.0
Compiling version_check v0.9.2
Compiling nb v1.0.0
Compiling void v1.0.2
Compiling autocfg v1.0.1
Compiling cortex-m v0.7.1
Compiling proc-macro2 v1.0.24
Compiling vcell v0.1.3
Compiling unicode-xid v0.2.1
Compiling stable_deref_trait v1.2.0
Compiling syn v1.0.60
Compiling bitfield v0.13.2
Compiling cortex-m v0.6.7
Compiling cortex-m-rt v0.6.13
Compiling r0 v0.2.2
Compiling stm32-usbd v0.5.1
Compiling stm32f3 v0.12.1
Compiling usb-device v0.2.7
Compiling cfg-if v1.0.0
Compiling paste v1.0.4
Compiling stm32f3-discovery v0.6.0
Compiling embedded-dma v0.1.2
Compiling volatile-register v0.2.0
Compiling nb v0.1.3
Compiling embedded-hal v0.2.4
Compiling semver v0.9.0
Compiling generic-array v0.14.4
Compiling switch-hal v0.3.2
Compiling num-traits v0.2.14
Compiling num-integer v0.1.44
Compiling rustc_version v0.2.3
Compiling bare-metal v0.2.5
Compiling cast v0.2.3
Compiling quote v1.0.9
Compiling generic-array v0.13.2
Compiling generic-array v0.12.3
Compiling generic-array v0.11.1
Compiling panic-itm v0.4.2
Compiling lsm303dlhc v0.2.0
Compiling as-slice v0.1.4
Compiling micromath v1.1.0
Compiling accelerometer v0.12.0
Compiling chrono v0.4.19
Compiling aligned v0.3.4
Compiling rtcc v0.2.0
Compiling cortex-m-rt-macros v0.1.8
Compiling stm32f3xx-hal v0.6.1
Compiling aux5 v0.2.0 (~/.cargo/registry/src/05-led-roulette/auxiliary)
Compiling led-roulette v0.2.0 (~/.cargo/registry/src/05-led-roulette)
Finished dev [unoptimized + debuginfo] target(s) in 17.91s
```

NOTE Be sure to compile this crate *without* optimizations. The provided Cargo.toml file and build command above will ensure optimizations are off.

OK, now we have produced an executable. This executable won't blink any LEDs, it's just a simplified version that we will build upon later in the chapter. As a sanity check, let's verify that the produced

```
cargo readobj --target thumbv7em-none-eabihf --bin led-roulette -- --file-header
```

The `cargo readobj ..` above is equivalent to `readelf -h target/thumbv7em-none-eabihf/debug/led-roulette` and should produce something similar to:

```
$ cargo readobj --target thumbv7em-none-eabihf --bin led-roulette -- --file-header
  Finished dev [unoptimized + debuginfo] target(s) in 0.02s
ELF Header:
  Magic:   7f 45 4c 46 01 01 01 00 00 00 00 00 00 00 00 00
  Class:                               ELF32
  Data:                                   2's complement, little endian
  Version:                             1 (current)
  OS/ABI:                               UNIX - System V
  ABI Version:                          0
  Type:                                  EXEC (Executable file)
  Machine:                              ARM
  Version:                              0x1
  Entry point address:                  0x8000195
  Start of program headers:             52 (bytes into file)
  Start of section headers:            818328 (bytes into file)
  Flags:                                0x5000400
  Size of this header:                  52 (bytes)
  Size of program headers:              32 (bytes)
  Number of program headers:            4
  Size of section headers:              40 (bytes)
  Number of section headers:            22
  Section header string table index: 20
```

Next, we'll flash the program into our microcontroller.

Flash it

Flashing is the process of moving our program into the microcontroller's (persistent) memory. Once flashed, the microcontroller will execute the flashed program every time it is powered on.

In this case, our `led-roulette` program will be the *only* program in the microcontroller memory. By this I mean that there's nothing else running on the microcontroller: no OS, no "daemon", nothing.

`led-roulette` has full control over the device.

Onto the actual flashing. First thing we need to do is launch OpenOCD. We did that in the previous section but this time we'll run the command inside a temporary directory (`/tmp` on *nix; `%TEMP%` on Windows).

Make sure the F3 is connected to your computer and run the following commands in a **new terminal**.

For *nix & MacOS:

```
cd /tmp
openocd -f interface/stlink-v2-1.cfg -f target/stm32f3x.cfg
```

For Windows Note: substitute C: for the actual OpenOCD path:

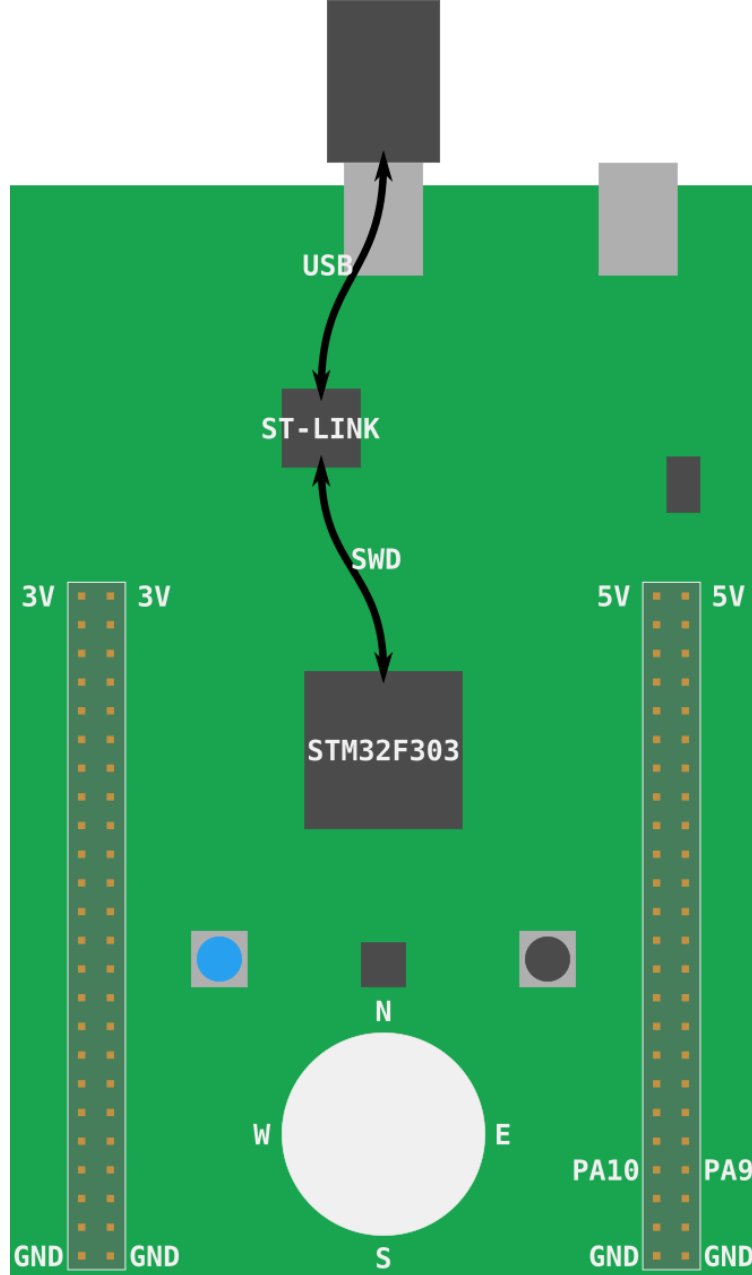
```
cd %TEMP%
openocd -s C:\share\scripts -f interface/stlink-v2-1.cfg -f target/stm32f3x.cfg
```

NOTE Older revisions of the board need to pass slightly different arguments to `openocd`. Review [this section](#) for the details.

The program will block; leave that terminal open.

Now it's a good time to explain what the `openocd` command is actually doing.

I mentioned that the STM32F3DISCOVERY (aka F3) actually has two microcontrollers. One of them is used as a programmer/debugger. The part of the board that's used as a programmer is called ST-LINK (that's what STMicroelectronics decided to call it). This ST-LINK is connected to the target microcontroller using a Serial Wire Debug (SWD) interface (this interface is an ARM standard so you'll run into it when dealing with other Cortex-M based microcontrollers). This SWD interface can be used to flash and debug a microcontroller. The ST-LINK is connected to the "USB ST-LINK" port and will appear as a USB device when you connect the F3 to your computer.



As for OpenOCD, it's software that provides some services like a *GDB server* on top of USB devices that expose a debugging protocol like SWD or JTAG.

Onto the actual command: those `.cfg` files we are using instruct OpenOCD to look for a ST-LINK USB device (`interface/stlink-v2-1.cfg`) and to expect a STM32F3XX microcontroller (`target/stm32f3x.cfg`) to be connected to the ST-LINK.

The OpenOCD output looks like this:

```
$ openocd -f interface/stlink-v2-1.cfg -f target/stm32f3x.cfg
Open On-Chip Debugger 0.10.0
Licensed under GNU GPL v2
For bug reports, read
    http://openocd.org/doc/doxygen/bugs.html
Info : auto-selecting first available session transport "hla_swd". To override use
'transport select <transport>'.
adapter speed: 1000 kHz
adapter_nsrst_delay: 100
Info : The selected transport took over low-level target control. The results might
differ compared to plain JTAG/SWD
none separate
Info : Unable to match requested speed 1000 kHz, using 950 kHz
Info : Unable to match requested speed 1000 kHz, using 950 kHz
Info : clock speed 950 kHz
Info : STLINK v2 JTAG v37 API v2 SWIM v26 VID 0x0483 PID 0x374B
Info : using stlink api v2
Info : Target voltage: 2.888183
Info : stm32f3x.cpu: hardware has 6 breakpoints, 4 watchpoints
```

The “6 breakpoints, 4 watchpoints” part indicates the debugging features the processor has available.

Leave that `openocd` process running, and in the previous terminal or a new terminal **make sure that you are inside the project's `src/05-led-roulette/` directory.**

I mentioned that OpenOCD provides a GDB server so let's connect to that right now:

Execute GDB

First, we need to determine what version of `gdb` you have that is capable of debugging ARM binaries.

This could be any one of the commands below, try each one:

```
arm-none-eabi-gdb -q -ex "target remote :3333" target/thumbv7em-none-eabihf/debug/led-roulette
```

```
gdb-multiarch -q -ex "target remote :3333" target/thumbv7em-none-eabihf/debug/led-roulette
```

```
gdb -q -ex "target remote :3333" target/thumbv7em-none-eabihf/debug/led-roulette
```

NOTE: If you are getting `target/thumbv7em-none-eabihf/debug/led-roulette: No such file or directory` error, try adding `../../` to the file path, for example:

```
$ gdb -q -ex "target remote :3333" ../../target/thumbv7em-none-eabihf/debug/led-roulette
```

This is caused by each example project being in a `workspace` that contains the entire book, and workspaces have a single `target` directory. Check out [Workspaces chapter in Rust Book] for more.

Failing case

You can detect a failing case if there is a `warning` or `error` after the `Remote debugging using :3333` line:

```
$ gdb -q -ex "target remote :3333" target/thumbv7em-none-eabihf/debug/led-roulette
Reading symbols from target/thumbv7em-none-eabihf/debug/led-roulette...
Remote debugging using :3333
warning: Architecture rejected target-supplied description
Truncated register 16 in remote 'g' packet
(gdb)
```

Successful case

Successful case 1:

```
$ arm-none-eabi-gdb -q -ex "target remote :3333" target/thumbv7em-none-eabihf/debug/led-roulette
Reading symbols from target/thumbv7em-none-eabihf/debug/led-roulette...
Remote debugging using :3333
cortex_m_rt::Reset () at ~/.cargo/registry/src/github.com-1ecc6299db9ec823/cortex-m-rt-0.6.13/src/lib.rs:497
497      pub unsafe extern "C" fn Reset() -> ! {
(gdb)
```

Successful case 2:

```
~/embedded-discovery/src/05-led-roulette (master)
$ arm-none-eabi-gdb -q -ex "target remote :3333" target/thumbv7em-none-eabihf/debug/led-roulette
Reading symbols from target/thumbv7em-none-eabihf/debug/led-roulette...
Remote debugging using :3333
0x00000000 in ?? ()
(gdb)
```

In both failing and successful cases you should see new output in the **OpenOCD terminal**, something like the following:

```
Info : stm32f3x.cpu: hardware has 6 breakpoints, 4 watchpoints
+Info : accepting 'gdb' connection on tcp/3333
+Info : device id = 0x10036422
+Info : flash size = 256kbytes
```

NOTE If you are getting an error like `undefined debug reason 7 - target needs reset`, you

By default OpenOCD's GDB server listens on TCP port 3333 (localhost). This command is connecting to that port.

Update `../.cargo/config.toml`

Now that you've successfully determined which debugger you need to use we need to change `../.cargo/config.toml` so that the `cargo run` command will succeed.

NOTE `cargo` is the Rust package manager and you can read about it [here](#).

Get back to the terminal prompt and look at `../.cargo/config.toml`:

```
~/embedded-discovery/src/05-led-roulette
$ cat ../.cargo/config.toml
# default runner starts a GDB session, which requires OpenOCD to be
# running, e.g.,
## openocd -f interface/stlink.cfg -f target/stm32f3x.cfg
# depending on your local GDB, pick one of the following
[target.thumbv7em-none-eabihf]
runner = "arm-none-eabi-gdb -q -x ../openocd.gdb"
# runner = "gdb-multiarch -q -x ../openocd.gdb"
# runner = "gdb -q -x ../openocd.gdb"
rustflags = [
  "-C", "link-arg=-Tlink.x",
]

[build]
target = "thumbv7em-none-eabihf"
```

Use your favorite editor to edit `../.cargo/config.toml` so that the `runner` line contains the correct name of that debugger:

```
nano ../.cargo/config.toml
```

For example, if your debugger was `gdb-multiarch` then after editing the `git diff` should be:

```
$ git diff ../.cargo/config.toml
diff --git a/f3discovery/src/.cargo/config.toml b/f3discovery/src/.cargo/config.toml
index 2f38f6b..95860a0 100644
--- a/f3discovery/src/.cargo/config.toml
+++ b/f3discovery/src/.cargo/config.toml
@@ -3,8 +3,8 @@
  ## openocd -f interface/stlink.cfg -f target/stm32f3x.cfg
  # depending on your local GDB, pick one of the following
  [target.thumbv7em-none-eabihf]
- runner = "arm-none-eabi-gdb -q -x ../openocd.gdb"
-# runner = "gdb-multiarch -q -x ../openocd.gdb"
+# runner = "arm-none-eabi-gdb -q -x ../openocd.gdb"
+runner = "gdb-multiarch -q -x ../openocd.gdb"
  # runner = "gdb -q -x ../openocd.gdb"
  rustflags = [
    "-C", "link-arg=-Tlink.x",
```

Now that you have `../.cargo/config.toml` setup let's test it using `cargo run` to start the debug session.

NOTE The `--target thumbv7em-none-eabihf` defines which architecture to build and run. In our `../.cargo/config.toml` file we have `target = "thumbv7em-none-eabihf"` so it is actually not necessary to specify `--target` we do it here just so you know that parameters on the command line can be used and they override those in `config.toml` files.

```
cargo run --target thumbv7em-none-eabihf
```

Results in:

```
~/embedded-discovery/src/05-led-roulette
$ cargo run --target thumbv7em-none-eabihf
    Finished dev [unoptimized + debuginfo] target(s) in 0.14s
    Running `gdb-multiarch -q -x ../openocd.gdb /home/adam/vc/rust-
training/discovery/f3discovery/target/thumbv7em-none-eabihf/debug/led-roulette`
Reading symbols from /home/adam/vc/rust-
training/discovery/f3discovery/target/thumbv7em-none-eabihf/debug/led-roulette...
0x08000230 in core::fmt::Arguments::new_v1 (pieces=..., args=...)
    at /rustc/d5a82bbd26e1ad8b7401f6a718a9c57c96905483/library/core/src/fmt/mod.rs:394
394      /rustc/d5a82bbd26e1ad8b7401f6a718a9c57c96905483/library/core/src/fmt/mod.rs:
No such file or directory.
Loading section .vector_table, size 0x194 lma 0x80000000
Loading section .text, size 0x1ad8 lma 0x8000194
Loading section .rodata, size 0x5a4 lma 0x8001c6c
Start address 0x08000194, load size 8720
Transfer rate: 12 KB/sec, 2906 bytes/write.
Breakpoint 1 at 0x80001e8: file src/05-led-roulette/src/main.rs, line 7.
Note: automatically using hardware breakpoints for read-only addresses.
Breakpoint 2 at 0x800020a: file src/lib.rs, line 570.
Breakpoint 3 at 0x8001c5a: file src/lib.rs, line 560.

Breakpoint 1, led_roulette::__cortex_m_rt_main_trampoline () at src/05-led-
roulette/src/main.rs:7
7      #[entry]
halted: PC: 0x080001ee
led_roulette::__cortex_m_rt_main () at src/05-led-roulette/src/main.rs:10
10          let x = 42;
```

Bravo, we will be modifying `../.cargo/config.toml` in future. **But**, since this file is shared with all of the chapters those changes should be made with that in mind. If you want or we need to make changes that only pertain to a particular chapter then create a `.cargo/config.toml` local to that chapter directory.

Flash the device

Assuming you have GDB running, if not start it as suggested in the previous section.

NOTE The `-x ../openocd.gdb` arguments to `gdb` is already setup to flash the device, so explicitly flashing the project code to the device is normally handled with a simple `cargo run`. We'll cover the openocd configuration script in the next section.

Now use the `load` command in `gdb` to actually flash the program into the device:

```
(gdb) load
Loading section .vector_table, size 0x194 lma 0x80000000
Loading section .text, size 0x20ec lma 0x8000194
Loading section .rodata, size 0x514 lma 0x8002280
Start address 0x08000194, load size 10132
Transfer rate: 17 KB/sec, 3377 bytes/write.
```

You'll also see new output in the OpenOCD terminal, something like:

```
Info : flash size = 256kbytes
+Info : Unable to match requested speed 1000 kHz, using 950 kHz
+Info : Unable to match requested speed 1000 kHz, using 950 kHz
+adapter speed: 950 kHz
+target halted due to debug-request, current mode: Thread
+xPSR: 0x01000000 pc: 0x08000194 msp: 0x2000a000
+Info : Unable to match requested speed 8000 kHz, using 4000 kHz
+Info : Unable to match requested speed 8000 kHz, using 4000 kHz
+adapter speed: 4000 kHz
+target halted due to breakpoint, current mode: Thread
+xPSR: 0x61000000 pc: 0x2000003a msp: 0x2000a000
+Info : Unable to match requested speed 1000 kHz, using 950 kHz
+Info : Unable to match requested speed 1000 kHz, using 950 kHz
+adapter speed: 950 kHz
+target halted due to debug-request, current mode: Thread
+xPSR: 0x01000000 pc: 0x08000194 msp: 0x2000a000
```

Our program is loaded, let's debug it!

Debug it

We are already inside a debugging session so let's debug our program.

After the `load` command, our program is stopped at its *entry point*. This is indicated by the “Start address 0x8000XXX” part of GDB's output. The entry point is the part of a program that a processor / CPU will execute first.

The starter project I've provided to you has some extra code that runs *before* the `main` function. At this time, we are not interested in that “pre-main” part so let's skip right to the beginning of the `main` function. We'll do that using a breakpoint. Issue `break main` at the `(gdb)` prompt:

NOTE For these GDB commands I generally won't provide a copyable code block as these are short and it's faster just to type them yourself. In addition most can be shortened. For instance `b` for `break` or `s` for `step`, see [GDB Quick Reference](#) for more info or use Google to find your others. In addition, you can use tab completion by typing the first few letters than one tab to complete or two tabs to see all possible commands.

Finally, `help xxxx` where `xxxx` is the command will provide short names and other info:

```
(gdb) help s
step, s
Step program until it reaches a different source line.
Usage: step [N]
Argument N means step N times (or till program stops for another reason).
```

```
(gdb) break main
Breakpoint 1 at 0x80001f0: file src/05-led-roulette/src/main.rs, line 7.
Note: automatically using hardware breakpoints for read-only addresses.
```

Next issue a `continue` command:

```
(gdb) continue
Continuing.

Breakpoint 1, led_roulette::__cortex_m_rt_main_trampoline () at src/05-led-roulette/src/main.rs:7
7          #[entry]
```

Breakpoints can be used to stop the normal flow of a program. The `continue` command will let the program run freely *until* it reaches a breakpoint. In this case, until it reaches `#[entry]` which is a trampoline to the main function and where `break main` sets the breakpoint.

Note that GDB output says “Breakpoint 1”. Remember that our processor can only use six of these breakpoints so it's a good idea to pay attention to these messages.

OK. Since we are stopped at `#[entry]` and using the `disassemble /m` we see the code for entry, which is a trampoline to main. That means it sets up the stack and then invokes a subroutine call to the `main` function using an ARM branch and link instruction, `bl`.

```
(gdb) disassemble /m
Dump of assembler code for function main:
7      #[entry]
      0x080001ec <+0>:      push    {r7, lr}
      0x080001ee <+2>:      mov     r7, sp
=> 0x080001f0 <+4>:      bl      0x80001f6
<_ZN12led_roulette18__cortex_m_rt_main17he61ef18c060014a5E>
      0x080001f4 <+8>:      udf     #254      ; 0xfe

End of assembler dump.
```

Next we need to issue a `step` GDB command which will advance the program statement by statement stepping into functions/procedures. So after this first `step` command we're inside `main` and are positioned at the first executable `rust` statement, line 10, but it is **not** executed:

```
(gdb) step
led_roulette::__cortex_m_rt_main () at src/05-led-roulette/src/main.rs:10
10      let x = 42;
```

Next we'll issue a second `step` which executes line 10 and stops at line `11` `_y = x;`, again line 11 is **not** executed.

NOTE We could have pressed enter at the second `(gdb)` prompt and it would have reissued the previous statement, `step`, but for clarity in this tutorial we'll generally retype the command.

```
(gdb) step
11      _y = x;
```

As you can see, in this mode, on each `step` command GDB will print the current statement along with its line number. As you'll see later in the TUI mode you'll not see the statement in the command area.

We are now "on" the `_y = x` statement; that statement hasn't been executed yet. This means that `x` is initialized but `_y` is not. Let's inspect those stack/local variables using the `print` command, `p` for short:

```
(gdb) print x
$1 = 42
(gdb) p &x
$2 = (*mut i32) 0x20009fe0
(gdb) p _y
$3 = 536870912
(gdb) p &_y
$4 = (*mut i32) 0x20009fe4
```

As expected, `x` contains the value `42`. `_y`, however, contains the value `536870912` (?). This is because `_y` has not been initialized yet, it contains some garbage value.

The command `print &x` prints the address of the variable `x`. The interesting bit here is that GDB output shows the type of the reference: `*mut i32`, a mutable pointer to an `i32` value. Another interesting thing is that the addresses of `x` and `_y` are very close to each other: their addresses are just `4` bytes apart.

Instead of printing the local variables one by one, you can also use the `info locals` command:

```
(gdb) info locals
x = 42
_y = 536870912
```

OK. With another `step`, we'll be on top of the `loop {}` statement:

```
(gdb) step
14      loop {}
```

And `_y` should now be initialized.

```
(gdb) print _y
$5 = 42
```

If we use `step` again on top of the `loop {}` statement, we'll get stuck because the program will never pass that statement.

NOTE If you used the `step` or any other command by mistake and GDB gets stuck, you can get it unstuck by hitting `Ctrl+C`.

As introduced above the `disassemble /m` command can be used to disassemble the program around the line you are currently at. You might also want to `set print asm-demangle on` so the names are demangled, this only needs to be done once a debug session. Later this and other commands will be placed in an initialization file which will simplify starting a debug session.

```
(gdb) set print asm-demangle on
(gdb) disassemble /m
Dump of assembler code for function
_ZN12led_roulette18__cortex_m_rt_main17h51e7c3daad2af251E:
8      fn main() -> ! {
      0x080001f6 <+0>:      sub      sp, #8
      0x080001f8 <+2>:      movs     r0, #42 ; 0x2a

9          let _y;
10         let x = 42;
      0x080001fa <+4>:      str      r0, [sp, #0]

11         _y = x;
      0x080001fc <+6>:      str      r0, [sp, #4]

12
13         // infinite loop; just so we don't leave this stack frame
14         loop {}
=> 0x080001fe <+8>:      b.n      0x8000200 <led_roulette::__cortex_m_rt_main+10>
      0x08000200 <+10>:    b.n      0x8000200 <led_roulette::__cortex_m_rt_main+10>

End of assembler dump.
```

See the fat arrow `=>` on the left side? It shows the instruction the processor will execute next.

Also, as mentioned above if you were to execute the `step` command GDB gets stuck because it is executing a branch instruction to itself and never gets past it. So you need to use `Ctrl+C` to regain control. An alternative is to use the `stepi` (`si`) GDB command, which steps one asm instruction, and GDB will print the address **and** line number of the statement the processor will execute next and it won't get stuck.

```
(gdb) stepi
0x08000194      14      loop {}

(gdb) si
0x08000194      14      loop {}
```

One last trick before we move to something more interesting. Enter the following commands into GDB:

```
(gdb) monitor reset halt
Unable to match requested speed 1000 kHz, using 950 kHz
Unable to match requested speed 1000 kHz, using 950 kHz
adapter speed: 950 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x01000000 pc: 0x08000194 msp: 0x2000a000
```

```
(gdb) continue
Continuing.
```

```
Breakpoint 1, led_roulette::__cortex_m_rt_main_trampoline () at src/05-led-roulette/src/main.rs:7
7      #[entry]
```

```
(gdb) disassemble /m
Dump of assembler code for function main:
7      #[entry]
    0x080001ec <+0>:    push    {r7, lr}
    0x080001ee <+2>:    mov     r7, sp
=> 0x080001f0 <+4>:    bl      0x80001f6 <led_roulette::__cortex_m_rt_main>
    0x080001f4 <+8>:    udf     #254      ; 0xfe

End of assembler dump.
```

We are now back at the beginning of `#[entry]` !

`monitor reset halt` will reset the microcontroller and stop it right at the beginning of the program. The `continue` command will then let the program run freely until it reaches a breakpoint, in this case it is the breakpoint at `#[entry]` .

This combo is handy when you, by mistake, skipped over a part of the program that you were interested in inspecting. You can easily roll back the state of your program back to its very beginning.

The fine print: This `reset` command doesn't clear or touch RAM. That memory will retain its values from the previous run. That shouldn't be a problem though, unless your program behavior depends of the value of *uninitialized* variables but that's the definition of Undefined Behavior (UB).

We are done with this debug session. You can end it with the `quit` command.

```
(gdb) quit
A debugging session is active.
```

```
Inferior 1 [Remote target] will be detached.
```

```
Quit anyway? (y or n) y
Detaching from program: $PWD/target/thumbv7em-none-eabihf/debug/led-roulette, Remote target
Ending remote debugging.
```

For a nicer debugging experience, you can use GDB's Text User Interface (TUI). To enter into that mode enter one of the following commands in the GDB shell:

```
(gdb) layout src
(gdb) layout asm
(gdb) layout split
```

NOTE Apologies to Windows users, the GDB shipped with the GNU ARM Embedded Toolchain may not support this TUI mode :-(.

Below is an example of setting up for a `layout split` by executing the follow commands. As you can see we've dropped passing the `--target` parameter:

```
$ cargo run
(gdb) target remote :3333
(gdb) load
(gdb) set print asm-demangle on
(gdb) set style sources off
(gdb) break main
(gdb) continue
```

Here is a command line with the above commands as `-ex` parameters to save you some typing, shortly we'll be providing an easier way to execute the initial set of commands:

```
cargo run -- -q -ex 'target remote :3333' -ex 'load' -ex 'set print asm-demangle on' -
ex 'set style sources off' -ex 'b main' -ex 'c' target/thumbv7em-none-
eabihf/debug/led-roulette
```

And below is the result:

```
wink@3900x:~/prgs/rust/tutorial/embedded-discovery/src/05-led-roulette (New-05-LED-roulette)
$ cargo run --target thumbv7em-none-eabihf
    Finished dev [unoptimized + debuginfo] target(s) in 0.01s
    Running `arm-none-eabi-gdb -q /home/wink/prgs/rust/tutorial/embedded-discovery/target/thumbv7em-none-eabihf/debug/led-roulette`
Reading symbols from /home/wink/prgs/rust/tutorial/embedded-discovery/target/thumbv7em-none-eabihf/debug/led-roulette...
(gdb) target remote :3333
Remote debugging using :3333
cortex_m_rt::Reset () at /home/wink/.cargo/registry/src/github.com-1ecc6299db9ec823/cortex-m-rt-0.6.13/src/lib.rs:497
497     pub unsafe extern "C" fn Reset() -> ! {
(gdb) load
Loading section .vector_table, size 0x194 lma 0x8000000
Loading section .text, size 0x21cc lma 0x8000194
Loading section .rodata, size 0x594 lma 0x8002360
Start address 0x08000194, load size 10484
Transfer rate: 17 KB/sec, 3494 bytes/write.
(gdb) set print asm-demangle on
(gdb) set style sources off
(gdb) break main
Breakpoint 1 at 0x80001f0: file src/05-led-roulette/src/main.rs, line 7.
Note: automatically using hardware breakpoints for read-only addresses.
(gdb) continue
Continuing.

Breakpoint 1, led_roulette::__cortex_m_rt_main_trampoline () at src/05-led-roulette/src/main.rs:7
7      #[entry]
(gdb) layout split
```

Now we'll scroll the top source window down so we see the entire file and execute `layout split` and then `step`:

```

src/05-led-roulette/src/main.rs
1      #![deny(unsafe_code)]
2      #![no_main]
3      #![no_std]
4
5      use aux5::entry;
6
B+ 7      #[entry]
8      fn main() -> ! {
9          let _y;
>10         let x = 42;
11         _y = x;
12
13         // infinite loop; just so we don't leave this stack frame
14         loop {}
15     }

```

```

0x80001ec <led_roulette::__cortex_m_rt_main_trampoline>    push    {r7, lr}
0x80001ee <led_roulette::__cortex_m_rt_main_trampoline+2>    mov     r7, sp
B+ 0x80001f0 <led_roulette::__cortex_m_rt_main_trampoline+4>    bl      0x80001f6 <led_roulette::__cortex_m_rt_main>
0x80001f4 <led_roulette::__cortex_m_rt_main_trampoline+8>    udf     #254      ; 0xfe
0x80001f6 <led_roulette::__cortex_m_rt_main>                sub     sp, #8
0x80001f8 <led_roulette::__cortex_m_rt_main+2>              movs    r0, #42 ; 0x2a
>0x80001fa <led_roulette::__cortex_m_rt_main+4>              str     r0, [sp, #0]
0x80001fc <led_roulette::__cortex_m_rt_main+6>              str     r0, [sp, #4]
0x80001fe <led_roulette::__cortex_m_rt_main+8>              b.n     0x8000200 <led_roulette::__cortex_m_rt_main+10>
0x8000200 <led_roulette::__cortex_m_rt_main+10>            b.n     0x8000200 <led_roulette::__cortex_m_rt_main+10>
0x8000202 <cortex_m_rt::Reset::trampoline>                  push    {r7, lr}
0x8000204 <cortex_m_rt::Reset::trampoline+2>                  mov     r7, sp
0x8000206 <cortex_m_rt::Reset::trampoline+4>                  bl      0x80001ec <led_roulette::__cortex_m_rt_main_trampoline>
0x800020a <cortex_m_rt::Reset::trampoline+8>                  udf     #254      ; 0xfe
0x800020c <cortex_m_rt::DefaultHandler_>                    push    {r7, lr}
0x800020e <cortex_m_rt::DefaultHandler_+2>                        mov     r7, sp
0x8000210 <cortex_m_rt::DefaultHandler_+4>                        sub     sp, #8
0x8000212 <cortex_m_rt::DefaultHandler_+6>                  b.n     0x8000214 <cortex_m_rt::DefaultHandler_+8>
0x8000214 <cortex_m_rt::DefaultHandler_+8>                    movs    r0, #4

```

```

remote Remote target In: led_roulette::__cortex_m_rt_main      L10    PC: 0x80001fa
(gdb) step
led_roulette::__cortex_m_rt_main () at src/05-led-roulette/src/main.rs:10
(gdb)

```

Then we'll execute a few `info locals` and `step`'s:

```

(gdb) info locals
(gdb) step
(gdb) info locals
(gdb) step
(gdb) info locals

```

```

src/05-led-roulette/src/main.rs
1      #[deny(unsafe_code)]
2      #[no_main]
3      #[no_std]
4
5      use aux5::entry;
6
B+ 7      #[entry]
8      fn main() -> ! {
9          let _y;
10         let x = 42;
11         _y = x;
12
13         // infinite loop; just so we don't leave this stack frame
>14         loop {}
15     }

```

```

0x80001ec <led_roulette::__cortex_m_rt_main_trampoline>    push    {r7, lr}
0x80001ee <led_roulette::__cortex_m_rt_main_trampoline+2>    mov     r7, sp
B+ 0x80001f0 <led_roulette::__cortex_m_rt_main_trampoline+4>    bl      0x80001f6 <led_roulette::__cortex_m_rt_main>
0x80001f4 <led_roulette::__cortex_m_rt_main_trampoline+8>    udf     #254      ; 0xfe
0x80001f6 <led_roulette::__cortex_m_rt_main>                sub     sp, #8
0x80001f8 <led_roulette::__cortex_m_rt_main+2>              movs    r0, #42 ; 0x2a
0x80001fa <led_roulette::__cortex_m_rt_main+4>              str     r0, [sp, #0]
0x80001fc <led_roulette::__cortex_m_rt_main+6>              str     r0, [sp, #4]
>0x80001fe <led_roulette::__cortex_m_rt_main+8>              b.n     0x8000200 <led_roulette::__cortex_m_rt_main+10>
0x8000200 <led_roulette::__cortex_m_rt_main+10>            b.n     0x8000200 <led_roulette::__cortex_m_rt_main+10>
0x8000202 <cortex_m_rt::Reset::trampoline>                  push    {r7, lr}
0x8000204 <cortex_m_rt::Reset::trampoline+2>                  mov     r7, sp
0x8000206 <cortex_m_rt::Reset::trampoline+4>                  bl      0x80001ec <led_roulette::__cortex_m_rt_main_trampoline>
0x800020a <cortex_m_rt::Reset::trampoline+8>                  udf     #254      ; 0xfe
0x800020c <cortex_m_rt::DefaultHandler_>                    push    {r7, lr}
0x800020e <cortex_m_rt::DefaultHandler_+2>                      mov     r7, sp
0x8000210 <cortex_m_rt::DefaultHandler_+4>                      sub     sp, #8
0x8000212 <cortex_m_rt::DefaultHandler_+6>                  b.n     0x8000214 <cortex_m_rt::DefaultHandler_+8>
0x8000214 <cortex_m_rt::DefaultHandler_+8>                  movs    r0, #4

```

```

remote Remote target In: led_roulette::__cortex_m_rt_main L14 PC: 0x80001fe
(gdb) step
led_roulette::__cortex_m_rt_main () at src/05-led-roulette/src/main.rs:10
(gdb) info locals
_y = 536870912
(gdb) step
(gdb) info locals
x = 42
_y = 536870912
(gdb) step
(gdb) info locals
x = 42
_y = 42
(gdb)

```

At any point you can leave the TUI mode using the following command:

```
(gdb) tui disable
```

```
wink@3900x:~/prgs/rust/tutorial/embedded-discovery/src/05-led-roulette (New-05-LED-roulette)
$ cargo run --target thumbv7em-none-eabihf
   Finished dev [unoptimized + debuginfo] target(s) in 0.01s
   Running `arm-none-eabi-gdb -q /home/wink/prgs/rust/tutorial/embedded-discovery/target/thumbv7em-none-eabihf/debug/led-roulette`
Reading symbols from /home/wink/prgs/rust/tutorial/embedded-discovery/target/thumbv7em-none-eabihf/debug/led-roulette...
(gdb) target remote :3333
Remote debugging using :3333
led_roulette::__cortex_m_rt_main () at src/05-led-roulette/src/main.rs:14
14      loop {}
(gdb) load
Loading section .vector_table, size 0x194 lma 0x8000000
Loading section .text, size 0x21cc lma 0x8000194
Loading section .rodata, size 0x594 lma 0x8002360
Start address 0x08000194, load size 10484
Transfer rate: 16 KB/sec, 3494 bytes/write.
(gdb) set print asm-demangle on
(gdb) set style sources off
(gdb) break main
Breakpoint 1 at 0x80001f0: file src/05-led-roulette/src/main.rs, line 7.
Note: automatically using hardware breakpoints for read-only addresses.
(gdb) continue
Continuing.

Breakpoint 1, led_roulette::__cortex_m_rt_main_trampoline () at src/05-led-roulette/src/main.rs:7
7      #[entry]
(gdb) layout split
(gdb)
```

NOTE If the default GDB CLI is not to your liking check out [gdb-dashboard](#). It uses Python to turn the default GDB CLI into a dashboard that shows registers, the source view, the assembly view and other things.

Don't close OpenOCD though! We'll use it again and again later on. It's better just to leave it running. If you want to learn more about what GDB can do, check out the section [How to use GDB](#).

What's next? The high level API I promised.

The Led and Delay abstractions

Now, I'm going to introduce two high level abstractions that we'll use to implement the LED roulette application.

The auxiliary crate, `aux5`, exposes an initialization function called `init`. When called this function returns two values packed in a tuple: a `Delay` value and a `LedArray` value.

`Delay` can be used to block your program for a specified amount of milliseconds.

`LedArray` is an array of eight `Led`s. Each `Led` represents one of the LEDs on the F3 board, and exposes two methods: `on` and `off` which can be used to turn the LED on or off, respectively.

Let's try out these two abstractions by modifying the starter code to look like this:

```
#![deny(unsafe_code)]
#![no_main]
#![no_std]

use aux5::{entry, Delay, DelayMs, LedArray, OutputSwitch};

#[entry]
fn main() -> ! {
    let (mut delay, mut leds): (Delay, LedArray) = aux5::init();

    let half_period = 500_u16;

    loop {
        leds[0].on().ok();
        delay.delay_ms(half_period);

        leds[0].off().ok();
        delay.delay_ms(half_period);
    }
}
```

Now build it:

```
cargo build
```

NOTE: It's possible to forget to rebuild the program *before* starting a GDB session; this omission can lead to very confusing debug sessions. To avoid this problem you can call just `cargo run` instead of `cargo build`. The `cargo run` command will build *and* start a debug session ensuring you never forget to recompile your program.

Now we'll run and repeat the flashing procedure as we did in the previous section but with the new program. I'll let you type in the `cargo run`, *this will get easier shortly.* :)

NOTE: Don't forget to start `openocd` (debugger) on a separate terminal. Otherwise `target`

```
$ cargo run
  Finished dev [unoptimized + debuginfo] target(s) in 0.01s
  Running `arm-none-eabi-gdb -q ~/embedded-discovery/target/thumbv7em-none-eabihf/debug/led-roulette`
Reading symbols from ~/embedded-discovery/target/thumbv7em-none-eabihf/debug/led-roulette...

(gdb) target remote :3333
Remote debugging using :3333
led_roulette:::__cortex_m_rt_main_trampoline () at ~/embedded-discovery/src/05-led-roulette/src/main.rs:7
7          #[entry]

(gdb) load
Loading section .vector_table, size 0x194 lma 0x80000000
Loading section .text, size 0x52c0 lma 0x8000194
Loading section .rodata, size 0xb50 lma 0x8005454
Start address 0x08000194, load size 24484
Transfer rate: 21 KB/sec, 6121 bytes/write.

(gdb) break main
Breakpoint 1 at 0x8000202: file ~/embedded-discovery/src/05-led-roulette/src/main.rs, line 7.
Note: automatically using hardware breakpoints for read-only addresses.

(gdb) continue
Continuing.

Breakpoint 1, led_roulette:::__cortex_m_rt_main_trampoline ()
  at ~/embedded-discovery/src/05-led-roulette/src/main.rs:7
7          #[entry]

(gdb) step
led_roulette:::__cortex_m_rt_main () at ~/embedded-discovery/src/05-led-roulette/src/main.rs:9
9          let (mut delay, mut leds): (Delay, LedArray) = aux5::init();

(gdb)
```

OK. Let's step through the code. This time, we'll use the `next` command instead of `step`. The difference is that the `next` command will step over function calls instead of going inside them.

```
(gdb) next
11          let half_period = 500_u16;

(gdb) next
13          loop {

(gdb) next
14              leds[0].on().ok();

(gdb) next
15              delay.delay_ms(half_period);
```

After executing the `leds[0].on().ok()` statement, you should see a red LED, the one pointing North, turn on.

```
(gdb) next
17          leds[0].off().ok();

(gdb) next
18          delay.delay_ms(half_period);
```

The `delay_ms` call will block the program for half a second but you may not notice because the `next` command also takes some time to execute. However, after stepping over the `leds[0].off()` statement you should see the red LED turn off.

You can already guess what this program does. Let it run uninterrupted using the `continue` command.

```
(gdb) continue
Continuing.
```

Now, let's do something more interesting. We are going to modify the behavior of our program using GDB.

First, let's stop the infinite loop by hitting `Ctrl+C`. You'll probably end up somewhere inside `Led::on`, `Led::off` or `delay_ms`:

```
^C
Program received signal SIGINT, Interrupt.
0x08003434 in core::ptr::read_volatile<u32> (src=0xe000e010)
    at ~/.rustup/toolchains/stable-x86_64-unknown-linux-
gnu/lib/rustlib/src/rust/library/core/src/ptr/mod.rs:1053
```

In my case, the program stopped its execution inside a `read_volatile` function. GDB output shows some interesting information about that: `core::ptr::read_volatile (src=0xe000e010)`. This means that the function comes from the `core` crate and that it was called with argument `src = 0xe000e010`.

Just so you know, a more explicit way to show the arguments of a function is to use the `info args` command:

```
(gdb) info args
src = 0xe000e010
```

Regardless of where your program may have stopped you can always look at the output of the `backtrace` command (`bt` for short) to learn how it got there:

```
(gdb) backtrace
#0  0x08003434 in core::ptr::read_volatile<u32> (src=0xe000e010)
    at ~/.rustup/toolchains/stable-x86_64-unknown-linux-
gnu/lib/rustlib/src/rust/library/core/src/ptr/mod.rs:1053
#1  0x08002d66 in vcell::VolatileCell<u32>::get<u32> (self=0xe000e010) at
~/.cargo/registry/src/github.com-1ecc6299db9ec823/vcell-0.1.3/src/lib.rs:33
#2  volatile_register::RW<u32>::read<u32> (self=0xe000e010) at
~/.cargo/registry/src/github.com-1ecc6299db9ec823/volatile-register-
0.2.0/src/lib.rs:75
#3  cortex_m::peripheral::SYST::has_wrapped (self=0x20009fa4)
    at ~/.cargo/registry/src/github.com-1ecc6299db9ec823/cortex-m-
0.6.4/src/peripheral/syst.rs:136
#4  0x08003004 in stm32f3xx_hal::delay::{{impl}}::delay_us (self=0x20009fa4,
us=500000)
    at ~/.cargo/registry/src/github.com-1ecc6299db9ec823/stm32f3xx-hal-
0.5.0/src/delay.rs:58
#5  0x08002f3e in stm32f3xx_hal::delay::{{impl}}::delay_ms (self=0x20009fa4, ms=500)
    at ~/.cargo/registry/src/github.com-1ecc6299db9ec823/stm32f3xx-hal-
0.5.0/src/delay.rs:32
#6  0x08002f80 in stm32f3xx_hal::delay::{{impl}}::delay_ms (self=0x20009fa4, ms=500)
    at ~/.cargo/registry/src/github.com-1ecc6299db9ec823/stm32f3xx-hal-
0.5.0/src/delay.rs:38
#7  0x0800024c in led_roulette::__cortex_m_rt_main () at src/05-led-
roulette/src/main.rs:15
#8  0x08000206 in led_roulette::__cortex_m_rt_main_trampoline () at src/05-led-
roulette/src/main.rs:7
```

backtrace will print a trace of function calls from the current function down to main.

Back to our topic. To do what we are after, first, we have to return to the **main** function. We can do that using the **finish** command. This command resumes the program execution and stops it again right after the program returns from the current function. We'll have to call it several times.

```
(gdb) finish
Run till exit from #0  0x08003434 in core::ptr::read_volatile<u32> (src=0xe000e010)
  at ~/.rustup/toolchains/stable-x86_64-unknown-linux-
gnu/lib/rustlib/src/rust/library/core/src/ptr/mod.rs:1053
cortex_m::peripheral::SYST::has_wrapped (self=0x20009fa4)
  at ~/.cargo/registry/src/github.com-1ecc6299db9ec823/cortex-m-
0.6.4/src/peripheral/syst.rs:136
136          self.csr.read() & SYST_CSR_COUNTFLAG != 0
Value returned is $1 = 5

(..)
```

```
(gdb) finish
Run till exit from #0  0x08002f3e in stm32f3xx_hal::delay::{impl}::delay_ms
(self=0x20009fa4, ms=500)
  at ~/.cargo/registry/src/github.com-1ecc6299db9ec823/stm32f3xx-hal-
0.5.0/src/delay.rs:32
0x08002f80 in stm32f3xx_hal::delay::{impl}::delay_ms (self=0x20009fa4, ms=500)
  at ~/.cargo/registry/src/github.com-1ecc6299db9ec823/stm32f3xx-hal-
0.5.0/src/delay.rs:38
38          self.delay_ms(u32(ms));
```

```
(gdb) finish
Run till exit from #0  0x08002f80 in stm32f3xx_hal::delay::{impl}::delay_ms
(self=0x20009fa4, ms=500)
  at ~/.cargo/registry/src/github.com-1ecc6299db9ec823/stm32f3xx-hal-
0.5.0/src/delay.rs:38
0x0800024c in led_roulette::__cortex_m_rt_main () at src/05-led-
roulette/src/main.rs:15
15          delay.delay_ms(half_period);
```

We are back in `main`. We have a local variable in here: `half_period`

```
(gdb) print half_period
$3 = 500
```

Now, we are going to modify this variable using the `set` command:

```
(gdb) set half_period = 100

(gdb) print half_period
$5 = 100
```

If you let program run free again using the `continue` command, you **might** see that the LED will blink at a much faster rate now, but more likely the blink rate didn't change. **What happened?**

Let's stop the program with `Ctrl+C` and then set a break point at `main:14`.

```
(gdb) continue
Continuing.
^C
Program received signal SIGINT, Interrupt.
core::cell::UnsafeCell<u32>::get<u32> (self=0x20009fa4)
  at ~/.rustup/toolchains/stable-x86_64-unknown-linux-
gnu/lib/rustlib/src/rust/library/core/src/cell.rs:1711
1711      pub const fn get(&self) -> *mut T {
```

```
(gdb) break main.rs:14
Breakpoint 2 at 0x8000236: file src/05-led-roulette/src/main.rs, line 14.
(gdb) continue
Continuing.

Breakpoint 2, led_roulette::__cortex_m_rt_main () at src/05-led-
roulette/src/main.rs:14
14             leds[0].on().ok();
```

Now open your terminal window so it's about 80 lines long an 170 characters wide if possible.

NOTE: If you can't open the terminal that large, no problem you'll just see `--Type <RET> for more, q to quit, c to continue without paging--` so just type return until you see the `(gdb)` prompt. Then scroll your terminal window to see the results.

```
(gdb) disassemble /m
```

```
Dump of assembler code for function
```

```
_ZN12led_roulette18__cortex_m_rt_main17h51e7c3daad2af251E:
```

```

8      fn main() -> ! {
    0x08000208 <+0>:      push    {r7, lr}
    0x0800020a <+2>:      mov     r7, sp
    0x0800020c <+4>:      sub     sp, #64 ; 0x40
    0x0800020e <+6>:      add     r0, sp, #32

9          let (mut delay, mut leds): (Delay, LedArray) = aux5::init();
    0x08000210 <+8>:      bl      0x8000302 <aux5::init>
    0x08000214 <+12>:     b.n     0x8000216 <led_roulette::__cortex_m_rt_main+14>
    0x08000216 <+14>:     add     r0, sp, #32
    0x08000218 <+16>:     add     r1, sp, #4
    0x0800021a <+18>:     ldmbia.w r0, {r2, r3, r4, r12, lr}
    0x0800021e <+22>:     stmbia.w r1, {r2, r3, r4, r12, lr}
    0x08000222 <+26>:     ldr     r0, [sp, #52] ; 0x34
    0x08000224 <+28>:     ldr     r1, [sp, #56] ; 0x38
    0x08000226 <+30>:     str     r1, [sp, #28]
    0x08000228 <+32>:     str     r0, [sp, #24]
    0x0800022a <+34>:     mov.w   r0, #500 ; 0x1f4

10
11          let half_period = 500_u16;
    0x0800022e <+38>:     strh.w  r0, [r7, #-2]

12
13          loop {
    0x08000232 <+42>:     b.n     0x8000234 <led_roulette::__cortex_m_rt_main+44>
    0x08000234 <+44>:     add     r0, sp, #24
    0x08000268 <+96>:     b.n     0x8000234 <led_roulette::__cortex_m_rt_main+44>

14          leds[0].on().ok();
=> 0x08000236 <+46>:     bl      0x80001ec <switch_hal::output::
{{impl}}::on<stm32f3xx_hal::gpio::gpioe::PEx<stm32f3xx_hal::gpio::Output<stm32f3xx_hal
::gpio::PushPull>>>>
    0x0800023a <+50>:     b.n     0x800023c <led_roulette::__cortex_m_rt_main+52>
    0x0800023c <+52>:     bl      0x8000594 <core::result::Result<(),
core::convert::Infallible>::ok<(),core::convert::Infallible>>
    0x08000240 <+56>:     b.n     0x8000242 <led_roulette::__cortex_m_rt_main+58>
    0x08000242 <+58>:     add     r0, sp, #4
    0x08000244 <+60>:     mov.w   r1, #500 ; 0x1f4

15          delay.delay_ms(half_period);
    0x08000248 <+64>:     bl      0x8002f5c <stm32f3xx_hal::delay::{{impl}}::delay_ms>
    0x0800024c <+68>:     b.n     0x800024e <led_roulette::__cortex_m_rt_main+70>
    0x0800024e <+70>:     add     r0, sp, #24

16
17          leds[0].off().ok();
    0x08000250 <+72>:     bl      0x800081a <switch_hal::output::
{{impl}}::off<stm32f3xx_hal::gpio::gpioe::PEx<stm32f3xx_hal::gpio::Output<stm32f3xx_ha
l::gpio::PushPull>>>>
    0x08000254 <+76>:     b.n     0x8000256 <led_roulette::__cortex_m_rt_main+78>
    0x08000256 <+78>:     bl      0x8000594 <core::result::Result<(),
core::convert::Infallible>::ok<(),core::convert::Infallible>>
    0x0800025a <+82>:     b.n     0x800025c <led_roulette::__cortex_m_rt_main+84>
    0x0800025c <+84>:     add     r0, sp, #4
    0x0800025e <+86>:     mov.w   r1, #500 ; 0x1f4

18          delay.delay_ms(half_period);

```

```
0x08000262 <+90>:    bl      0x8002f5c <stm32f3xx_hal::delay::{impl}::delay_ms>
0x08000266 <+94>:    b.n     0x8000268 <led_roulette::__cortex_m_rt_main+96>
```

End of assembler dump.

In the above dump the reason the delay didn't change was because the compiler recognized that `half_period` didn't change and instead in the two places where `delay.delay_ms(half_period);` is called we see `mov.w r1, #500`. So changing the value of `half_period` does nothing!

```
0x08000244 <+60>:    mov.w    r1, #500          ; 0x1f4
15          delay.delay_ms(half_period);
0x08000248 <+64>:    bl      0x8002f5c <stm32f3xx_hal::delay::{impl}::delay_ms>
(..)
0x0800025e <+86>:    mov.w    r1, #500          ; 0x1f4
18          delay.delay_ms(half_period);
0x08000262 <+90>:    bl      0x8002f5c <stm32f3xx_hal::delay::{impl}::delay_ms>
```

One solution to the problem is to wrap `half_period` in a `Volatile` as shown below.

```
#![deny(unsafe_code)]
#![no_main]
#![no_std]

use volatile::Volatile;
use aux5::{Delay, DelayMs, LedArray, OutputSwitch, entry};

#[entry]
fn main() -> ! {
    let (mut delay, mut leds): (Delay, LedArray) = aux5::init();

    let mut half_period = 500_u16;
    let v_half_period = Volatile::new(&mut half_period);

    loop {
        leds[0].on().ok();
        delay.delay_ms(v_half_period.read());

        leds[0].off().ok();
        delay.delay_ms(v_half_period.read());
    }
}
```

Edit `Cargo.toml` adding `volatile = "0.4.3"` in the `[dependencies]` section.

```
[dependencies]
aux5 = { path = "auxiliary" }
volatile = "0.4.3"
```

With the above code using `Volatile` you can now change `half_period` and you'll be able to experiment with different values. Here is the list of commands followed by an explanation; `# xxxx` to demonstrate.

```

$ cargo run --target thumbv7em-none-eabihf # Compile and load the program into gdb
(gdb) target remote :3333 # Connect to STM32F3DISCOVERY board from PC
(gdb) load # Flash program
(gdb) break main.rs:16 # Set breakpoint 1 at top of loop
(gdb) continue # Continue, will stop at main.rs:16
(gdb) disable 1 # Disable breakpoint 1
(gdb) set print asm-demangle on # Enable asm-demangle
(gdb) disassemble /m # Disassemble main function
(gdb) continue # Led blinking on for 1/2 sec then off 1/2 sec
^C # Stop with Ctrl+C
(gdb) enable 1 # Enable breakpoint 1
(gdb) continue # Continue, will stop at main.rs:16
(gdb) print half_period # Print half_period result is 500
(gdb) set half_period = 2000 # Set half_period to 2000ms
(gdb) print half_period # Print half_period and result is 2000
(gdb) disable 1 # Disable breakpoint 1
(gdb) continue # Led blinking on for 2 secs then off 2 sec
^C # Stop with Ctrl+C
(gdb) quit # Quit gdb

```

The critical changes are at lines 13, 17 and 20 in the source code which you can see in the disassembly. At 13 we create `v_half_period` and then `read()` its value in lines 17 and 20. This means that when we `set half_period = 2000` the led will now be on for 2 seconds then off for 2 seconds.

```
$ cargo run --target thumbv7em-none-eabihf
Compiling led-roulette v0.2.0 (~/.embedded-discovery/src/05-led-roulette)
Finished dev [unoptimized + debuginfo] target(s) in 0.18s
Running `arm-none-eabi-gdb -q ~/.embedded-discovery/target/thumbv7em-none-eabihf/debug/led-roulette`
Reading symbols from ~/.embedded-discovery/target/thumbv7em-none-eabihf/debug/led-roulette...
```

```
(gdb) target remote :3333
Remote debugging using :3333
led_roulette::~__cortex_m_rt_main () at src/05-led-roulette/src/main.rs:16
16          leds[0].on().ok();
```

```
(gdb) load
Loading section .vector_table, size 0x194 lma 0x80000000
Loading section .text, size 0x5258 lma 0x8000194
Loading section .rodata, size 0xbd8 lma 0x80053ec
Start address 0x08000194, load size 24516
Transfer rate: 21 KB/sec, 6129 bytes/write.
```

```
(gdb) break main.rs:16
Breakpoint 1 at 0x8000246: file src/05-led-roulette/src/main.rs, line 16.
Note: automatically using hardware breakpoints for read-only addresses.
```

```
(gdb) continue
Continuing.
```

```
Breakpoint 1, led_roulette::~__cortex_m_rt_main () at src/05-led-roulette/src/main.rs:16
16          leds[0].on().ok();
```

```
(gdb) disable 1
```

```
(gdb) set print asm-demangle on
```

```
(gdb) disassemble /m
Dump of assembler code for function
_ZN12led_roulette18__cortex_m_rt_main17he1f2bc7990b13731E:
```

```
9      fn main() -> ! {
0x0800020e <+0>:      push    {r7, lr}
0x08000210 <+2>:      mov     r7, sp
0x08000212 <+4>:      sub     sp, #72 ; 0x48
0x08000214 <+6>:      add     r0, sp, #36      ; 0x24

10          let (mut delay, mut leds): (Delay, LedArray) = aux5::init();
0x08000216 <+8>:      bl      0x800036a <aux5::init>
0x0800021a <+12>:     b.n     0x800021c <led_roulette::~__cortex_m_rt_main+14>
0x0800021c <+14>:     add     r0, sp, #36      ; 0x24
0x0800021e <+16>:     add     r1, sp, #8
0x08000220 <+18>:     ldmia.w r0, {r2, r3, r4, r12, lr}
0x08000224 <+22>:     stmia.w r1, {r2, r3, r4, r12, lr}
0x08000228 <+26>:     ldr     r0, [sp, #56]      ; 0x38
0x0800022a <+28>:     ldr     r1, [sp, #60]      ; 0x3c
0x0800022c <+30>:     str     r1, [sp, #32]
0x0800022e <+32>:     str     r0, [sp, #28]
0x08000230 <+34>:     mov.w   r0, #500          ; 0x1f4
```

```
11
12          let mut half_period = 500_u16;
0x08000234 <+38>:     strh.w  r0, [r7, #-6]
0x08000238 <+42>:     subs    r0, r7, #6
```

```

13      let v_half_period = Volatile::new(&mut half_period);
      0x0800023a <+44>:    bl      0x800033e <volatile::Volatile<&mut u16,
volatile::access::ReadWrite>::new<&mut u16>>
      0x0800023e <+48>:    str      r0, [sp, #68]    ; 0x44
      0x08000240 <+50>:    b.n      0x8000242 <led_roulette::__cortex_m_rt_main+52>

14
15      loop {
      0x08000242 <+52>:    b.n      0x8000244 <led_roulette::__cortex_m_rt_main+54>
      0x08000244 <+54>:    add      r0, sp, #28
      0x08000288 <+122>:   b.n      0x8000244 <led_roulette::__cortex_m_rt_main+54>

16      leds[0].on().ok();
=> 0x08000246 <+56>:    bl      0x800032c <switch_hal::output::
{{impl}}::on<stm32f3xx_hal::gpio::gpioe::PEx<stm32f3xx_hal::gpio::Output<stm32f3xx_hal
::gpio::PushPull>>>>
      0x0800024a <+60>:    b.n      0x800024c <led_roulette::__cortex_m_rt_main+62>
      0x0800024c <+62>:    bl      0x80005fc <core::result::Result<(),
core::convert::Infallible>::ok<(),core::convert::Infallible>>
      0x08000250 <+66>:    b.n      0x8000252 <led_roulette::__cortex_m_rt_main+68>
      0x08000252 <+68>:    add      r0, sp, #68      ; 0x44

17      delay.delay_ms(v_half_period.read());
      0x08000254 <+70>:    bl      0x800034a <volatile::Volatile<&mut u16,
volatile::access::ReadWrite>::read<&mut u16,u16,volatile::access::ReadWrite>>
      0x08000258 <+74>:    str      r0, [sp, #4]
      0x0800025a <+76>:    b.n      0x800025c <led_roulette::__cortex_m_rt_main+78>
      0x0800025c <+78>:    add      r0, sp, #8
      0x0800025e <+80>:    ldr      r1, [sp, #4]
      0x08000260 <+82>:    bl      0x8002fc4 <stm32f3xx_hal::delay::{{impl}}::delay_ms>
      0x08000264 <+86>:    b.n      0x8000266 <led_roulette::__cortex_m_rt_main+88>
      0x08000266 <+88>:    add      r0, sp, #28

18
19      leds[0].off().ok();
      0x08000268 <+90>:    bl      0x8000882 <switch_hal::output::
{{impl}}::off<stm32f3xx_hal::gpio::gpioe::PEx<stm32f3xx_hal::gpio::Output<stm32f3xx_ha
l::gpio::PushPull>>>>
      0x0800026c <+94>:    b.n      0x800026e <led_roulette::__cortex_m_rt_main+96>
      0x0800026e <+96>:    bl      0x80005fc <core::result::Result<(),
core::convert::Infallible>::ok<(),core::convert::Infallible>>
      0x08000272 <+100>:   b.n      0x8000274 <led_roulette::__cortex_m_rt_main+102>
      0x08000274 <+102>:   add      r0, sp, #68      ; 0x44

20      delay.delay_ms(v_half_period.read());
      0x08000276 <+104>:   bl      0x800034a <volatile::Volatile<&mut u16,
volatile::access::ReadWrite>::read<&mut u16,u16,volatile::access::ReadWrite>>
      0x0800027a <+108>:   str      r0, [sp, #0]
      0x0800027c <+110>:   b.n      0x800027e <led_roulette::__cortex_m_rt_main+112>
      0x0800027e <+112>:   add      r0, sp, #8
      0x08000280 <+114>:   ldr      r1, [sp, #0]
      0x08000282 <+116>:   bl      0x8002fc4 <stm32f3xx_hal::delay::{{impl}}::delay_ms>
      0x08000286 <+120>:   b.n      0x8000288 <led_roulette::__cortex_m_rt_main+122>

```

End of assembler dump.

(gdb) continue

Continuing.

^C

Program received signal SIGINT, Interrupt.

0x080037b2 in core::cell::UnsafeCell<u32>::get<u32> (self=0x20009fa0) at

```
~/.rustup/toolchains/stable-x86_64-unknown-linux-  
gnu/lib/rustlib/src/rust/library/core/src/cell.rs:1716  
1716      }
```

```
(gdb) enable 1
```

```
(gdb) continue  
Continuing.
```

```
Breakpoint 1, led_roulette::__cortex_m_rt_main () at src/05-led-  
roulette/src/main.rs:16  
16      leds[0].on().ok();
```

```
(gdb) print half_period  
$2 = 500
```

```
(gdb) disable 1
```

```
(gdb) continue  
Continuing.  
^C
```

```
Program received signal SIGINT, Interrupt.  
0x08003498 in core::ptr::read_volatile<u32> (src=0xe000e010) at  
~/.rustup/toolchains/stable-x86_64-unknown-linux-  
gnu/lib/rustlib/src/rust/library/core/src/ptr/mod.rs:1052  
1052      unsafe { intrinsics::volatile_load(src) }
```

```
(gdb) enable 1
```

```
(gdb) continue  
Continuing.
```

```
Breakpoint 1, led_roulette::__cortex_m_rt_main () at src/05-led-  
roulette/src/main.rs:16  
16      leds[0].on().ok();
```

```
(gdb) print half_period  
$3 = 500
```

```
(gdb) set half_period = 2000
```

```
(gdb) print half_period  
$4 = 2000
```

```
(gdb) disable 1
```

```
(gdb) continue  
Continuing.  
^C
```

```
Program received signal SIGINT, Interrupt.  
0x0800348e in core::ptr::read_volatile<u32> (src=0xe000e010) at  
~/.rustup/toolchains/stable-x86_64-unknown-linux-  
gnu/lib/rustlib/src/rust/library/core/src/ptr/mod.rs:1046  
1046      pub unsafe fn read_volatile<T>(src: *const T) -> T {
```

```
(gdb) q
```

```
Detaching from program: ~/embedded-discovery/target/thumbv7em-none-eabi-hf/debug/led-  
roulette, Remote target  
Ending remote debugging.  
[Inferior 1 (Remote target) detached]
```

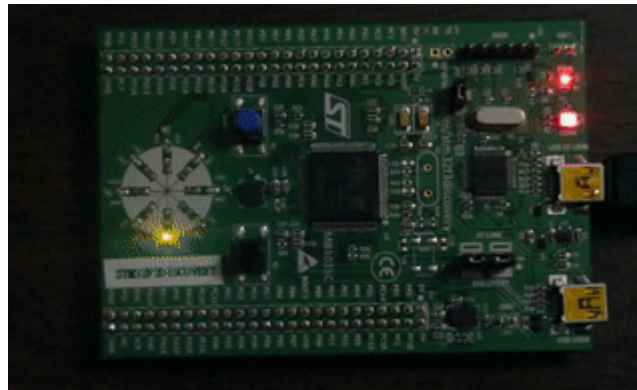
Question! What happens if you start lowering the value of `half_period`? At what value of `half_period` you can no longer see the LED blink?

Now, it's your turn to write a program.

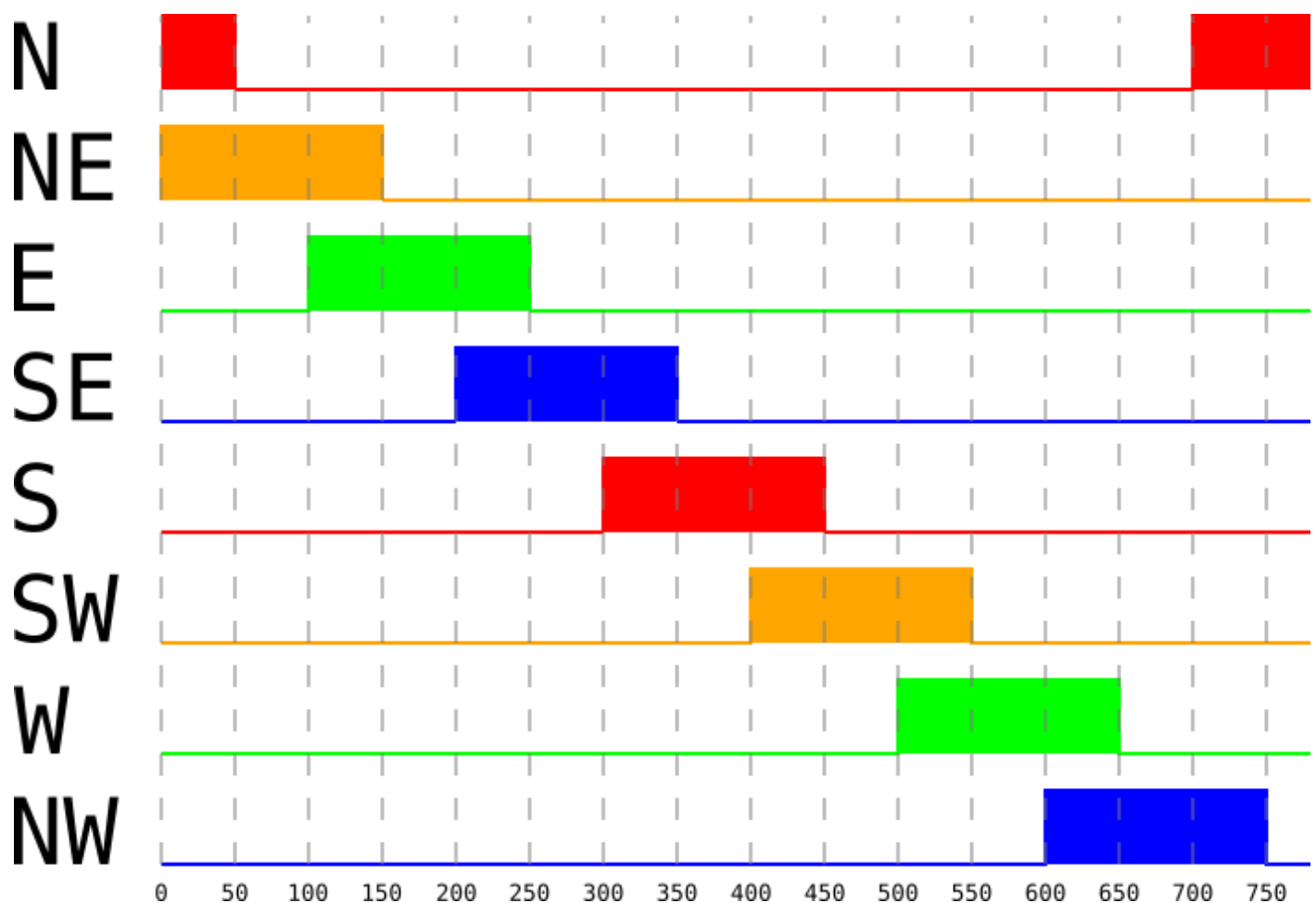
The challenge

You are now well armed to face a challenge! Your task will be to implement the application I showed you at the beginning of this chapter.

Here's the GIF again:



Also, this may help:



This is a timing diagram. It indicates which LED is on at any given instant of time and for how long each LED should be on. On the X axis we have the time in milliseconds. The timing diagram shows a single period. This pattern will repeat itself every 800 ms. The Y axis labels each LED with a cardinal point: North, East, etc. As part of the challenge you'll have to figure out how each element in the `Leds` array maps to these cardinal points (hint: `cargo doc --open ;-`).

Before you attempt this challenge, let me give you one additional tip. Our GDB sessions always involve entering the same commands at the beginning. We can use a `.gdb` file to execute some commands right after GDB is started. This way you can save yourself the effort of having to enter them manually on each GDB session.

As it turns out we've already created `../openocd.gdb` and you can see it's doing pretty much what we did in the previous section plus a few other commands. Look at the comments for additional information:

```
$ cat ../openocd.gdb
# Connect to gdb remote server
target remote :3333

# Load will flash the code
load

# Enable demangling asm names on disassembly
set print asm-demangle on

# Enable pretty printing
set print pretty on

# Disable style sources as the default colors can be hard to read
set style sources off

# Initialize monitoring so iprintln! macro output
# is sent from the itm port to itm.txt
monitor tpiu config internal itm.txt uart off 8000000

# Turn on the itm port
monitor itm port 0 on

# Set a breakpoint at main, aka entry
break main

# Set a breakpoint at DefaultHandler
break DefaultHandler

# Set a breakpoint at HardFault
break HardFault

# Continue running and until we hit the main breakpoint
continue

# Step from the trampoline code in entry into main
step
```

Now we need to modify the `../.cargo/config.toml` file to execute `../openocd.gdb`

```
nano ../.cargo/config.toml
```

Edit your `runner` command `-x ../openocd.gdb`. Assuming you're using `arm-none-eabi-gdb` the diff is:

```
~/embedded-discovery/src/05-led-roulette
$ git diff ../.cargo/config.toml
diff --git a/src/.cargo/config.toml b/src/.cargo/config.toml
index ddff17f..02ac952 100644
--- a/src/.cargo/config.toml
+++ b/src/.cargo/config.toml
@@ -1,5 +1,5 @@
 [target.thumbv7em-none-eabihf]
-runner = "arm-none-eabi-gdb -q"
+runner = "arm-none-eabi-gdb -q -x ../openocd.gdb"
 # runner = "gdb-multiarch -q"
 # runner = "gdb -q"
 rustflags = [
```

And the full contents of `../.cargo/config.toml`, again assuming `arm-none-eabi-gdb`, is:

```
[target.thumbv7em-none-eabihf]
runner = "arm-none-eabi-gdb -q -x ../openocd.gdb"
# runner = "gdb-multiarch -q"
# runner = "gdb -q"
rustflags = [
  "-C", "link-arg=-Tlink.x",
]

[build]
target = "thumbv7em-none-eabihf"
```

With that in place, you can now use a simple `cargo run` command which will build the ARM version of the code and run the `gdb` session. The `gdb` session will automatically flash the program and jump to the beginning of `main` as it `step`'s through the entry trampoline:

```
cargo run
```

```
~/embedded-discovery/src/05-led-roulette (Update-05-led-roulette-WIP)
```

```
$ cargo run
```

```
Finished dev [unoptimized + debuginfo] target(s) in 0.01s
```

```
Running `arm-none-eabi-gdb -q -x openocd.gdb ~/embedded-  
discovery/target/thumbv7em-none-eabihf/debug/led-roulette`
```

```
Reading symbols from ~/embedded-discovery/target/thumbv7em-none-eabihf/debug/led-  
roulette...
```

```
led_roulette::__cortex_m_rt_main_trampoline () at ~/embedded-discovery/src/05-led-  
roulette/src/main.rs:7
```

```
7      #[entry]
```

```
Loading section .vector_table, size 0x194 lma 0x80000000
```

```
Loading section .text, size 0x52c0 lma 0x8000194
```

```
Loading section .rodata, size 0xb50 lma 0x8005454
```

```
Start address 0x08000194, load size 24484
```

```
Transfer rate: 21 KB/sec, 6121 bytes/write.
```

```
Breakpoint 1 at 0x8000202: file ~/embedded-discovery/src/05-led-roulette/src/main.rs,  
line 7.
```

```
Note: automatically using hardware breakpoints for read-only addresses.
```

```
Breakpoint 1, led_roulette::__cortex_m_rt_main_trampoline ()
```

```
at ~/embedded-discovery/src/05-led-roulette/src/main.rs:7
```

```
7      #[entry]
```

```
led_roulette::__cortex_m_rt_main () at ~/embedded-discovery/src/05-led-  
roulette/src/main.rs:9
```

```
9          let (mut delay, mut leds): (Delay, LedArray) = aux5::init();
```

Fork the discovery book

If you haven't already ready, it's probably a good idea to fork the [embedded discovery book](#) so you can save your changes in your own branch of your fork. We suggest creating your own branch and leaving the `master` branch alone so the `master` branch of your fork can stay in sync with the upstream repo. Also, it allows you to more easily create PR's and improve this book, **thank you in advance!**

My solution

What solution did you come up with?

Here's mine:

```
#![deny(unsafe_code)]
#![no_main]
#![no_std]

use aux5::{Delay, DelayMs, LedArray, OutputSwitch, entry};

#[entry]
fn main() -> ! {
    let (mut delay, mut leds): (Delay, LedArray) = aux5::init();

    let ms = 50_u8;
    loop {
        for curr in 0..8 {
            let next = (curr + 1) % 8;

            leds[next].on().ok();
            delay.delay_ms(ms);
            leds[curr].off().ok();
            delay.delay_ms(ms);
        }
    }
}
```

One more thing! Check that your solution also works when compiled in “release” mode:

```
$ cargo build --target thumbv7em-none-eabihf --release
```

You can test it with this `gdb` command:

```
$ # or, you could simply call `cargo run --target thumbv7em-none-eabihf --release`
$ arm-none-eabi-gdb target/thumbv7em-none-eabihf/release/led-roulette
$ #
```

Binary size is something we should always keep an eye on! How big is your solution? You can check that using the `size` command on the release binary:

```
$ # equivalent to size target/thumbv7em-none-eabihf/debug/led-roulette
```

```
$ cargo size --target thumbv7em-none-eabihf --bin led-roulette -- -A
```

```
Finished dev [unoptimized + debuginfo] target(s) in 0.02s
```

```
led-roulette :
```

section	size	addr
.vector_table	404	0x80000000
.text	21144	0x8000194
.rodata	3144	0x800542c
.data	0	0x20000000
.bss	4	0x20000000
.uninit	0	0x20000004
.debug_abbrev	19160	0x0
.debug_info	471239	0x0
.debug_aranges	18376	0x0
.debug_ranges	102536	0x0
.debug_str	508618	0x0
.debug_pubnames	76975	0x0
.debug_pubtypes	112797	0x0
.ARM.attributes	58	0x0
.debug_frame	55848	0x0
.debug_line	282067	0x0
.debug_loc	845	0x0
.comment	147	0x0
Total	1673362	

```
$ cargo size --target thumbv7em-none-eabihf --bin led-roulette --release -- -A
```

```
Finished release [optimized + debuginfo] target(s) in 0.03s
```

```
led-roulette :
```

section	size	addr
.vector_table	404	0x80000000
.text	5380	0x8000194
.rodata	564	0x8001698
.data	0	0x20000000
.bss	4	0x20000000
.uninit	0	0x20000004
.debug_loc	9994	0x0
.debug_abbrev	1821	0x0
.debug_info	74974	0x0
.debug_aranges	600	0x0
.debug_ranges	6848	0x0
.debug_str	52828	0x0
.debug_pubnames	20821	0x0
.debug_pubtypes	18891	0x0
.ARM.attributes	58	0x0
.debug_frame	1088	0x0
.debug_line	15307	0x0
.comment	19	0x0
Total	209601	

NOTE The Cargo project is already configured to build the release binary using LTO.

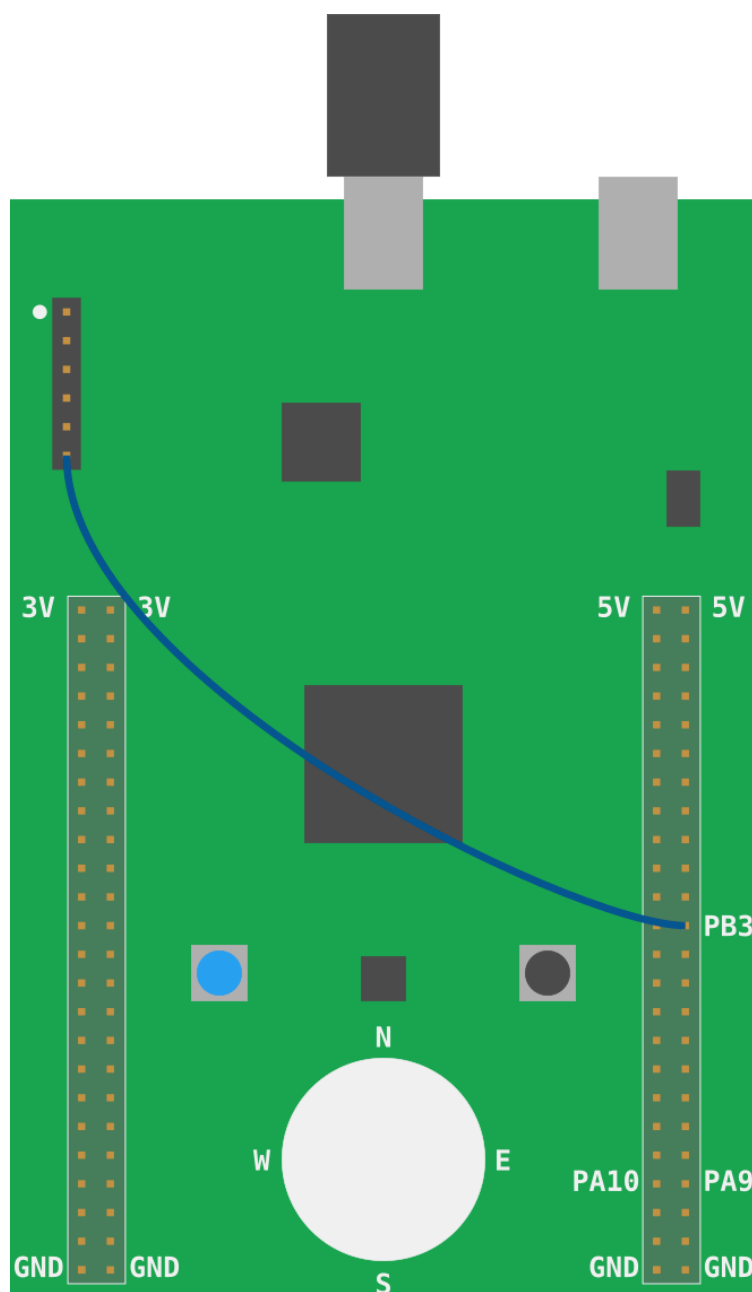
Know how to read this output? The `text` section contains the program instructions. It's around 5.25KB in my case. On the other hand, the `data` and `bss` sections contain variables statically allocated in RAM (`static` variables). A `static` variable is being used in `aux5::init`; that's why it shows 4 bytes of `bss`.

One final thing! We have been running our programs from within GDB but our programs doesn't depend on GDB at all. You can confirm this by closing both GDB and OpenOCD and then resetting the board by pressing the black button on the board. The LED roulette application will run without intervention of GDB.

Hello, world!

HEADS UP The “solder bridge” SB10 (see back of the board) on the STM32F3DISCOVERY, which is required to use the ITM and the **iprint!** macros shown below, is **not** soldered by default (see page 21 of the [User Manual](#)). (To be more accurate: this actually depends on the board revision. If you have an old version of the board as the [old User Manual](#) said, the SB10 was soldered. Check your board to decide whether you need to fix it.)

TL;DR You have two options to fix this: Either **solder** the solder bridge SB10 or connect a female to female jumper wire between SWO and PB3 as shown in the picture below.



Just a little more of helpful magic before we start doing low level stuff.

Blinking an LED is like the “Hello, world” of the embedded world.

But in this section, we'll run a proper "Hello, world" program that prints stuff to your computer console.

Go to the `06-hello-world` directory. There's some starter code in it:

```
#![deny(unsafe_code)]
#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux6::{entry, iprint, iprintln};

#[entry]
fn main() -> ! {
    let mut itm = aux6::init();

    iprintln!(&mut itm.stim[0], "Hello, world!");

    loop {}
}
```

The `iprintln` macro will format messages and output them to the microcontroller's *ITM*. ITM stands for Instrumentation Trace Macrocell and it's a communication protocol on top of SWD (Serial Wire Debug) which can be used to send messages from the microcontroller to the debugging host. This communication is only *one way*: the debugging host can't send data to the microcontroller.

OpenOCD, which is managing the debug session, can receive data sent through this ITM *channel* and redirect it to a file.

The ITM protocol works with *frames* (you can think of them as Ethernet frames). Each frame has a header and a variable length payload. OpenOCD will receive these frames and write them directly to a file without parsing them. So, if the microcontroller sends the string "Hello, world!" using the `iprintln` macro, OpenOCD's output file won't exactly contain that string.

To retrieve the original string, OpenOCD's output file will have to be parsed. We'll use the `itmdump` program to perform the parsing as new data arrives.

You should have already installed the `itmdump` program during the [installation chapter](#).

In a new terminal, run this command inside the `/tmp` directory, if you are using a *nix OS, or from within the `%TEMP%` directory, if you are running Windows. This should be the same directory from where you are running OpenOCD.

NOTE It's very important that both `itmdump` and `openocd` are running from the same directory!

```
$ # itmdump terminal

$ # *nix
$ cd /tmp && touch itm.txt

$ # Windows
$ cd %TEMP% && type nul >> itm.txt

$ # both
$ itmdump -F -f itm.txt
```

This command will block as `itmdump` is now watching the `itm.txt` file. Leave this terminal open.

Make sure that the STM32F3DISCOVERY board is connected to your computer. Open another terminal from `/tmp` directory (on Windows `%TEMP%`) to launch OpenOCD similar as described in [chapter 3](#).

Alright. Now, let's build the starter code and flash it into the microcontroller.

We will now build and run the application, `cargo run`. And step through it using `next`. Since `openocd.gdb` contains the `monitor` commands in `openocd.gdb` OpenOCD will redirect the ITM output to `itm.txt` and `itmdump` will write it to its terminal window. Also, it setup break points and stepped through the trampoline we are at the first executable statement in `fn main()`:

```
~/embedded-discovery/src/06-hello-world
$ cargo run
    Finished dev [unoptimized + debuginfo] target(s) in 0.01s
    Running `arm-none-eabi-gdb -q -x ../openocd.gdb ~/embedded-
discovery/target/thumbv7em-none-eabihf/debug/hello-world`
Reading symbols from ~/embedded-discovery/target/thumbv7em-none-eabihf/debug/hello-
world...
hello_world::__cortex_m_rt_main () at ~/embedded-discovery/src/06-hello-
world/src/main.rs:14
14      loop {}
Loading section .vector_table, size 0x194 lma 0x8000000
Loading section .text, size 0x2828 lma 0x8000194
Loading section .rodata, size 0x638 lma 0x80029bc
Start address 0x08000194, load size 12276
Transfer rate: 18 KB/sec, 4092 bytes/write.
Breakpoint 1 at 0x80001f0: file ~/embedded-discovery/src/06-hello-world/src/main.rs,
line 8.
Note: automatically using hardware breakpoints for read-only addresses.
Breakpoint 2 at 0x800092a: file /home/wink/.cargo/registry/src/github.com-
1ecc6299db9ec823/cortex-m-rt-0.6.13/src/lib.rs, line 570.
Breakpoint 3 at 0x80029a8: file /home/wink/.cargo/registry/src/github.com-
1ecc6299db9ec823/cortex-m-rt-0.6.13/src/lib.rs, line 560.

Breakpoint 1, hello_world::__cortex_m_rt_main_trampoline () at ~/embedded-
discovery/src/06-hello-world/src/main.rs:8
8      #[entry]
hello_world::__cortex_m_rt_main () at ~/embedded-discovery/src/06-hello-
world/src/main.rs:10
10      let mut itm = aux6::init();

(gdb)
```

Now issue a `next` command which will execute `aux6::init()` and stop at the next executable statement in `main.rs`, which positions us at line 12:

```
(gdb) next
12      iprintln!(&mut itm.stim[0], "Hello, world!");
```

Then issue another `next` command which will execute line 12, executing the `iprintln` and stop at line 14:

```
(gdb) next
14      loop {}
```

Now since `iprintln` has been executed the output on the `itmdump` terminal window should be the `Hello, world!` string:

```
$ itmdump -F -f itm.txt
(...)
Hello, world!
```

Awesome, right? Feel free to use `iprintln` as a logging tool in the coming sections.

Next: That's not all! The `iprint!` macros are not the only thing that uses the ITM. `:-)`

panic!

The `panic!` macro also sends its output to the ITM!

Change the `main` function to look like this:

```
#[entry]
fn main() -> ! {
    panic!("Hello, world!");
}
```

Before running one other suggestion, I find it inconvenient to have to confirm when quitting gdb. Add the following file in your home directory `~/.gdbinit` so that it quits immediately:

```
$ cat ~/.gdbinit
define hook-quit
    set confirm off
end
```

OK, now use `cargo run` and it stops at the first line of `fn main()`:

```
$ cargo run
   Compiling hello-world v0.2.0 (~/.embedded-discovery/src/06-hello-world)
   Finished dev [unoptimized + debuginfo] target(s) in 0.11s
   Running `arm-none-eabi-gdb -q -x ../openocd.gdb ~/.embedded-
discovery/target/thumbv7em-none-eabihf/debug/hello-world`
Reading symbols from ~/.embedded-discovery/target/thumbv7em-none-eabihf/debug/hello-
world...
hello_world::__cortex_m_rt_main () at ~/.embedded-discovery/src/06-hello-
world/src/main.rs:10
10      panic!("Hello, world!");
Loading section .vector_table, size 0x194 lma 0x8000000
Loading section .text, size 0x20fc lma 0x8000194
Loading section .rodata, size 0x554 lma 0x8002290
Start address 0x08000194, load size 10212
Transfer rate: 17 KB/sec, 3404 bytes/write.
Breakpoint 1 at 0x80001f0: file ~/.embedded-discovery/src/06-hello-world/src/main.rs,
line 8.
Note: automatically using hardware breakpoints for read-only addresses.
Breakpoint 2 at 0x8000222: file ~/.cargo/registry/src/github.com-
1ecc6299db9ec823/cortex-m-rt-0.6.13/src/lib.rs, line 570.
Breakpoint 3 at 0x800227a: file ~/.cargo/registry/src/github.com-
1ecc6299db9ec823/cortex-m-rt-0.6.13/src/lib.rs, line 560.

Breakpoint 1, hello_world::__cortex_m_rt_main_trampoline () at ~/.embedded-
discovery/src/06-hello-world/src/main.rs:8
8      #[entry]
hello_world::__cortex_m_rt_main () at ~/.embedded-discovery/src/06-hello-
world/src/main.rs:10
10      panic!("Hello, world!");
(gdb)
```

We'll use short command names to save typing, enter `c` then the `Enter` or `Return` key:

```
(gdb) c
Continuing.
```

If all is well you'll see some new output in the `itmdump` terminal.

```
$ # itmdump terminal
(..)
panicked at 'Hello, world!', src/06-hello-world/src/main.rs:10:5
```

Then type `Ctrl-c` which breaks out of a loop in the runtime:

```
^C
Program received signal SIGINT, Interrupt.
0x0800115c in panic_itm::panic (info=0x20009fa0) at ~/.cargo/registry/src/github.com-
1ecc6299db9ec823/panic-itm-0.4.2/src/lib.rs:57
57          atomic::compiler_fence(Ordering::SeqCst);
```

Ultimately, `panic!` is just another function call so you can see it leaves behind a trace of function calls. This allows you to use `backtrace` or just `bt` and to see call stack that caused the panic:

```
(gdb) bt
#0  panic_itm::panic (info=0x20009fa0) at ~/.cargo/registry/src/github.com-
1ecc6299db9ec823/panic-itm-0.4.2/src/lib.rs:47
#1  0x080005c2 in core::panicking::panic_fmt () at library/core/src/panicking.rs:92
#2  0x0800055a in core::panicking::panic () at library/core/src/panicking.rs:50
#3  0x08000210 in hello_world::__cortex_m_rt_main () at src/06-hello-
world/src/main.rs:10
#4  0x080001f4 in hello_world::__cortex_m_rt_main_trampoline () at src/06-hello-
world/src/main.rs:8
```

Another thing we can do is catch the panic *before* it does the logging. So we'll do several things; reset to the beginning, disable breakpoint 1, set a new breakpoint at `rust_begin_unwind`, list the breakpoints and then continue:

```
(gdb) monitor reset halt
Unable to match requested speed 1000 kHz, using 950 kHz
Unable to match requested speed 1000 kHz, using 950 kHz
adapter speed: 950 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x01000000 pc: 0x08000194 msp: 0x2000a000
```

```
(gdb) disable 1
```

```
(gdb) break rust_begin_unwind
Breakpoint 4 at 0x800106c: file ~/.cargo/registry/src/github.com-1ecc6299db9ec823/panic-itm-0.4.2/src/lib.rs, line 47.
```

```
(gdb) info break
```

Num	Type	Disp	Enb	Address	What
1	breakpoint	keep	n	0x080001f0	in hello_world::__cortex_m_rt_main_trampoline
at ~/prgs/rust/tutorial/embedded-discovery/src/06-hello-world/src/main.rs:8					
breakpoint already hit 1 time					
2	breakpoint	keep	y	0x08000222	in cortex_m_rt::DefaultHandler_
at ~/.cargo/registry/src/github.com-1ecc6299db9ec823/cortex-m-rt-0.6.13/src/lib.rs:570					
3	breakpoint	keep	y	0x0800227a	in cortex_m_rt::HardFault_
at ~/.cargo/registry/src/github.com-1ecc6299db9ec823/cortex-m-rt-0.6.13/src/lib.rs:560					
4	breakpoint	keep	y	0x0800106c	in panic_itm::panic
at ~/.cargo/registry/src/github.com-1ecc6299db9ec823/panic-itm-0.4.2/src/lib.rs:47					

```
(gdb) c
Continuing.
```

```
Breakpoint 4, panic_itm::panic (info=0x20009fa0) at ~/.cargo/registry/src/github.com-1ecc6299db9ec823/panic-itm-0.4.2/src/lib.rs:47
47          interrupt::disable();
```

You'll notice that nothing got printed on the `itmdump` console this time. If you resume the program using `continue` then a new line will be printed.

In a later section we'll look into other simpler communication protocols.

Finally, enter the `q` command to quit and it quits immediately without asking for confirmation:

```
(gdb) q
Detaching from program: ~/prgs/rust/tutorial/embedded-discovery/target/thumbv7em-none-eabihf/debug/hello-world, Remote target
Ending remote debugging.
[Inferior 1 (Remote target) detached]
```

As an even shorter sequence you can type `Ctrl-d`, which eliminates one keystroke!

NOTE In this case the `(gdb)` prompt is overwritten with `quit`)

```
quit)
Detaching from program: ~/prgs/rust/tutorial/embedded-discovery/target/thumbv7em-none-
eabihf/debug/hello-world, Remote target
Ending remote debugging.
[Inferior 1 (Remote target) detached]
```


Registers

It's time to explore what the `Led` API does under the hood.

In a nutshell, it just writes to some special memory regions. Go into the `07-registers` directory and let's run the starter code statement by statement.

```
#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux7::{entry, iprint, iprintln};

#[entry]
fn main() -> ! {
    aux7::init();

    unsafe {
        // A magic address!
        const GPIOE_BSRR: u32 = 0x48001018;

        // Turn on the "North" LED (red)
        *(GPIOE_BSRR as *mut u32) = 1 << 9;

        // Turn on the "East" LED (green)
        *(GPIOE_BSRR as *mut u32) = 1 << 11;

        // Turn off the "North" LED
        *(GPIOE_BSRR as *mut u32) = 1 << (9 + 16);

        // Turn off the "East" LED
        *(GPIOE_BSRR as *mut u32) = 1 << (11 + 16);
    }

    loop {}
}
```

What's this magic?

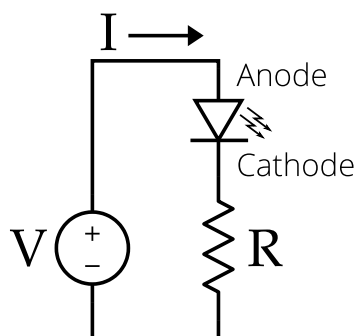
The address `0x48001018` points to a *register*. A register is a special region of memory that controls a *peripheral*. A peripheral is a piece of electronics that sits right next to the processor within the microcontroller package and provides the processor with extra functionality. After all, the processor, on its own, can only do math and logic.

This particular register controls General Purpose Input/Output (GPIO) *pins* (GPIO *is* a peripheral) and can be used to *drive* each of those pins *low* or *high*.

An aside: LEDs, digital outputs and voltage levels

Drive? Pin? Low? High?

A pin is an electrical contact. Our microcontroller has several of them and some of them are connected to LEDs. An LED, a Light Emitting Diode, will only emit light when voltage is applied to it with a certain polarity.



Luckily for us, the microcontroller's pins are connected to the LEDs with the right polarity. All that we have to do is *output* some non-zero voltage through the pin to turn the LED on. The pins attached to the LEDs are configured as *digital outputs* and can only output two different voltage levels: “low”, 0 Volts, or “high”, 3 Volts. A “high” (voltage) level will turn the LED on whereas a “low” (voltage) level will turn it off.

These “low” and “high” states map directly to the concept of digital logic. “low” is `0` or `false` and “high” is `1` or `true`. This is why this pin configuration is known as digital output.

OK. But how can one find out what this register does? Time to RTRM (Read the Reference Manual)!

RTRM: Reading The Reference Manual

I mentioned that the microcontroller has several pins. For convenience, these pins are grouped in *ports* of 16 pins. Each port is named with a letter: Port A, Port B, etc. and the pins within each port are named with numbers from 0 to 15.

The first thing we have to find out is which pin is connected to which LED. This information is in the STM32F3DISCOVERY [User Manual](#) (You downloaded a copy, right?). In this particular section:

Section 6.4 LEDs - Page 18

The manual says:

- **LD3**, the North LED, is connected to the pin **PE9**. **PE9** is the short form of: Pin 9 on Port E.
- **LD7**, the East LED, is connected to the pin **PE11**.

Up to this point, we know that we want to change the state of the pins PE9 and PE11 to turn the North/East LEDs on/off. These pins are part of Port E so we'll have to deal with the **GPIOE** peripheral.

Each peripheral has a register *block* associated to it. A register block is a collection of registers allocated in contiguous memory. The address at which the register block starts is known as its base address. We need to figure out what's the base address of the **GPIOE** peripheral. That information is in the following section of the microcontroller [Reference Manual](#):

Section 3.2.2 Memory map and register boundary addresses - Page 51

The table says that base address of the **GPIOE** register block is **0x4800_1000**.

Each peripheral also has its own section in the documentation. Each of these sections ends with a table of the registers that the peripheral's register block contains. For the **GPIO** family of peripheral, that table is in:

Section 11.4.12 GPIO register map - Page 243

'BSRR' is the register which we will be using to set/reset. Its offset value is '0x18' from the base address of the 'GPIOE'. We can look up BSRR in the reference manual. GPIO Registers -> GPIO port bit set/reset register (GPIOx_BSRR).

Now we need to jump to the documentation of that particular register. It's a few pages above in:

Section 11.4.7 GPIO port bit set/reset register (GPIOx_BSRR) - Page 240

Finally!

This is the register we were writing to. The documentation says some interesting things. First, this register is write only ... so let's try reading its value :-).

We'll use GDB's `examine` command: `x`.

```
(gdb) next
16          *(GPIOE_BSRR as *mut u32) = 1 << 9;

(gdb) x 0x48001018
0x48001018: 0x00000000

(gdb) # the next command will turn the North LED on
(gdb) next
19          *(GPIOE_BSRR as *mut u32) = 1 << 11;

(gdb) x 0x48001018
0x48001018: 0x00000000
```

Reading the register returns `0`. That matches what the documentation says.

The other thing that the documentation says is that the bits 0 to 15 can be used to *set* the corresponding pin. That is bit 0 sets the pin 0. Here, *set* means outputting a *high* value on the pin.

The documentation also says that bits 16 to 31 can be used to *reset* the corresponding pin. In this case, the bit 16 resets the pin number 0. As you may guess, *reset* means outputting a *low* value on the pin.

Correlating that information with our program, all seems to be in agreement:

- Writing `1 << 9` (`BS9 = 1`) to `BSRR` sets `PE9` *high*. That turns the North LED *on*.
- Writing `1 << 11` (`BS11 = 1`) to `BSRR` sets `PE11` *high*. That turns the East LED *on*.
- Writing `1 << 25` (`BR9 = 1`) to `BSRR` sets `PE9` *low*. That turns the North LED *off*.
- Finally, writing `1 << 27` (`BR11 = 1`) to `BSRR` sets `PE11` *low*. That turns the East LED *off*.

(mis)Optimization

Reads/writes to registers are quite special. I may even dare to say that they are embodiment of side effects. In the previous example we wrote four different values to the same register. If you didn't know that address was a register, you may have simplified the logic to just write the final value `1 << (11 + 16)` into the register.

Actually, LLVM, the compiler's backend / optimizer, does not know we are dealing with a register and will merge the writes thus changing the behavior of our program. Let's check that really quick.

```
$ cargo run --release
(..)
Breakpoint 1, registers::__cortex_m_rt_main_trampoline () at src/07-registers/src/main.rs:7
7      #[entry]

(gdb) step
registers::__cortex_m_rt_main () at src/07-registers/src/main.rs:9
9      aux7::init();

(gdb) next
25      *(GPIOE_BSRR as *mut u32) = 1 << (11 + 16);

(gdb) disassemble /m
Dump of assembler code for function
_ZN9registers18__cortex_m_rt_main17h45b1ef53e18aa8d0E:
8      fn main() -> ! {
    0x08000248 <+0>:      push    {r7, lr}
    0x0800024a <+2>:      mov     r7, sp

9          aux7::init();
    0x0800024c <+4>:      bl      0x8000260 <aux7::init>
    0x08000250 <+8>:      movw   r0, #4120          ; 0x1018
    0x08000254 <+12>:     mov.w  r1, #134217728   ; 0x80000000
    0x08000258 <+16>:     movt   r0, #18432         ; 0x4800

10
11      unsafe {
12          // A magic address!
13          const GPIOE_BSRR: u32 = 0x48001018;
14
15          // Turn on the "North" LED (red)
16          *(GPIOE_BSRR as *mut u32) = 1 << 9;
17
18          // Turn on the "East" LED (green)
19          *(GPIOE_BSRR as *mut u32) = 1 << 11;
20
21          // Turn off the "North" LED
22          *(GPIOE_BSRR as *mut u32) = 1 << (9 + 16);
23
24          // Turn off the "East" LED
25          *(GPIOE_BSRR as *mut u32) = 1 << (11 + 16);
=> 0x0800025c <+20>:     str     r1, [r0, #0]
    0x0800025e <+22>:     b.n    0x800025e <registers::__cortex_m_rt_main+22>

End of assembler dump.
```

The state of the LEDs didn't change this time! The `str` instruction is the one that writes a value to the register. Our *debug* (unoptimized) program had four of them, one for each write to the register, but the *release* (optimized) program only has one.

We can check that using `objdump` and capture the output to `out.asm`:

```
# same as cargo objdump -- -d --no-show-raw-insn --print-imm-hex --source
target/thumbv7em-none-eabihf/debug/registers
cargo objdump --bin registers -- -d --no-show-raw-insn --print-imm-hex --source >
debug.txt
```

Then examine `debug.txt` looking for `main` and we see the 4 `str` instructions:

```
080001ec <main>:
; #[entry]
80001ec:      push    {r7, lr}
80001ee:      mov     r7, sp
80001f0:      bl      #0x2
80001f4:      trap

080001f6 <registers::__cortex_m_rt_main::hc2e3436fa38cd6f2>:
; fn main() -> ! {
80001f6:      push    {r7, lr}
80001f8:      mov     r7, sp
;      aux7::init();
80001fa:      bl      #0x3e
80001fe:      b       #-0x2 <registers::__cortex_m_rt_main::hc2e3436fa38cd6f2+0xa>
;      *(GPIOE_BSRR as *mut u32) = 1 << 9;
8000200:      movw    r0, #0x2640
8000204:      movt    r0, #0x800
8000208:      ldr     r0, [r0]
800020a:      movw    r1, #0x1018
800020e:      movt    r1, #0x4800
8000212:      str     r0, [r1]
;      *(GPIOE_BSRR as *mut u32) = 1 << 11;
8000214:      movw    r0, #0x2648
8000218:      movt    r0, #0x800
800021c:      ldr     r0, [r0]
800021e:      str     r0, [r1]
;      *(GPIOE_BSRR as *mut u32) = 1 << (9 + 16);
8000220:      movw    r0, #0x2650
8000224:      movt    r0, #0x800
8000228:      ldr     r0, [r0]
800022a:      str     r0, [r1]
;      *(GPIOE_BSRR as *mut u32) = 1 << (11 + 16);
800022c:      movw    r0, #0x2638
8000230:      movt    r0, #0x800
8000234:      ldr     r0, [r0]
8000236:      str     r0, [r1]
;      loop {}
8000238:      b       #-0x2 <registers::__cortex_m_rt_main::hc2e3436fa38cd6f2+0x44>
800023a:      b       #-0x4 <registers::__cortex_m_rt_main::hc2e3436fa38cd6f2+0x44>
(..)
```

How do we prevent LLVM from misoptimizing our program? We use *volatile* operations instead of plain reads/writes:

```

#![no_main]
#![no_std]

use core::ptr;

#[allow(unused_imports)]
use aux7::entry;

#[entry]
fn main() -> ! {
    aux7::init();

    unsafe {
        // A magic address!
        const GPIOE_BSRR: u32 = 0x48001018;

        // Turn on the "North" LED (red)
        ptr::write_volatile(GPIOE_BSRR as *mut u32, 1 << 9);

        // Turn on the "East" LED (green)
        ptr::write_volatile(GPIOE_BSRR as *mut u32, 1 << 11);

        // Turn off the "North" LED
        ptr::write_volatile(GPIOE_BSRR as *mut u32, 1 << (9 + 16));

        // Turn off the "East" LED
        ptr::write_volatile(GPIOE_BSRR as *mut u32, 1 << (11 + 16));
    }

    loop {}
}

```

Generate `release.txt` using with `--release` mode.

```

cargo objdump --release --bin registers -- -d --no-show-raw-insn --print-imm-hex --
source > release.txt

```

Now find the `main` routine in `release.txt` and we see the 4 `str` instructions.

```

0800023e <main>:
; #[entry]
800023e:      push    {r7, lr}
8000240:      mov     r7, sp
8000242:      bl      #0x2
8000246:      trap

08000248 <registers::__cortex_m_rt_main::h45b1ef53e18aa8d0>:
; fn main() -> ! {
8000248:      push    {r7, lr}
800024a:      mov     r7, sp
;      aux7::init();
800024c:      bl      #0x22
8000250:      movw    r0, #0x1018
8000254:      mov.w   r1, #0x200
8000258:      movt    r0, #0x4800
;      intrinsics::volatile_store(dst, src);
800025c:      str     r1, [r0]
800025e:      mov.w   r1, #0x800
8000262:      str     r1, [r0]
8000264:      mov.w   r1, #0x20000000
8000268:      str     r1, [r0]
800026a:      mov.w   r1, #0x80000000
800026e:      str     r1, [r0]
8000270:      b      #-0x4 <registers::__cortex_m_rt_main::h45b1ef53e18aa8d0+0x28>
(..)

```

We see that the four writes (`str` instructions) are preserved. If you run it using `gdb` you'll also see that we get the expected behavior.

NB: The last `next` will endlessly execute `loop {}`, use `Ctrl-c` to get back to the `(gdb)` prompt.

```
$ cargo run --release
(..)
```

```
Breakpoint 1, registers::__cortex_m_rt_main_trampoline () at src/07-
registers/src/main.rs:9
9      #[entry]
```

```
(gdb) step
registers::__cortex_m_rt_main () at src/07-registers/src/main.rs:11
11      aux7::init();
```

```
(gdb) next
18      ptr::write_volatile(GPIOE_BSRR as *mut u32, 1 << 9);
```

```
(gdb) next
21      ptr::write_volatile(GPIOE_BSRR as *mut u32, 1 << 11);
```

```
(gdb) next
24      ptr::write_volatile(GPIOE_BSRR as *mut u32, 1 << (9 + 16));
```

```
(gdb) next
27      ptr::write_volatile(GPIOE_BSRR as *mut u32, 1 << (11 + 16));
```

```
(gdb) next
^C
```

```
Program received signal SIGINT, Interrupt.
```

```
0x08000270 in registers::__cortex_m_rt_main ()
    at ~/.rustup/toolchains/stable-x86_64-unknown-linux-
gnu/lib/rustlib/src/rust/library/core/src/ptr/mod.rs:1124
```

```
1124      intrinsics::volatile_store(dst, src);
(gdb)
```


0xBAAAAAAD address

Not all the peripheral memory can be accessed. Look at this program.

```
#![no_main]
#![no_std]

use core::ptr;

#[allow(unused_imports)]
use aux7::{entry, iprint, iprintln};

#[entry]
fn main() -> ! {
    aux7::init();

    unsafe {
        ptr::read_volatile(0x4800_1800 as *const u32);
    }

    loop {}
}
```

This address is close to the `GPIOE_BSRR` address we used before but this address is *invalid*. Invalid in the sense that there's no register at this address.

Now, let's try it.

```
$ cargo run
(..)
Breakpoint 1, registers::__cortex_m_rt_main_trampoline () at src/07-
registers/src/main.rs:9
9      #[entry]

(gdb) continue
Continuing.

Breakpoint 3, cortex_m_rt::HardFault_ (ef=0x20009fb0)
at ~/.cargo/registry/src/github.com-1ecc6299db9ec823/cortex-m-rt-
0.6.13/src/lib.rs:560
560      loop {

(gdb)
```

We tried to do an invalid operation, reading memory that doesn't exist, so the processor raised an *exception*, a *hardware* exception.

In most cases, exceptions are raised when the processor attempts to perform an invalid operation. Exceptions break the normal flow of a program and force the processor to execute an *exception handler*, which is just a function/subroutine.

There are different kind of exceptions. Each kind of exception is raised by different conditions and each one is handled by a different exception handler.

The `aux7` crate depends on the `cortex-m-rt` crate which defines a default *hard fault* handler, named `HardFault`, that handles the “invalid memory address” exception. `openocd.gdb` placed a breakpoint on `HardFault`; that’s why the debugger halted your program while it was executing the exception handler. We can get more information about the exception from the debugger. Let’s see:

```
(gdb) list
555     #[allow(unused_variables)]
556     #[doc(hidden)]
557     #[link_section = ".HardFault.default"]
558     #[no_mangle]
559     pub unsafe extern "C" fn HardFault_(ef: &ExceptionFrame) -> ! {
560         loop {
561             // add some side effect to prevent this from turning into a UDF
instruction
562             // see rust-lang/rust#28728 for details
563             atomic::compiler_fence(Ordering::SeqCst);
564         }
```

`ef` is a snapshot of the program state right before the exception occurred. Let’s inspect it:

```
(gdb) print/x *ef
$1 = cortex_m_rt::ExceptionFrame {
  r0: 0x48001800,
  r1: 0x80036b0,
  r2: 0x1,
  r3: 0x80000000,
  r12: 0xb,
  lr: 0x800020d,
  pc: 0x8001750,
  xpsr: 0xa1000200
}
```

There are several fields here but the most important one is `pc`, the Program Counter register. The address in this register points to the instruction that generated the exception. Let’s disassemble the program around the bad instruction.

```
(gdb) disassemble /m ef.pc
Dump of assembler code for function core::ptr::read_volatile<u32>:
1046   pub unsafe fn read_volatile<T>(src: *const T) -> T {
      0x0800174c <+0>:      sub     sp, #12
      0x0800174e <+2>:      str     r0, [sp, #4]

1047       if cfg!(debug_assertions) && !is_aligned_and_not_null(src) {
1048           // Not panicking to keep codegen impact smaller.
1049           abort();
1050       }
1051       // SAFETY: the caller must uphold the safety contract for `volatile_load`.
1052       unsafe { intrinsics::volatile_load(src) }
      0x08001750 <+4>:      ldr     r0, [r0, #0]
      0x08001752 <+6>:      str     r0, [sp, #8]
      0x08001754 <+8>:      ldr     r0, [sp, #8]
      0x08001756 <+10>:     str     r0, [sp, #0]
      0x08001758 <+12>:     b.n     0x800175a <core::ptr::read_volatile<u32>+14>

1053   }
      0x0800175a <+14>:     ldr     r0, [sp, #0]
      0x0800175c <+16>:     add     sp, #12
      0x0800175e <+18>:     bx      lr

End of assembler dump.
```

The exception was caused by the `ldr r0, [r0, #0]` instruction, a read instruction. The instruction tried to read the memory at the address indicated by the `r0` register. By the way, `r0` is a CPU (processor) register not a memory mapped register; it doesn't have an associated address like, say, `GPIO_BSRR`.

Wouldn't it be nice if we could check what the value of the `r0` register was right at the instant when the exception was raised? Well, we already did! The `r0` field in the `ef` value we printed before is the value of `r0` register had when the exception was raised. Here it is again:

```
(gdb) print/x *ef
$1 = cortex_m_rt::ExceptionFrame {
  r0: 0x48001800,
  r1: 0x80036b0,
  r2: 0x1,
  r3: 0x80000000,
  r12: 0xb,
  lr: 0x800020d,
  pc: 0x8001750,
  xpsr: 0xa1000200
}
```

`r0` contains the value `0x4800_1800` which is the invalid address we called the `read_volatile` function with.

Spooky action at a distance

`BSRR` is not the only register that can control the pins of Port E. The `ODR` register also lets you change the value of the pins. Furthermore, `ODR` also lets you retrieve the current output status of Port E.

`ODR` is documented in:

Section 11.4.6 GPIO port output data register - Page 239

Let's look at this program. The key to this program is `fn iprint_odr`. This function prints the current value in `ODR` to the `ITM` console

```

#![no_main]
#![no_std]

use core::ptr;

#[allow(unused_imports)]
use aux7::{entry, iprintln, ITM};

// Print the current contents of odr
fn iprint_odr(itm: &mut ITM) {
    const GPIOE_ODR: u32 = 0x4800_1014;

    unsafe {
        iprintln!(
            &mut itm.stim[0],
            "ODR = 0x{:04x}",
            ptr::read_volatile(GPIOE_ODR as *const u16)
        );
    }
}

#[entry]
fn main() -> ! {
    let mut itm = aux7::init().0;

    unsafe {
        // A magic addresses!
        const GPIOE_BSRR: u32 = 0x4800_1018;

        // Print the initial contents of ODR
        iprint_odr(&mut itm);

        // Turn on the "North" LED (red)
        ptr::write_volatile(GPIOE_BSRR as *mut u32, 1 << 9);
        iprint_odr(&mut itm);

        // Turn on the "East" LED (green)
        ptr::write_volatile(GPIOE_BSRR as *mut u32, 1 << 11);
        iprint_odr(&mut itm);

        // Turn off the "North" LED
        ptr::write_volatile(GPIOE_BSRR as *mut u32, 1 << (9 + 16));
        iprint_odr(&mut itm);

        // Turn off the "East" LED
        ptr::write_volatile(GPIOE_BSRR as *mut u32, 1 << (11 + 16));
        iprint_odr(&mut itm);
    }

    loop {}
}

```

If you run this program

```
$ cargo run
(..)
Breakpoint 1, registers::__cortex_m_rt_main_trampoline () at src/07-
registers/src/main.rs:22
22      #[entry]

(gdb) continue
Continuing.
```

You'll see on itmdump's console:

```
$ # itmdump's console
(..)
ODR = 0x0000
ODR = 0x0200
ODR = 0x0a00
ODR = 0x0800
ODR = 0x0000
```

Side effects! Although we are reading the same address multiple times without actually modifying it, we still see its value change every time **BSRR** is written to.

Type safe manipulation

The last register we were working with, `ODR`, had this in its documentation:

Bits 31:16 Reserved, must be kept at reset value

We are not supposed to write to those bits of the register or Bad Stuff May Happen.

There's also the fact the registers have different read/write permissions. Some of them are write only, others can be read and written to and there must be others that are read only.

Finally, directly working with hexadecimal addresses is error prone. You already saw that trying to access an invalid memory address causes an exception which disrupts the execution of our program.

Wouldn't it be nice if we had an API to manipulate registers in a "safe" manner? Ideally, the API should encode these three points I've mentioned: No messing around with the actual addresses, should respect read/write permissions and should prevent modification of the reserved parts of a register.

Well, we do! `aux7::init()` actually returns a value that provides a type safe API to manipulate the registers of the `GPIOE` peripheral.

As you may remember: a group of registers associated to a peripheral is called register block, and it's located in a contiguous region of memory. In this type safe API each register block is modeled as a `struct` where each of its fields represents a register. Each register field is a different newtype over e.g. `u32` that exposes a combination of the following methods: `read`, `write` or `modify` according to its read/write permissions. Finally, these methods don't take primitive values like `u32`, instead they take yet another newtype that can be constructed using the builder pattern and that prevent the modification of the reserved parts of the register.

The best way to get familiar with this API is to port our running example to it.

```

#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux7::{entry, iprintln, ITM, RegisterBlock};

#[entry]
fn main() -> ! {
    let gpioe = aux7::init().1;

    // Turn on the North LED
    gpioe.bsrr.write(|w| w.bs9().set_bit());

    // Turn on the East LED
    gpioe.bsrr.write(|w| w.bs11().set_bit());

    // Turn off the North LED
    gpioe.bsrr.write(|w| w.br9().set_bit());

    // Turn off the East LED
    gpioe.bsrr.write(|w| w.br11().set_bit());

    loop {}
}

```

First thing you notice: There are no magic addresses involved. Instead we use a more human friendly way, for example `gpioe.bsrr`, to refer to the `BSRR` register in the `GPIOE` register block.

Then we have this `write` method that takes a closure. If the identity closure (`|w| w`) is used, this method will set the register to its *default* (reset) value, the value it had right after the microcontroller was powered on / reset. That value is `0x0` for the `BSRR` register. Since we want to write a non-zero value to the register, we use builder methods like `bs9` and `br9` to set some of the bits of the default value.

Let's run this program! There's some interesting stuff we can do *while* debugging the program.

`gpioe` is a reference to the `GPIOE` register block. `print gpioe` will return the base address of the register block.

```

$ cargo run
(..)

Breakpoint 1, registers::__cortex_m_rt_main_trampoline () at src/07-
registers/src/main.rs:7
7      #[entry]

(gdb) step
registers::__cortex_m_rt_main () at src/07-registers/src/main.rs:9
9      let gpioe = aux7::init().1;

(gdb) next
12      gpioe.bsrr.write(|w| w.bs9().set_bit());

(gdb) print gpioe
$1 = (*mut stm32f3::stm32f303::gpioc::RegisterBlock) 0x48001000

```

But if we instead `print *gpioe`, we'll get a *full view* of the register block: the value of each of its registers will be printed.

```
(gdb) print *gpioe
$2 = stm32f3::stm32f303::gpioe::RegisterBlock {
  moder: stm32f3::generic::Reg<u32, stm32f3::stm32f303::gpioe::_MODER> {
    register: vcell::VolatileCell<u32> {
      value: core::cell::UnsafeCell<u32> {
        value: 1431633920
      }
    },
    _marker: core::marker::PhantomData<stm32f3::stm32f303::gpioe::_MODER>
  },
  otyper: stm32f3::generic::Reg<u32, stm32f3::stm32f303::gpioe::_OTYPER> {
    register: vcell::VolatileCell<u32> {
      value: core::cell::UnsafeCell<u32> {
        value: 0
      }
    },
    _marker: core::marker::PhantomData<stm32f3::stm32f303::gpioe::_OTYPER>
  },
  ospeedr: stm32f3::generic::Reg<u32, stm32f3::stm32f303::gpioe::_OSPEEDR> {
    register: vcell::VolatileCell<u32> {
      value: core::cell::UnsafeCell<u32> {
        value: 0
      }
    },
    _marker: core::marker::PhantomData<stm32f3::stm32f303::gpioe::_OSPEEDR>
  },
  pupdr: stm32f3::generic::Reg<u32, stm32f3::stm32f303::gpioe::_PUPDR> {
    register: vcell::VolatileCell<u32> {
      value: core::cell::UnsafeCell<u32> {
        value: 0
      }
    },
    _marker: core::marker::PhantomData<stm32f3::stm32f303::gpioe::_PUPDR>
  },
  idr: stm32f3::generic::Reg<u32, stm32f3::stm32f303::gpioe::_IDR> {
    register: vcell::VolatileCell<u32> {
      value: core::cell::UnsafeCell<u32> {
        value: 204
      }
    },
    _marker: core::marker::PhantomData<stm32f3::stm32f303::gpioe::_IDR>
  },
  odr: stm32f3::generic::Reg<u32, stm32f3::stm32f303::gpioe::_ODR> {
    register: vcell::VolatileCell<u32> {
      value: core::cell::UnsafeCell<u32> {
        value: 0
      }
    },
    _marker: core::marker::PhantomData<stm32f3::stm32f303::gpioe::_ODR>
  },
  bsrr: stm32f3::generic::Reg<u32, stm32f3::stm32f303::gpioe::_BSRR> {
    register: vcell::VolatileCell<u32> {
      value: core::cell::UnsafeCell<u32> {
        value: 0
      }
    },
    _marker: core::marker::PhantomData<stm32f3::stm32f303::gpioe::_BSRR>
  },
  lckr: stm32f3::generic::Reg<u32, stm32f3::stm32f303::gpioe::_LCKR> {
    register: vcell::VolatileCell<u32> {
      value: core::cell::UnsafeCell<u32> {
```

```

        value: 0
    },
    _marker: core::marker::PhantomData<stm32f3::stm32f303::gpioc::_LCKR>
},
afrl: stm32f3::generic::Reg<u32, stm32f3::stm32f303::gpioc::_AFRL> {
    register: vcell::VolatileCell<u32> {
        value: core::cell::UnsafeCell<u32> {
            value: 0
        }
    },
    _marker: core::marker::PhantomData<stm32f3::stm32f303::gpioc::_AFRL>
},
afrh: stm32f3::generic::Reg<u32, stm32f3::stm32f303::gpioc::_AFRH> {
    register: vcell::VolatileCell<u32> {
        value: core::cell::UnsafeCell<u32> {
            value: 0
        }
    },
    _marker: core::marker::PhantomData<stm32f3::stm32f303::gpioc::_AFRH>
},
brr: stm32f3::generic::Reg<u32, stm32f3::stm32f303::gpioc::_BRR> {
    register: vcell::VolatileCell<u32> {
        value: core::cell::UnsafeCell<u32> {
            value: 0
        }
    },
    _marker: core::marker::PhantomData<stm32f3::stm32f303::gpioc::_BRR>
}
}

```

All these newtypes and closures sound like they'd generate large, bloated programs but, if you actually compile the program in release mode with [LTO](#) enabled, you'll see that it produces exactly the same instructions that the "unsafe" version that used `write_volatile` and hexadecimal addresses did!

Use `cargo objdump` to grab the assembler code to `release.txt`:

```
cargo objdump --bin registers --release -- -d --no-show-raw-insn --print-imm-hex >
release.txt
```

Then search for `main` in `release.txt`

```

0800023e <main>:
800023e:      push    {r7, lr}
8000240:      mov     r7, sp
8000242:      bl      #0x2
8000246:      trap

08000248 <registers::__cortex_m_rt_main::h199f1359501d5c71>:
8000248:      push    {r7, lr}
800024a:      mov     r7, sp
800024c:      bl      #0x22
8000250:      movw    r0, #0x1018
8000254:      mov.w   r1, #0x200
8000258:      movt    r0, #0x4800
800025c:      str     r1, [r0]
800025e:      mov.w   r1, #0x800
8000262:      str     r1, [r0]
8000264:      mov.w   r1, #0x20000000
8000268:      str     r1, [r0]
800026a:      mov.w   r1, #0x80000000
800026e:      str     r1, [r0]
8000270:      b      #-0x4 <registers::__cortex_m_rt_main::h199f1359501d5c71+0x28>

```

The best part of all this is that nobody had to write a single line of code to implement the GPIOE API. All the code was automatically generated from a System View Description (SVD) file using the [svd2rust](#) tool. This SVD file is actually an XML file that microcontroller vendors provide and that contains the register maps of their microcontrollers. The file contains the layout of register blocks, the base addresses, the read/write permissions of each register, the layout of the registers, whether a register has reserved bits and lots of other useful information.

LEDs, again

In the last section, I gave you *initialized* (configured) peripherals (I initialized them in `aux7::init`). That's why just writing to `BSRR` was enough to control the LEDs. But, peripherals are not *initialized* right after the microcontroller boots.

In this section, you'll have more fun with registers. I won't do any initialization and you'll have to initialize and configure `GPIOE` pins as digital outputs pins so that you'll be able to drive LEDs again.

This is the starter code.

```
#![deny(unsafe_code)]
#![no_main]
#![no_std]

use aux8::entry;

#[entry]
fn main() -> ! {
    let (gpioe, rcc) = aux8::init();

    // TODO initialize GPIOE

    // Turn on all the LEDs in the compass
    gpioe.odr.write(|w| {
        w.odr8().set_bit();
        w.odr9().set_bit();
        w.odr10().set_bit();
        w.odr11().set_bit();
        w.odr12().set_bit();
        w.odr13().set_bit();
        w.odr14().set_bit();
        w.odr15().set_bit()
    });

    aux8::bkpt();

    loop {}
}
```

If you run the starter code, you'll see that nothing happens this time. Furthermore, if you print the `GPIOE` register block, you'll see that every register reads as zero even after the `gpioe.odr.write` statement was executed!

```
$ cargo run
Breakpoint 1, main () at src/08-leds-again/src/main.rs:9
9         let (gpioe, rcc) = aux8::init();
```

```
(gdb) continue
Continuing.
```

```
Program received signal SIGTRAP, Trace/breakpoint trap.
0x08000f3c in __bkpt ()
```

```
(gdb) finish
Run till exit from #0 0x08000f3c in __bkpt ()
main () at src/08-leds-again/src/main.rs:25
25         aux8::bkpt();
```

```
(gdb) p/x *gpioe
$1 = stm32f30x::gpioc::RegisterBlock {
  moder: stm32f30x::gpioc::MODER {
    register: vcell::VolatileCell<u32> {
      value: core::cell::UnsafeCell<u32> {
        value: 0x0
      }
    }
  },
  otyper: stm32f30x::gpioc::OTYPER {
    register: vcell::VolatileCell<u32> {
      value: core::cell::UnsafeCell<u32> {
        value: 0x0
      }
    }
  },
  ospeedr: stm32f30x::gpioc::OSPEEDR {
    register: vcell::VolatileCell<u32> {
      value: core::cell::UnsafeCell<u32> {
        value: 0x0
      }
    }
  },
  pupdr: stm32f30x::gpioc::PUPDR {
    register: vcell::VolatileCell<u32> {
      value: core::cell::UnsafeCell<u32> {
        value: 0x0
      }
    }
  },
  idr: stm32f30x::gpioc::IDR {
    register: vcell::VolatileCell<u32> {
      value: core::cell::UnsafeCell<u32> {
        value: 0x0
      }
    }
  },
  odr: stm32f30x::gpioc::ODR {
    register: vcell::VolatileCell<u32> {
      value: core::cell::UnsafeCell<u32> {
        value: 0x0
      }
    }
  },
  bsrr: stm32f30x::gpioc::BSRR {
    register: vcell::VolatileCell<u32> {
```

```
        value: core::cell::UnsafeCell<u32> {
            value: 0x0
        }
    },
    lckr: stm32f30x::gpioc::LCKR {
        register: vcell::VolatileCell<u32> {
            value: core::cell::UnsafeCell<u32> {
                value: 0x0
            }
        }
    },
    afrl: stm32f30x::gpioc::AFRL {
        register: vcell::VolatileCell<u32> {
            value: core::cell::UnsafeCell<u32> {
                value: 0x0
            }
        }
    },
    afrh: stm32f30x::gpioc::AFRH {
        register: vcell::VolatileCell<u32> {
            value: core::cell::UnsafeCell<u32> {
                value: 0x0
            }
        }
    },
    brr: stm32f30x::gpioc::BRR {
        register: vcell::VolatileCell<u32> {
            value: core::cell::UnsafeCell<u32> {
                value: 0x0
            }
        }
    }
}
```


Power

Turns out that, to save power, most peripherals start in a powered off state – that's their state right after the microcontroller boots.

The Reset and Clock Control (**RCC**) peripheral can be used to power on or off every other peripheral.

You can find the list of registers in the **RCC** register block in:

Section 9.4.14 - RCC register map - Page 166 - Reference Manual

The registers that control the power status of other peripherals are:

- **AHBENR**
- **APB1ENR**
- **APB2ENR**

Each bit in these registers controls the power status of a single peripheral, including **GPIOE** .

Your task in this section is to power on the **GPIOE** peripheral. You'll have to:

- Figure out which of the three registers I mentioned before has the bit that controls the power status.
- Figure out what value that bit must be set to, **0** or **1** , to power on the **GPIOE** peripheral.
- Finally, you'll have to change the starter code to *modify* the right register to turn on the **GPIOE** peripheral.

If you are successful, you'll see that the **gpioe.odr.write** statement will now be able to modify the value of the **ODR** register.

Note that this won't be enough to actually turn on the LEDs.

Configuration

After turning on the GPIOE peripheral, it still needs to be configured. In this case, we want the pins to be configured as digital *outputs* so they can drive the LEDs; by default, most pins are configured as digital *inputs*.

You can find the list of registers in the **GPIOE** register block in:

Section 11.4.12 - GPIO registers - Page 243 - Reference Manual

The register we'll have to deal with is: **MODER**.

Your task for this section is to further update the starter code to configure the *right* **GPIOE** pins as digital outputs. You'll have to:

- Figure out *which* pins you need to configure as digital outputs. (hint: check Section 6.4 LEDs of the *User Manual* (page 18)).
- Read the documentation to understand what the bits in the **MODER** register do.
- Modify the **MODER** register to configure the pins as digital outputs.

If successful, you'll see the 8 LEDs turn on when you run the program.

The solution

```

#![deny(unsafe_code)]
#![no_main]
#![no_std]

use aux8::entry;

#[entry]
fn main() -> ! {
    let (gpioe, rcc) = aux8::init();

    // enable the GPIOE peripheral
    rcc.ahbenr.write(|w| w.iopeen().set_bit());

    // configure the pins as outputs
    gpioe.moder.write(|w| {
        w.moder8().output();
        w.moder9().output();
        w.moder10().output();
        w.moder11().output();
        w.moder12().output();
        w.moder13().output();
        w.moder14().output();
        w.moder15().output()
    });

    // Turn on all the LEDs in the compass
    gpioe.odr.write(|w| {
        w.odr8().set_bit();
        w.odr9().set_bit();
        w.odr10().set_bit();
        w.odr11().set_bit();
        w.odr12().set_bit();
        w.odr13().set_bit();
        w.odr14().set_bit();
        w.odr15().set_bit()
    });

    aux8::bkpt();

    loop {}
}

```


Clocks and timers

In this section, we'll re-implement the LED roulette application. I'm going to give you back the `Led` abstraction but this time I'm going to take away the `Delay` abstraction `:-)`.

Here's the starter code. The `delay` function is unimplemented so if you run this program the LEDs will blink so fast that they'll appear to always be on.

```
#![no_main]
#![no_std]

use aux9::{entry, switch_hal::OutputSwitch, tim6};

#[inline(never)]
fn delay(tim6: &tim6::RegisterBlock, ms: u16) {
    // TODO implement this
}

#[entry]
fn main() -> ! {
    let (leds, rcc, tim6) = aux9::init();
    let mut leds = leds.into_array();

    // TODO initialize TIM6

    let ms = 50;
    loop {
        for curr in 0..8 {
            let next = (curr + 1) % 8;

            leds[next].on().unwrap();
            delay(tim6, ms);
            leds[curr].off().unwrap();
            delay(tim6, ms);
        }
    }
}
```


for loop delays

The first challenge is to implement the `delay` function without using any peripheral and the obvious solution is to implement it as a `for` loop delay:

```
#[inline(never)]
fn delay(tim6: &tim6::RegisterBlock, ms: u16) {
    for _ in 0..1_000 {}
}
```

Of course, the above implementation is wrong because it always generates the same delay for any value of `ms`.

In this section, you'll have to:

- Fix the `delay` function to generate delays proportional to its input `ms`.
- Tweak the `delay` function to make the LED roulette spin at a rate of approximately 5 cycles in 4 seconds (800 milliseconds period).
- The processor inside the microcontroller is clocked at 72 MHz and executes most instructions in one “tick”, a cycle of its clock. How many (`for`) loops do you *think* the `delay` function must do to generate a delay of 1 second?
- How many `for` loops does `delay(1000)` actually do?
- What happens if compile your program in release mode and run it?

NOP

If in the previous section you compiled the program in release mode and actually looked at the disassembly, you probably noticed that the `delay` function is optimized away and never gets called from within `main`.

LLVM decided that the function wasn't doing anything worthwhile and just removed it.

There is a way to prevent LLVM from optimizing the `for` loop delay: add a *volatile* assembly instruction. Any instruction will do but NOP (No OPeration) is a particular good choice in this case because it has no side effect.

Your `for` loop delay would become:

```
#[inline(never)]
fn delay(_tim6: &tim6::RegisterBlock, ms: u16) {
    const K: u16 = 3; // this value needs to be tweaked
    for _ in 0..(K * ms) {
        aux9::nop()
    }
}
```

And this time `delay` won't be compiled away by LLVM when you compile your program in release mode:

```
$ cargo objdump --bin clocks-and-timers --release -- -d --no-show-raw-insn
clocks-and-timers:      file format ELF32-arm-little

Disassembly of section .text:
clocks_and_timers::delay::h711ce9bd68a6328f:
8000188:      push    {r4, r5, r7, lr}
800018a:      movs    r4, #0
800018c:      adds    r4, #1
800018e:      uxth    r5, r4
8000190:      bl      #4666
8000194:      cmp     r5, #150
8000196:      blo     #-14 <clocks_and_timers::delay::h711ce9bd68a6328f+0x4>
8000198:      pop     {r4, r5, r7, pc}
```

Now, test this: Compile the program in debug mode and run it, then compile the program in release mode and run it. What's the difference between them? What do you think is the main cause of the difference? Can you think of a way to make them equivalent or at least more similar again?

One-shot timer

I hope that, by now, I have convinced you that `for` loop delays are a poor way to implement delays.

Now, we'll implement delays using a *hardware timer*. The basic function of a (hardware) timer is ... to keep precise track of time. A timer is yet another peripheral that's available to the microcontroller; thus it can be controlled using registers.

The microcontroller we are using has several (in fact, more than 10) timers of different kinds (basic, general purpose, and advanced timers) available to it. Some timers have more *resolution* (number of bits) than others and some can be used for more than just keeping track of time.

We'll be using one of the *basic* timers: `TIM6`. This is one of the simplest timers available in our microcontroller. The documentation for basic timers is in the following section:

Section 22 Timers - Page 670 - Reference Manual

Its registers are documented in:

Section 22.4.9 TIM6/TIM7 register map - Page 682 - Reference Manual

The registers we'll be using in this section are:

- `SR`, the status register.
- `EGR`, the event generation register.
- `CNT`, the counter register.
- `PSC`, the prescaler register.
- `ARR`, the autoreload register.

We'll be using the timer as a *one-shot* timer. It will sort of work like an alarm clock. We'll set the timer to go off after some amount of time and then we'll wait until the timer goes off. The documentation refers to this mode of operation as *one pulse mode*.

Here's a description of how a basic timer works when configured in one pulse mode:

- The counter is enabled by the user (`CR1.CEN = 1`).
- The `CNT` register resets its value to zero and, on each tick, its value gets incremented by one.
- Once the `CNT` register has reached the value of the `ARR` register, the counter will be disabled by hardware (`CR1.CEN = 0`) and an *update event* will be raised (`SR.UIF = 1`).

`TIM6` is driven by the APB1 clock, whose frequency doesn't have to necessarily match the processor frequency. That is, the APB1 clock could be running faster or slower. The default, however, is that both APB1 and the processor are clocked at 8 MHz.

The tick mentioned in the functional description of the one pulse mode is *not* the same as one tick of the APB1 clock. The `CNT` register increases at a frequency of $\text{apb1} / (\text{psc} + 1)$ times per second, where `apb1` is the frequency of the APB1 clock and `psc` is the value of the prescaler register, `PSC`.

Initialization

As with every other peripheral, we'll have to initialize this timer before we can use it. And just as in the previous section, initialization is going to involve two steps: powering up the timer and then configuring it.

Powering up the timer is easy: We just have to set `TIM6EN` bit to 1. This bit is in the `APB1ENR` register of the `RCC` register block.

```
// Power on the TIM6 timer
rcc.apb1enr.modify(|_, w| w.tim6en().set_bit());
```

The configuration part is slightly more elaborate.

First, we'll have to configure the timer to operate in one pulse mode.

```
// OPM Select one pulse mode
// CEN Keep the counter disabled for now
tim6.cr1.write(|w| w.opm().set_bit().cen().clear_bit());
```

Then, we'll like to have the `CNT` counter operate at a frequency of 1 KHz because our `delay` function takes a number of milliseconds as arguments and 1 KHz produces a 1 millisecond period. For that we'll have to configure the prescaler.

```
// Configure the prescaler to have the counter operate at 1 KHz
tim6.psc.write(|w| w.psc().bits(psc));
```

I'm going to let you figure out the value of the prescaler, `psc`. Remember that the frequency of the counter is $\text{apb1} / (\text{psc} + 1)$ and that `apb1` is 8 MHz.

Busy waiting

The timer should now be properly initialized. All that's left is to implement the `delay` function using the timer.

First thing we have to do is set the autoreload register (`ARR`) to make the timer go off in `ms` milliseconds. Because the counter operates at 1 KHz, the autoreload value will be the same as `ms`.

```
// Set the timer to go off in `ms` ticks
// 1 tick = 1 ms
tim6.arr.write(|w| w.arr().bits(ms));
```

Next, we need to enable the counter. It will immediately start counting.

```
// CEN: Enable the counter
tim6.cr1.modify(|_, w| w.cen().set_bit());
```

Now we need to wait until the counter reaches the value of the autoreload register, `ms`, then we'll know that `ms` milliseconds have passed. That condition is known as an *update event* and its indicated by the `UIF` bit of the status register (`SR`).

```
// Wait until the alarm goes off (until the update event occurs)
while !tim6.sr.read().uif().bit_is_set() {}
```

This pattern of just waiting until some condition is met, in this case that `UIF` becomes `1`, is known as *busy waiting* and you'll see it a few more times in this text `:~`.

Finally, we must clear (set to `0`) this `UIF` bit. If we don't, next time we enter the `delay` function we'll think the update event has already happened and skip over the busy waiting part.

```
// Clear the update event flag
tim6.sr.modify(|_, w| w.uif().clear_bit());
```

Now, put this all together and check if it works as expected.

Putting it all together

```

#![no_main]
#![no_std]

use aux9::{entry, switch_hal::OutputSwitch, tim6};

#[inline(never)]
fn delay(tim6: &tim6::RegisterBlock, ms: u16) {
    // Set the timer to go off in `ms` ticks
    // 1 tick = 1 ms
    tim6.arr.write(|w| w.arr().bits(ms));

    // CEN: Enable the counter
    tim6.cr1.modify(|_, w| w.cen().set_bit());

    // Wait until the alarm goes off (until the update event occurs)
    while !tim6.sr.read().uif().bit_is_set() {}

    // Clear the update event flag
    tim6.sr.modify(|_, w| w.uif().clear_bit());
}

#[entry]
fn main() -> ! {
    let (leds, rcc, tim6) = aux9::init();
    let mut leds = leds.into_array();

    // Power on the TIM6 timer
    rcc.apb1enr.modify(|_, w| w.tim6en().set_bit());

    // OPM Select one pulse mode
    // CEN Keep the counter disabled for now
    tim6.cr1.write(|w| w.opm().set_bit().cen().clear_bit());

    // Configure the prescaler to have the counter operate at 1 KHz
    // APB1_CLOCK = 8 MHz
    // PSC = 7999
    // 8 MHz / (7999 + 1) = 1 KHz
    // The counter (CNT) will increase on every millisecond
    tim6.psc.write(|w| w.psc().bits(7_999));

    let ms = 50;
    loop {
        for curr in 0..8 {
            let next = (curr + 1) % 8;

            leds[next].on().unwrap();
            delay(tim6, ms);
            leds[curr].off().unwrap();
            delay(tim6, ms);
        }
    }
}

```


Serial communication



This is what we'll be using. I hope your computer has one!

Nah, don't worry. This connector, the DE-9, went out of fashion on PCs quite some time ago; it got replaced by the Universal Serial Bus (USB). We won't be dealing with the DE-9 connector itself but with the communication protocol that this cable is/was usually used for.

So what's this *serial communication*? It's an *asynchronous* communication protocol where two devices exchange data *serially*, as in one bit at a time, using two data lines (plus a common ground). The protocol is asynchronous in the sense that neither of the shared lines carries a clock signal. Instead both parties must agree on how fast data will be sent along the wire *before* the communication occurs. This protocol allows *duplex* communication as data can be sent from A to B and from B to A simultaneously.

We'll be using this protocol to exchange data between the microcontroller and your computer. In contrast to the ITM protocol we have used before, with the serial communication protocol you can send data from your computer to the microcontroller.

The next practical question you probably want to ask is: How fast can we send data through this protocol?

This protocol works with frames. Each frame has one *start* bit, 5 to 9 bits of payload (data) and 1 to 2 *stop bits*. The speed of the protocol is known as *baud rate* and it's quoted in bits per second (bps). Common baud rates are: 9600, 19200, 38400, 57600 and 115200 bps.

To actually answer the question: With a common configuration of 1 start bit, 8 bits of data, 1 stop bit and a baud rate of 115200 bps one can, in theory, send 11,520 frames per second. Since each one frame carries a byte of data that results in a data rate of 11.52 KB/s. In practice, the data rate will probably be lower because of processing times on the slower side of the communication (the microcontroller).

Today's computers don't support the serial communication protocol. So you can't directly connect your computer to the microcontroller. But that's where the serial module comes in. This module will sit between the two and expose a serial interface to the microcontroller and an USB interface to your computer. The microcontroller will see your computer as another serial device and your computer will see the microcontroller as a virtual serial device.

Now, let's get familiar with the serial module and the serial communication tools that your OS offers. 151

Pick a route:

- [*nix](#)
- [Windows](#)

*nix tooling

Newer revisions of the discovery board

With newer revisions, if you connect the discovery board to your computer you should see a new TTY device appear in `/dev`.

```
$ # Linux
$ dmesg | tail | grep -i tty
[13560.675310] cdc_acm 1-1.1:1.2: ttyACM0: USB ACM device
```

This is the USB <-> Serial device. On Linux, it's named `tty*` (usually `ttyACM*` or `ttyUSB*`).

If you don't see the device appear then you probably have an older revision of the board; check the next section, which contains instructions for older revisions. If you do have a newer revision skip the next section and move to the “minicom” section.

Older revisions of the discovery board / external serial module

Connect the serial module to your computer and let's find out what name the OS assigned to it.

NOTE On macs, the USB device will named like this: `/dev/cu.usbserial-*`. You won't find it using `dmesg`, instead use `ls -l /dev | grep cu.usb` and adjust the following commands accordingly!

```
$ dmesg | grep -i tty
(..)
[ +0.000155] usb 3-2: FTDI USB Serial Device converter now attached to ttyUSB0
```

But what's this `ttyUSB0` thing? It's a file of course! Everything is a file in *nix:

```
$ ls -l /dev/ttyUSB0
crw-rw-rw- 1 root uucp 188, 0 Oct 27 00:00 /dev/ttyUSB0
```

NOTE if the permissions above is `crw-rw----`, the udev rules have not been set correctly see [udev rules](#)

You can send out data by simply writing to this file:

```
$ echo 'Hello, world!' > /dev/ttyUSB0
```

You should see the TX (red) LED on the serial module blink, just once and very fast!

All revisions: minicom

Dealing with serial devices using `echo` is far from ergonomic. So, we'll use the program `minicom` to interact with the serial device using the keyboard.

We must configure `minicom` before we use it. There are quite a few ways to do that but we'll use a `.minirc.dfl` file in the home directory. Create a file in `~/.minirc.dfl` with the following contents:

```
$ cat ~/.minirc.dfl
pu baudrate 115200
pu bits 8
pu parity N
pu stopbits 1
pu rtscts No
pu xonxoff No
```

NOTE Make sure this file ends in a newline! Otherwise, `minicom` will fail to read it.

That file should be straightforward to read (except for the last two lines), but nonetheless let's go over it line by line:

- `pu baudrate 115200` . Sets baud rate to 115200 bps.
- `pu bits 8` . 8 bits per frame.
- `pu parity N` . No parity check.
- `pu stopbits 1` . 1 stop bit.
- `pu rtscts No` . No hardware control flow.
- `pu xonxoff No` . No software control flow.

Once that's in place, we can launch `minicom`.

```
$ # NOTE you may need to use a different device here
$ minicom -D /dev/ttyACM0 -b 115200
```

This tells `minicom` to open the serial device at `/dev/ttyACM0` and set its baud rate to 115200. A text-based user interface (TUI) will pop out.

```
Welcome to minicom 2.7.1

OPTIONS: I18n
Compiled on Jun  5 2018, 10:54:41.
Port /dev/ttyACM0, 11:14:32

Press CTRL-A Z for help on special keys
```

```
█
```

You can now send data using the keyboard! Go ahead and type something. Note that the TUI will *not* echo back what you type but, if you are using an external module, you *may* see some LED on the module blink with each keystroke.

minicom commands

`minicom` exposes commands via keyboard shortcuts. On Linux, the shortcuts start with `Ctrl+A`. On mac, the shortcuts start with the `Meta` key. Some useful commands below:

- `Ctrl+A` + `Z` . Minicom Command Summary
- `Ctrl+A` + `C` . Clear the screen
- `Ctrl+A` + `X` . Exit and reset
- `Ctrl+A` + `Q` . Quit with no reset

NOTE mac users: In the above commands, replace `Ctrl+A` with `Meta` .

Windows tooling

Start by unplugging your discovery board.

Before plugging the discovery board or the serial module, run the following command on the terminal:

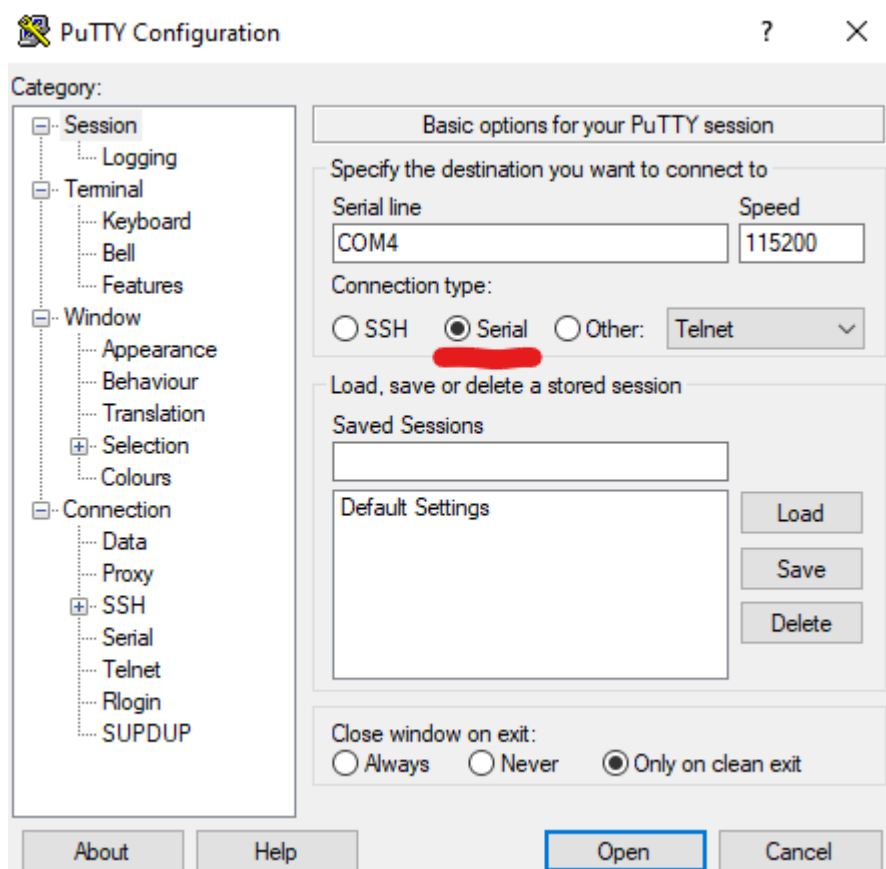
```
$ mode
```

It will print a list of devices that are connected to your computer. The ones that start with **COM** in their names are serial devices. This is the kind of device we'll be working with. Take note of all the **COM ports** **mode** outputs *before* plugging the serial module.

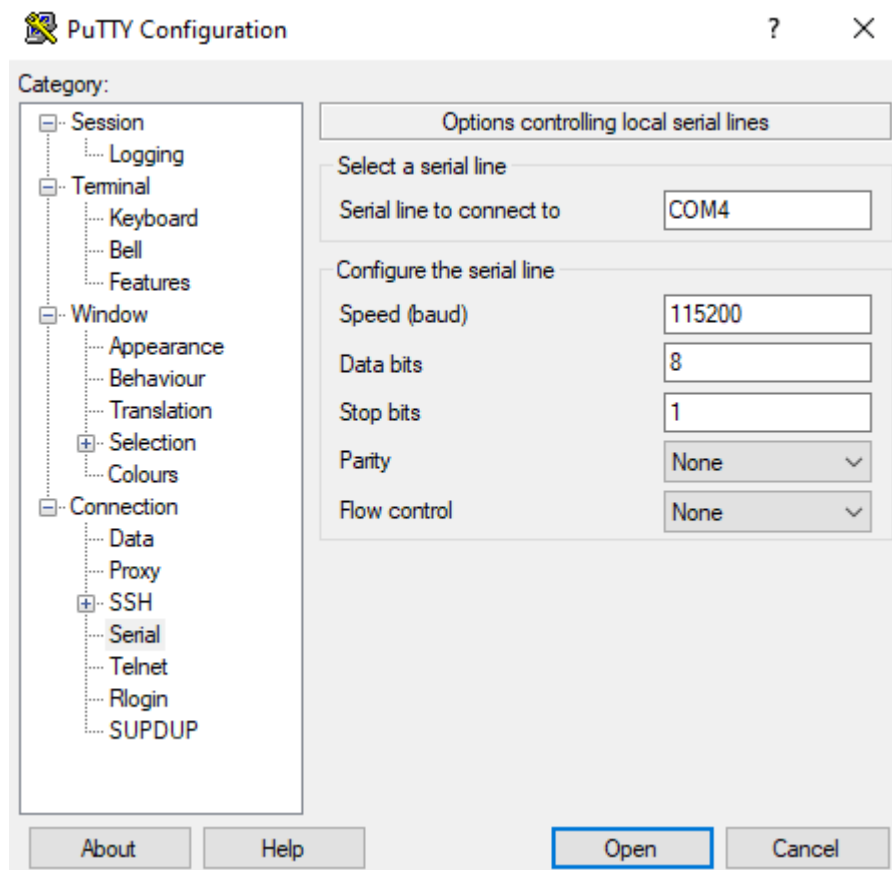
Now, plug the discovery board and run the **mode** command again. If you see a new **COM** port appear on the list then you have a newer revision of the discovery and that's the COM port assigned to the serial functionality on the discovery. You can skip the next paragraph.

If you didn't get a new COM port then you probably have an older revision of the discovery. Now plug the serial module; you should see new COM port appear; that's the COM port of the serial module.

Now launch **putty**. A GUI will pop out.



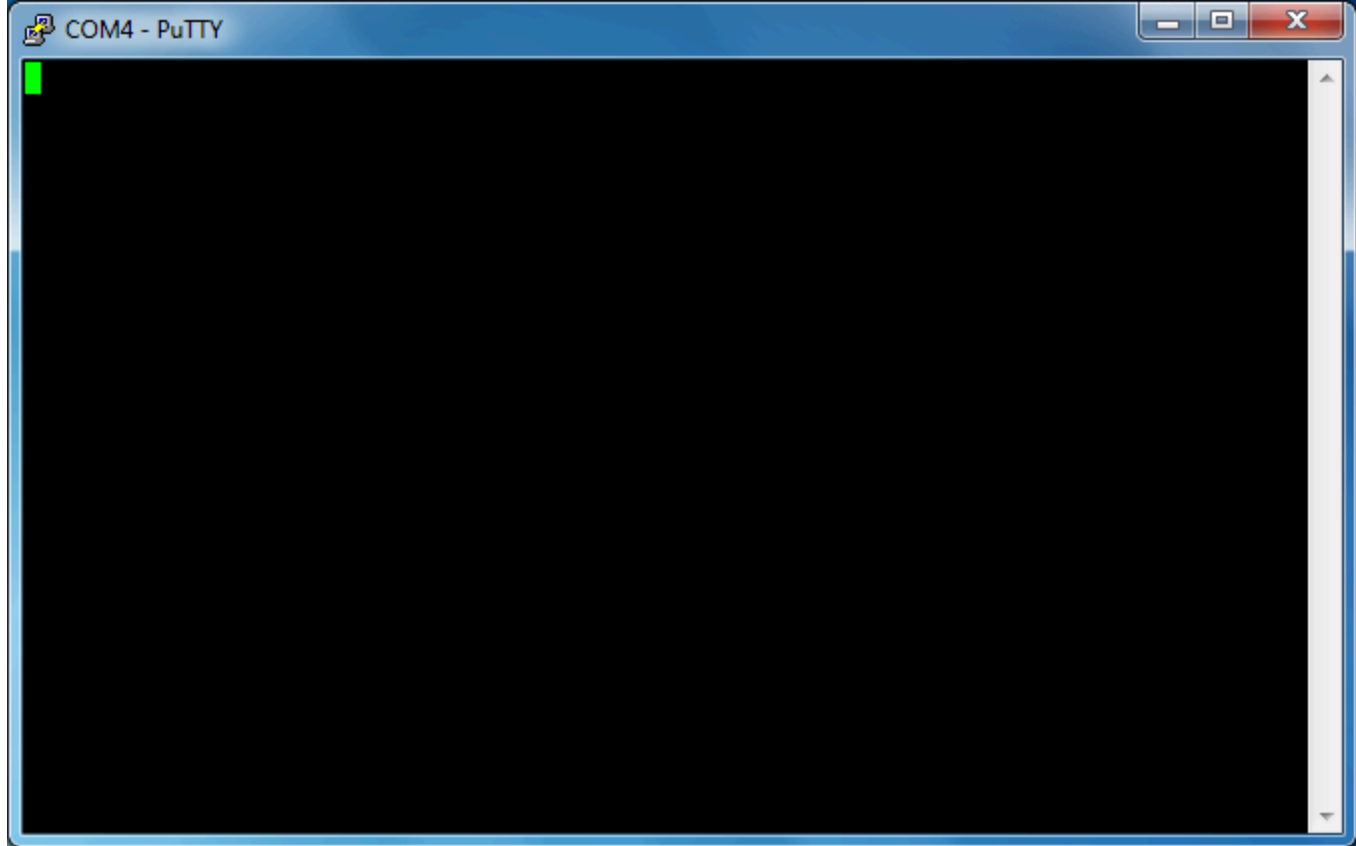
On the starter screen, which should have the “Session” category open, pick “Serial” as the “Connection type”. On the “Serial line” field enter the **COM** device you got on the previous step, for example **COM3**.



Next, pick the “Connection/Serial” category from the menu on the left. On this new view, make sure that the serial port is configured as follows:

- “Speed (baud)”: 115200
- “Data bits”: 8
- “Stop bits”: 1
- “Parity”: None
- “Flow control”: None

Finally, click the Open button. A console will show up now:

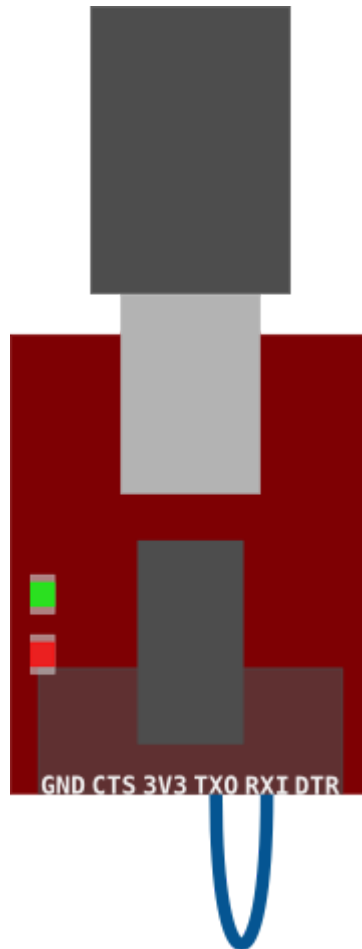


If you type on this console, the TX (red) LED on the Serial module should blink. Each key stroke should make the LED blink once. Note that the console won't echo back what you type so the screen will remain blank.

Loopbacks

We've tested sending data. It's time to test receiving it. Except that there's no other device that can send us some data ... or is there?

Enter: loopbacks



You can send data to yourself! Not very useful in production but very useful for debugging.

Older board revision / external serial module

Connect the **TX0** and the **RXI** pins of the serial module together using a male to male jumper wire as shown above.

Now enter some text into minicom/PuTTY and observe. What happens?

You should see three things:

- As before, the TX (red) LED blinks on each key press.
- But now the RX (green) LED blinks on each key press as well! This indicates that the serial module is receiving some data; the one it just sent.
- Finally, on the minicom/PuTTY console, you should see that what you type echoes back to the console.

If you have a newer revision of the board you can set up a loopback by shorting the PC4 and PC5 pins using a female to female jumper wire, like [you did for the SWO pin](#).

You should now be able to send data to yourself.

Now try to enter some text into minicom/PuTTY and observe.

NOTE: To rule out the possibility of the existing firmware doing weird things to the serial pins (PC4 and PC5) we recommend *holding* the reset button while you enter text into minicom/PuTTY.

If all is working you should see what you type echoed back to minicom/PuTTY console.

Now that you are familiar with sending and receiving data over serial port using minicom/PuTTY, let's make your microcontroller and your computer talk!

USART

The microcontroller has a peripheral called USART, which stands for Universal Synchronous/Asynchronous Receiver/Transmitter. This peripheral can be configured to work with several communication protocols like the serial communication protocol.

Throughout this chapter, we'll use serial communication to exchange information between the microcontroller and your computer. But before we do that we have to wire up everything.

I mentioned before that this protocol involves two data lines: TX and RX. TX stands for transmitter and RX stands for receiver. Transmitter and receiver are relative terms though; which line is the transmitter and which line is the receiver depends from which side of the communication you are looking at the lines.

Newer board revisions

If you have a newer revision of the board and are using the on-board USB <-> Serial functionality then the `auxiliary` crate will set pin `PC4` as the TX line and pin `PC5` as the RX line.

If you had previously connected the PC4 and PC4 pins in order to test the [loopback functionality](#) in the previous section, make sure to remove that wire, or the upcoming serial communication will fail silently.

Everything is already wired on the board so you don't need to wire anything yourself. You can move on to the [next section](#).

Older board revisions / external serial module

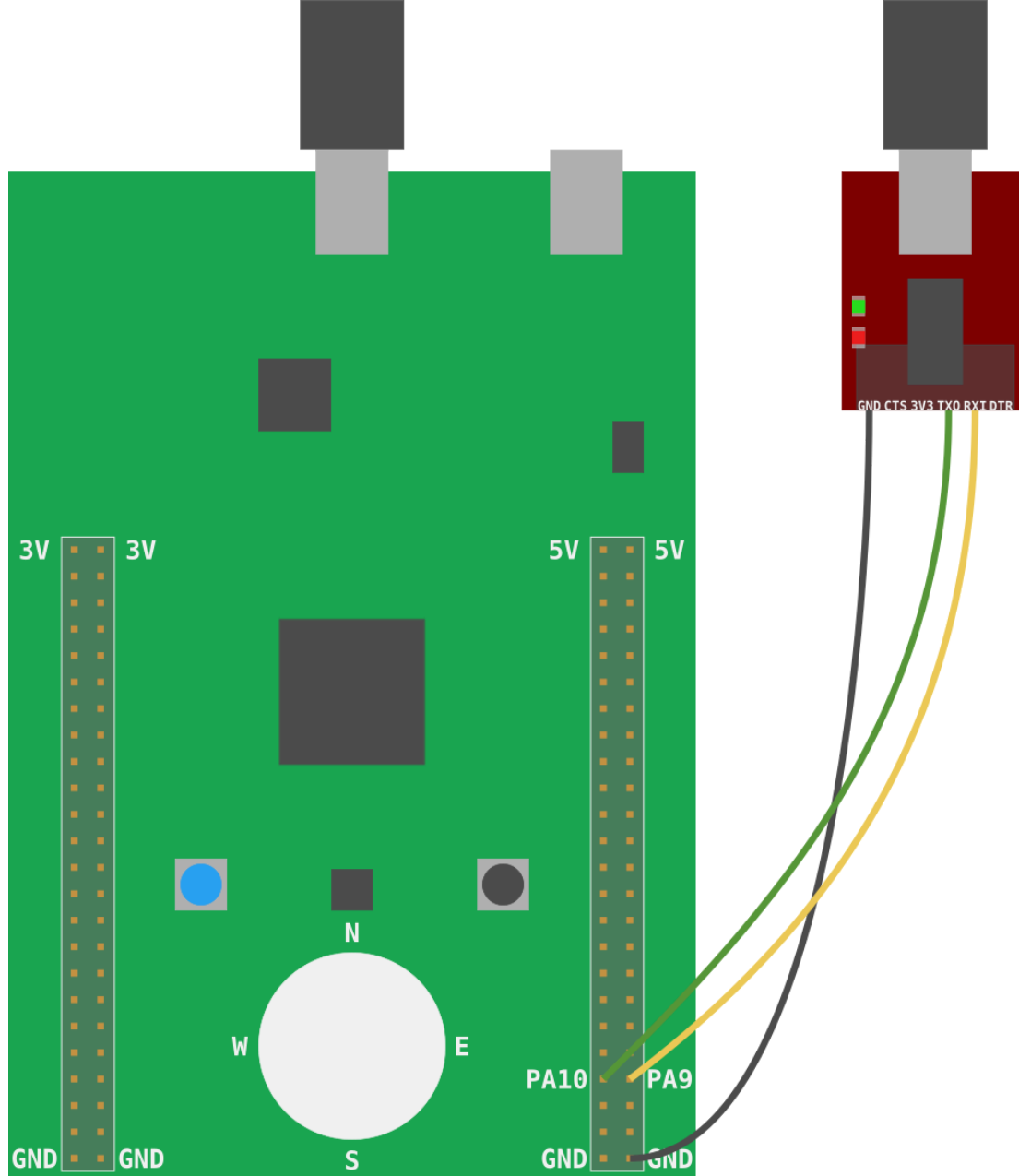
If you are using an external USB <-> Serial module then you will **need** to enable the `adapter` feature of the `aux11` crate dependency in `Cargo.toml`.

```
[dependencies.aux11]
path = "auxiliary"
# enable this if you are going to use an external serial adapter
features = ["adapter"] # <- uncomment this
```

We'll be using the pin `PA9` as the microcontroller's TX line and `PA10` as its RX line. In other words, the pin `PA9` outputs data onto its wire whereas the pin `PA10` listens for data on its wire.

We could have used a different pair of pins as the TX and RX pins. There's a table in page 44 of the [Data Sheet](#) that list all the other possible pins we could have used.

The serial module also has TX and RX pins. We'll have to *cross* these pins: that is connect the microcontroller's TX pin to the serial module's RX pin and the micro's RX pin to the serial module's TX pin. The wiring diagram below shows all the necessary connections.



These are the recommended steps to connect the microcontroller and the serial module:

- Close OpenOCD and `itmdump`
- Disconnect the USB cables from the F3 and the serial module.
- Connect one of F3 GND pins to the GND pin of the serial module using a female to male (F/M) wire. Preferably, a black one.
- Connect the PA9 pin on the back of the F3 to the RX1 pin of the serial module using a F/M wire.
- Connect the PA10 pin on the back of the F3 to the TX0 pin of the serial module using a F/M wire.
- Now connect the USB cable to the F3.
- Finally connect the USB cable to the Serial module.
- Re-launch OpenOCD and `itmdump`

Everything's wired up! Let's proceed to send data back and forth.

Send a single byte

Our first task will be to send a single byte from the microcontroller to the computer over the serial connection.

This time, I'm going to provide you with an already initialized USART peripheral. You'll only have to work with the registers that are in charge of sending and receiving data.

Go into the `11-usbart` directory and let's run the starter code therein. Make sure that you have minicom/PuTTY open.

```
#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux11::{entry, iprint, iprintln};

#[entry]
fn main() -> ! {
    let (usart1, _mono_timer, _itm) = aux11::init();

    // Send a single character
    usart1
        .tdr
        .write(|w| w.tdr().bits(u16::from(b'X')) );

    loop {}
}
```

This program writes to the `TDR` register. This causes the `USART` peripheral to send one byte of information through the serial interface.

On the receiving end, your computer, you should see show the character `X` appear on minicom/PuTTY's terminal.

Send a string

The next task will be to send a whole string from the microcontroller to your computer.

I want you to send the string `"The quick brown fox jumps over the lazy dog."` from the microcontroller to your computer.

It's your turn to write the program.

Execute your program inside the debugger, statement by statement. What do you see?

Then execute the program again but in *one go* using the `continue` command. What happens this time?

Finally, build the program in *release* mode and, again, run it in one go. What happens this time?

Overruns

If you wrote your program like this:

```
#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux11::{entry, iprint, iprintln};

#[entry]
fn main() -> ! {
    let (usart1, _mono_timer, _itm) = aux11::init();

    // Send a string
    for byte in b"The quick brown fox jumps over the lazy dog.".iter() {
        usart1
            .tdr
            .write(|w| w.tdr().bits(u16::from(*byte)));
    }

    loop {}
}
```

You probably received something like this on your computer when you executed the program compiled in debug mode.

```
$ # minicom's terminal
(..)
The uic brwn oxjums oer helaz do.
```

And if you compiled in release mode, you probably only got something like this:

```
$ # minicom's terminal
(..)
T
```

What went wrong?

You see, sending bytes over the wire takes a relatively large amount of time. I already did the math so let me quote myself:

With a common configuration of 1 start bit, 8 bits of data, 1 stop bit and a baud rate of 115200 bps one can, in theory, send 11,520 frames per second. Since each one frame carries a byte of data that results in a data rate of 11.52 KB/s

Our pangram has a length of 45 bytes. That means it's going to take, at least, 3,900 microseconds ($45 \text{ bytes} / (11,520 \text{ bytes/s}) = 3,906 \text{ us}$) to send the string. The processor is working at 8 MHz, where executing an instruction takes 125 nanoseconds, so it's likely going to be done with the `for` loop in less than 3,900 microseconds.

We can actually time how long it takes to execute the `for` loop. `aux11::init()` returns a `MonoTimer` (monotonic timer) value that exposes an `Instant` API that's similar to the one in `std::time`.

```
#![deny(unsafe_code)]
#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux11::{entry, iprint, iprintln};

#[entry]
fn main() -> ! {
    let (usart1, mono_timer, mut itm) = aux11::init();

    let instant = mono_timer.now();
    // Send a string
    for byte in b"The quick brown fox jumps over the lazy dog.".iter() {
        usart1.tdr.write(|w| w.tdr().bits(u16::from(*byte)));
    }
    let elapsed = instant.elapsed(); // in ticks

    iprintln!(
        &mut itm.stim[0],
        "`for` loop took {} ticks ({} us)",
        elapsed,
        elapsed as f32 / mono_timer.frequency().0 as f32 * 1e6
    );

    loop {}
}
```

In debug mode, I get:

```
$ # itmdump terminal
(..)
`for` loop took 22415 ticks (2801.875 us)
```

This is less than 3,900 microseconds but it's not that far off and that's why only a few bytes of information are lost.

In conclusion, the processor is trying to send bytes at a faster rate than what the hardware can actually handle and this results in data loss. This condition is known as buffer *overflow*.

How do we avoid this? The status register (`ISR`) has a flag, `TXE`, that indicates if it's "safe" to write to the `TDR` register without incurring in data loss.

Let's use that to slowdown the processor.

```

#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux11::{entry, iprint, iprintln};

#[entry]
fn main() -> ! {
    let (usart1, mono_timer, mut itm) = aux11::init();

    let instant = mono_timer.now();
    // Send a string
    for byte in b"The quick brown fox jumps over the lazy dog.".iter() {
        // wait until it's safe to write to TDR
        while usart1.isr.read().txe().bit_is_clear() {} // <- NEW!

        usart1
            .tdr
            .write(|w| w.tdr().bits(u16::from(*byte)));
    }
    let elapsed = instant.elapsed(); // in ticks

    iprintln!(
        &mut itm.stim[0],
        "`for` loop took {} ticks ({} us)",
        elapsed,
        elapsed as f32 / mono_timer.frequency().0 as f32 * 1e6
    );

    loop {}
}

```

This time, running the program in debug or release mode should result in a complete string on the receiving side.

```

$ # minicom/PuTTY's console
(..)
The quick brown fox jumps over the lazy dog.

```

The timing of the `for` loop should be closer to the theoretical 3,900 microseconds as well. The timing below is for the debug version.

```

$ # itmdump terminal
(..)
`for` loop took 30499 ticks (3812.375 us)

```


uprintln!

For the next exercise, we'll implement the `uprint!` family of macros. Your goal is to make this line of code work:

```
uprintln!(serial, "The answer is {}", 40 + 2);
```

Which must send the string `"The answer is 42"` through the serial interface.

How do we go about that? It's informative to look into the `std` implementation of `println!`.

```
// src/libstd/macros.rs
macro_rules! print {
    ($($arg:tt)*) => ($crate::io::_print(format_args!($($arg)*)));
}
```

Looks simple so far. We need the built-in `format_args!` macro (it's implemented in the compiler so we can't see what it actually does). We'll have to use that macro in the exact same way. What does this `_print` function do?

```
// src/libstd/io/stdio.rs
pub fn _print(args: fmt::Arguments) {
    let result = match LOCAL_STDOUT.state() {
        LocalKeyState::Uninitialized |
        LocalKeyState::Destroyed => stdout().write_fmt(args),
        LocalKeyState::Valid => {
            LOCAL_STDOUT.with(|s| {
                if s.borrow_state() == BorrowState::Unused {
                    if let Some(w) = s.borrow_mut().as_mut() {
                        return w.write_fmt(args);
                    }
                }
                stdout().write_fmt(args)
            })
        }
    };
    if let Err(e) = result {
        panic!("failed printing to stdout: {}", e);
    }
}
```

That *looks* complicated but the only part we are interested in is: `w.write_fmt(args)` and `stdout().write_fmt(args)`. What `print!` ultimately does is call the `fmt::Write::write_fmt` method with the output of `format_args!` as its argument.

Luckily we don't have to implement the `fmt::Write::write_fmt` method either because it's a default method. We only have to implement the `fmt::Write::write_str` method.

Let's do that.

This is what the macro side of the equation looks like. What's left to be done by you is provide the implementation of the `write_str` method.

Above we saw that `Write` is in `std::fmt`. We don't have access to `std` but `Write` is also available in `core::fmt`. 176

```
#![deny(unsafe_code)]
#![no_main]
#![no_std]

use core::fmt::{self, Write};

#[allow(unused_imports)]
use aux11::{entry, iprint, iprintln, usart1};

macro_rules! uprint {
    ($serial:expr, $($arg:tt)*) => {
        $serial.write_fmt(format_args!($($arg)*)).ok()
    };
}

macro_rules! uprintln {
    ($serial:expr, $fmt:expr) => {
        uprint!($serial, concat!($fmt, "\n"))
    };
    ($serial:expr, $fmt:expr, $($arg:tt)*) => {
        uprint!($serial, concat!($fmt, "\n"), $($arg)*)
    };
}

struct SerialPort {
    usart1: &'static mut usart1::RegisterBlock,
}

impl fmt::Write for SerialPort {
    fn write_str(&mut self, s: &str) -> fmt::Result {
        // TODO implement this
        // hint: this will look very similar to the previous program
        Ok(())
    }
}

#[entry]
fn main() -> ! {
    let (usart1, _mono_timer, _itm) = aux11::init();

    let mut serial = SerialPort { usart1 };

    uprintln!(serial, "The answer is {}", 40 + 2);

    loop {}
}
```


Receive a single byte

So far we have sending data from the microcontroller to your computer. It's time to try the opposite: receiving data from your computer.

There's a **RDR** register that will be filled with the data that comes from the RX line. If we read that register, we'll retrieve the data that the other side of the channel sent. The question is: How do we know that we have received (new) data? The status register, **ISR**, has a bit for that purpose: **RXNE**. We can just busy wait on that flag.

```
#![deny(unsafe_code)]
#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux11::{entry, iprint, iprintln};

#[entry]
fn main() -> ! {
    let (usart1, _mono_timer, _itm) = aux11::init();

    loop {
        // Wait until there's data available
        while usart1.isr.read().rxne().bit_is_clear() {}

        // Retrieve the data
        let _byte = usart1.rdr.read().rdr().bits() as u8;

        aux11::bkpt();
    }
}
```

Let's try this program! Let it run free using **continue** and then type a single character in minicom/PuTTY's console. What happens? What are the contents of the **_byte** variable?

```
(gdb) continue
Continuing.

Program received signal SIGTRAP, Trace/breakpoint trap.
0x8003d48 in __bkpt ()

(gdb) finish
Run till exit from #0 0x8003d48 in __bkpt ()
usart::main () at src/11-usart/src/main.rs:19
19          aux11::bkpt();

(gdb) p/c _byte
$1 = 97 'a'
```


Echo server

Let's merge transmission and reception into a single program and write an echo server. An echo server sends back to the client the same text it sent. For this application, the microcontroller will be the server and you and your computer will be the client.

This should be straightforward to implement. (hint: do it byte by byte)

Reverse a string

Alright, next let's make the server more interesting by having it respond to the client with the reverse of the text that they sent. The server will respond to the client every time they press the ENTER key. Each server response will be in a new line.

This time you'll need a buffer; you can use `heapless::Vec`. Here's the starter code:

```
#![deny(unsafe_code)]
#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux11::{entry, iprint, iprintln};
use heapless::Vec;

#[entry]
fn main() -> ! {
    let (usart1, _mono_timer, _itm) = aux11::init();

    // A buffer with 32 bytes of capacity
    let mut buffer: Vec<u8, 32> = Vec::new();

    loop {
        buffer.clear();

        // TODO Receive a user request. Each user request ends with ENTER
        // NOTE `buffer.push` returns a `Result`. Handle the error by responding
        // with an error message.

        // TODO Send back the reversed string
    }
}
```


My solution

```

#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux11::{entry, iprint, iprintln};
use heapless::Vec;

#[entry]
fn main() -> ! {
    let (usart1, _mono_timer, _itm) = aux11::init();

    // A buffer with 32 bytes of capacity
    let mut buffer: Vec<u8, 32> = Vec::new();

    loop {
        buffer.clear();

        loop {
            while usart1.isr.read().rxne().bit_is_clear() {}
            let byte = usart1.rdr.read().rdr().bits() as u8;

            if buffer.push(byte).is_err() {
                // buffer full
                for byte in b"error: buffer full\n\r" {
                    while usart1.isr.read().txe().bit_is_clear() {}
                    usart1
                        .tdr
                        .write(|w| w.tdr().bits(u16::from(*byte)));
                }

                break;
            }

            // Carriage return
            if byte == 13 {
                // Respond
                for byte in buffer.iter().rev().chain(&[b'\n', b'\r']) {
                    while usart1.isr.read().txe().bit_is_clear() {}
                    usart1
                        .tdr
                        .write(|w| w.tdr().bits(u16::from(*byte)));
                }

                break;
            }
        }
    }
}

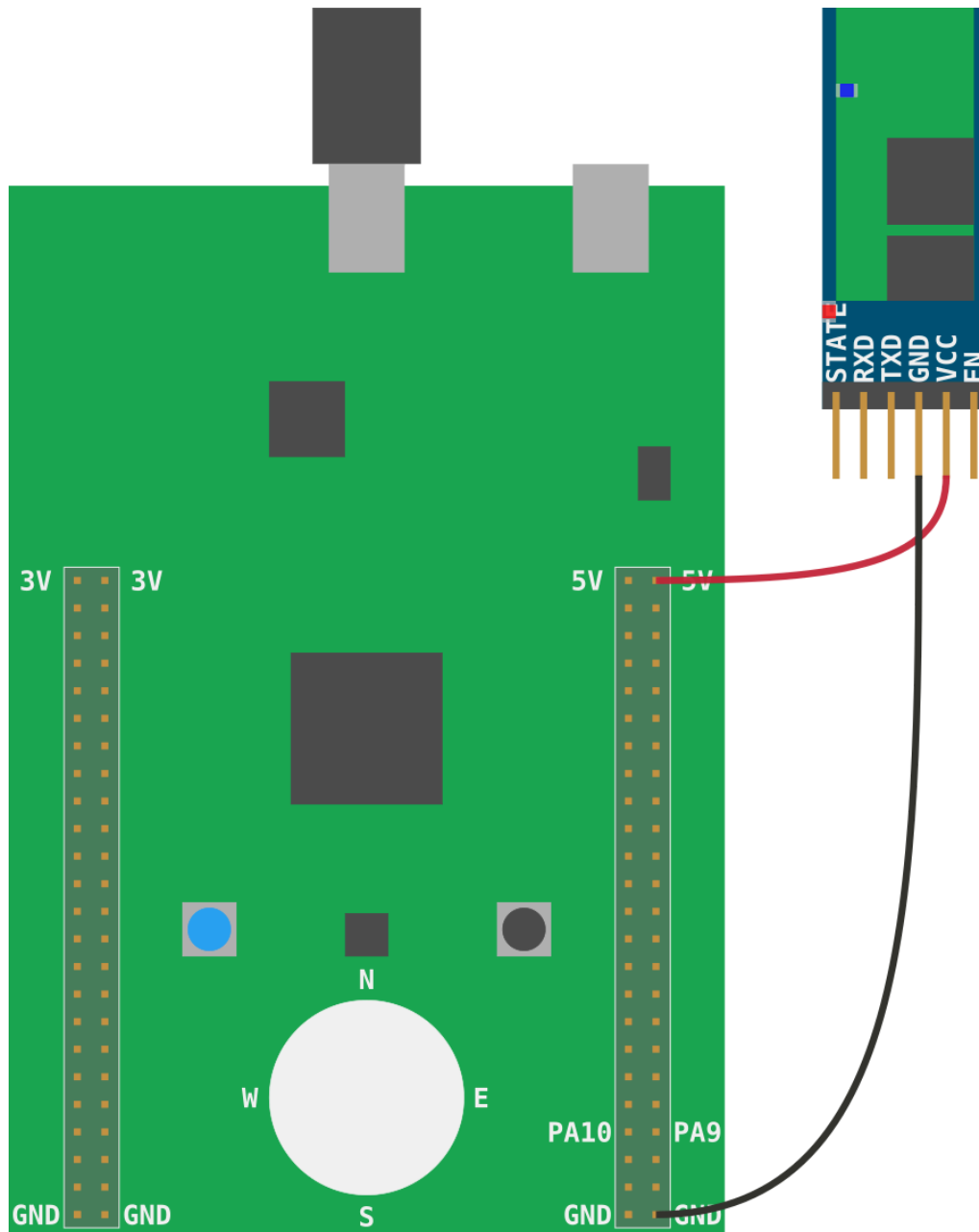
```


Bluetooth setup

It's time to get rid of some wires. Serial communication can not only be emulated on top of the USB protocol; it can also be emulated on top of the Bluetooth protocol. This serial over Bluetooth protocol is known as RFCOMM.

Before we use the Bluetooth module with the microcontroller, let's first interact with it using minicom/PuTTY.

The first thing we'll need to do is: turn on the Bluetooth module. We'll have to share some of the F3 power to it using the following connection:



The recommend steps to wire this up are:

- Close OpenOCD and `itmdump`
- Disconnect the USB cables from the F3 and the serial module.
- Connect F3's GND pin to the Bluetooth's GND pin using a female to female (F/F) wire. Preferably, a black one.
- Connect F3's 5V pin to the Bluetooth's VCC pin using a F/F wire. Preferably, a red one.

- Then, connect the USB cable back to the F3.
- Re-launch OpenOCD and `itmdump`

Two LEDs, a blue one and a red one, on the Bluetooth module should start blinking right after you power on the F3 board.

Next thing to do is pair your computer and the Bluetooth module. AFAIK, Windows and mac users can simply use their OS default Bluetooth manager to do the pairing. The Bluetooth module default pin is 1234.

Linux users will have to follow (some of) [these instructions](#).

Linux

If you have a graphical Bluetooth manager, you can use that to pair your computer to the Bluetooth module and skip most of these steps. You'll probably still have to [this step](#) though.

Power up

First, your computer's Bluetooth transceiver may be OFF. Check its status with `hciconfig` and turn it ON if necessary:

```
$ hciconfig
hci0:  Type: Primary  Bus: USB
      BD Address: 68:17:29:XX:XX:XX  ACL MTU: 310:10  SCO MTU: 64:8
      DOWN <--
      RX bytes:580 acl:0 sco:0 events:31 errors:0
      TX bytes:368 acl:0 sco:0 commands:30 errors:0

$ sudo hciconfig hci0 up

$ hciconfig
hci0:  Type: Primary  Bus: USB
      BD Address: 68:17:29:XX:XX:XX  ACL MTU: 310:10  SCO MTU: 64:8
      UP RUNNING <--
      RX bytes:1190 acl:0 sco:0 events:67 errors:0
      TX bytes:1072 acl:0 sco:0 commands:66 errors:0
```

Then you need to launch the BlueZ (Bluetooth) daemon:

- On systemd based Linux distributions, use:

```
$ sudo systemctl start bluetooth
```

- On Ubuntu (or upstart based Linux distributions), use:

```
$ sudo /etc/init.d/bluetooth start
```

You may also need to unblock your Bluetooth, depending on what `rfkill list` says:

```
$ rfkill list
9: hci0: Bluetooth
   Soft blocked: yes # <--
   Hard blocked: no

$ sudo rfkill unblock bluetooth

$ rfkill list
9: hci0: Bluetooth
   Soft blocked: no  # <--
   Hard blocked: no
```

```
$ hcitool scan
Scanning ...
      20:16:05:XX:XX:XX      Ferris
$ #                          ^^^^^^^
```

Pair

```
$ bluetoothctl
[bluetooth]# scan on
[bluetooth]# agent on
[bluetooth]# pair 20:16:05:XX:XX:XX
Attempting to pair with 20:16:05:XX:XX:XX
[CHG] Device 20:16:05:XX:XX:XX Connected: yes
Request PIN code
[agent] Enter PIN code: 1234
```

rfcomm device

We'll create a device file for our Bluetooth module in `/dev`. Then we'll be able to use it just like we used `/dev/ttyUSB0`.

```
$ sudo rfcomm bind 0 20:16:05:XX:XX:XX
```

Because we used `0` as an argument to `bind`, `/dev/rfcomm0` will be the device file assigned to our Bluetooth module.

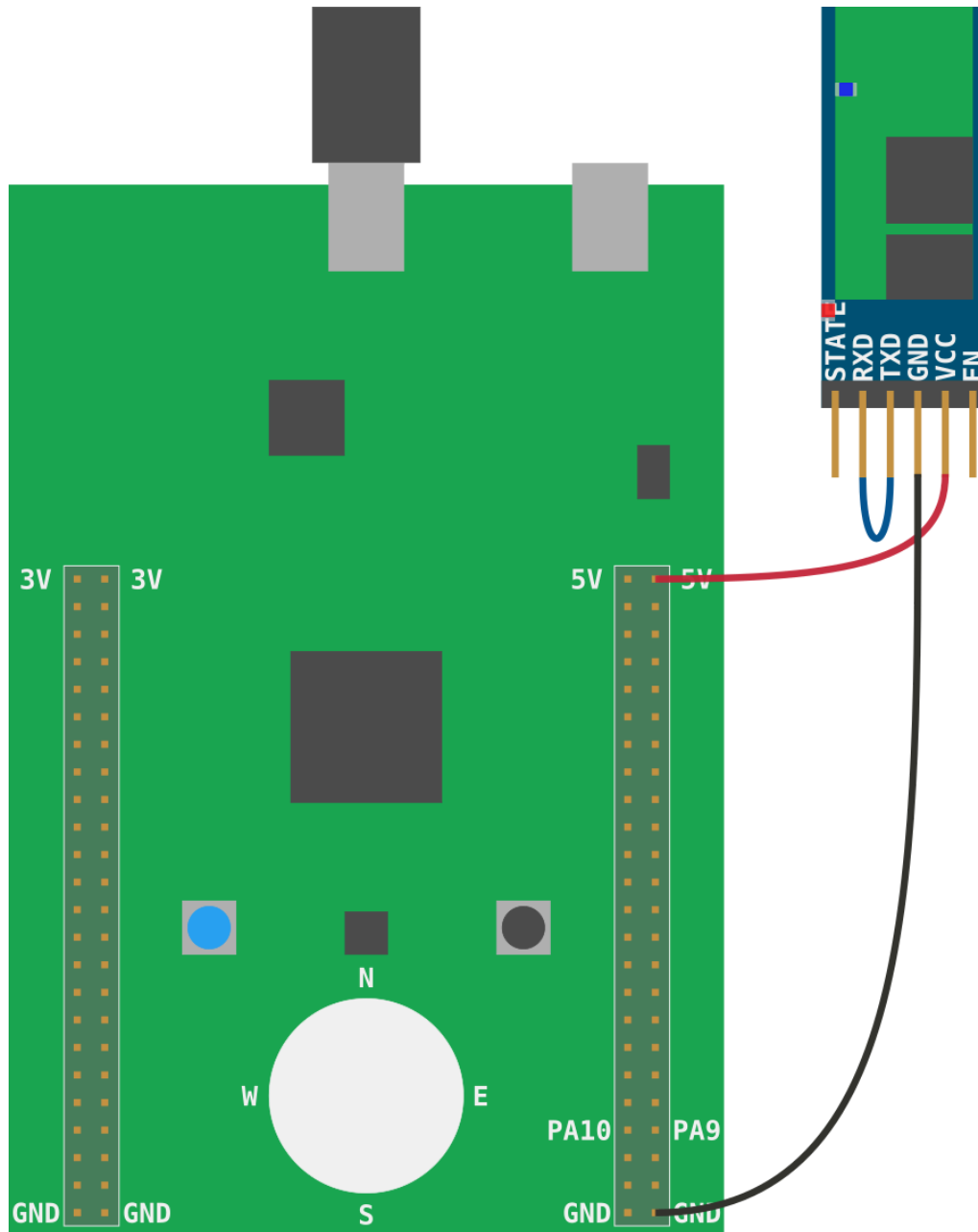
You can release (destroy) the device file at any time with the following command:

```
$ # Don't actually run this command right now!
$ sudo rfcomm release 0
```


Loopback, again

After pairing your computer to the Bluetooth module, your OS should have created a device file / COM port for you. On Linux, it should be `/dev/rfcomm*`; on mac, it should be `/dev/cu.*`; and on Windows, it should be a new COM port.

We can now test the Bluetooth module with minicom/PuTTY. Because this module doesn't have LED indicators for the transmission and reception events like the serial module did, we'll test the module using a loopback connection:



Just connect the module's TXD pin to its RXD pin using a F/F wire.

Now, connect to the device using `minicom` / `PuTTY` :

```
$ minicom -D /dev/rfcomm0
```

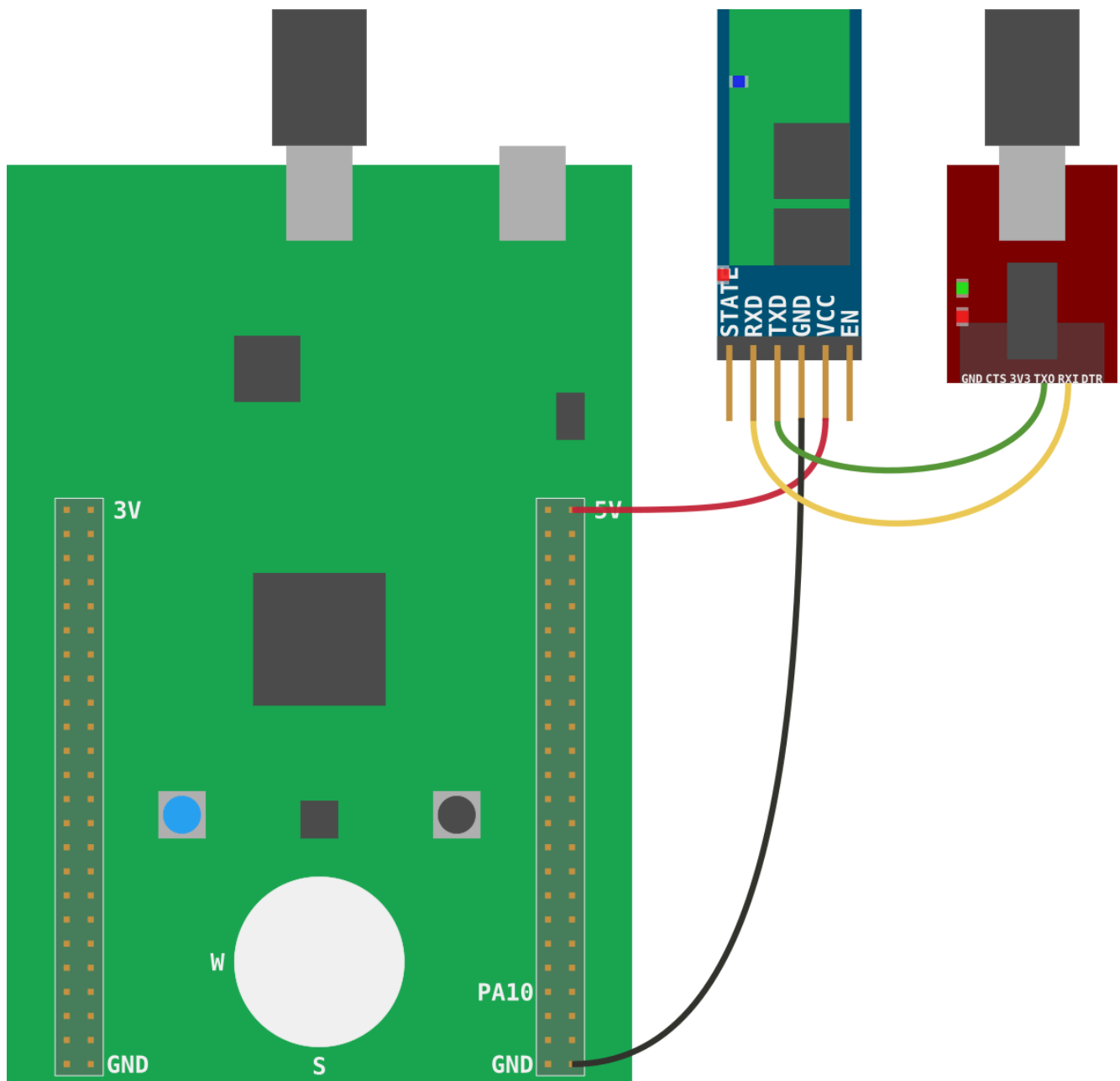
Upon connecting, the blinking pattern of the Bluetooth module should change to: long pause then blink twice quickly.

AT commands

AT commands

The Bluetooth module and the F3 need to be configured to communicate at the same baud rate. The tutorial code initializes the UART1 serial device to a baud rate of 115200. The HC-05 Bluetooth module is configured at a baud rate of 9600 by default.

The Bluetooth module supports an AT mode that allows you to examine and change its configuration and settings. To utilize the AT mode, connect the Bluetooth module to the F3 and FTDI as shown in the following diagram.



Recommended steps to enter AT mode:

- Disconnect the F3 and FTDI from your computer.
- Connect F3's GND pin to the Bluetooth's GND pin using a Female/Female (F/F) wire (preferably, a black one).

- Connect F3's 5V pin to the Bluetooth's VCC pin using a F/F wire (preferably, a red one).
- Connect the FTDI RXI pin to the Bluetooth's TXD pin using a Female/Male (F/M) wire.
- Connect the FTDI TXO pin to the Bluetooth's RXD pin using a Female/Male (F/M) wire.
- Now connect the FTDI to your computer via USB cable.
- Next connect the F3 to your computer via USB cable while simultaneously pressing and holding the button on the Bluetooth module (kinda tricky).
- Now, release the button and the Bluetooth module will enter AT mode. You can confirm this by observing that the red LED on the Bluetooth module is blinking in a slow pattern (approx 1-2 seconds on/off).

The AT mode always operates at a baud rate of 38400, so configure your terminal program for that baud rate and connect to the FTDI device.

When your serial connection is established, you may get a bunch of **ERROR: (0)** repeatedly being displayed. If this happens, just hit ENTER to stop the errors.

Sanity check

```
$ at
OK
OK
(etc...)
```

Answers **OK** repeatedly until you hit ENTER again.

Rename the device

```
$ at+name=ferris
OK
```

Query for the current baud rate of the Bluetooth module

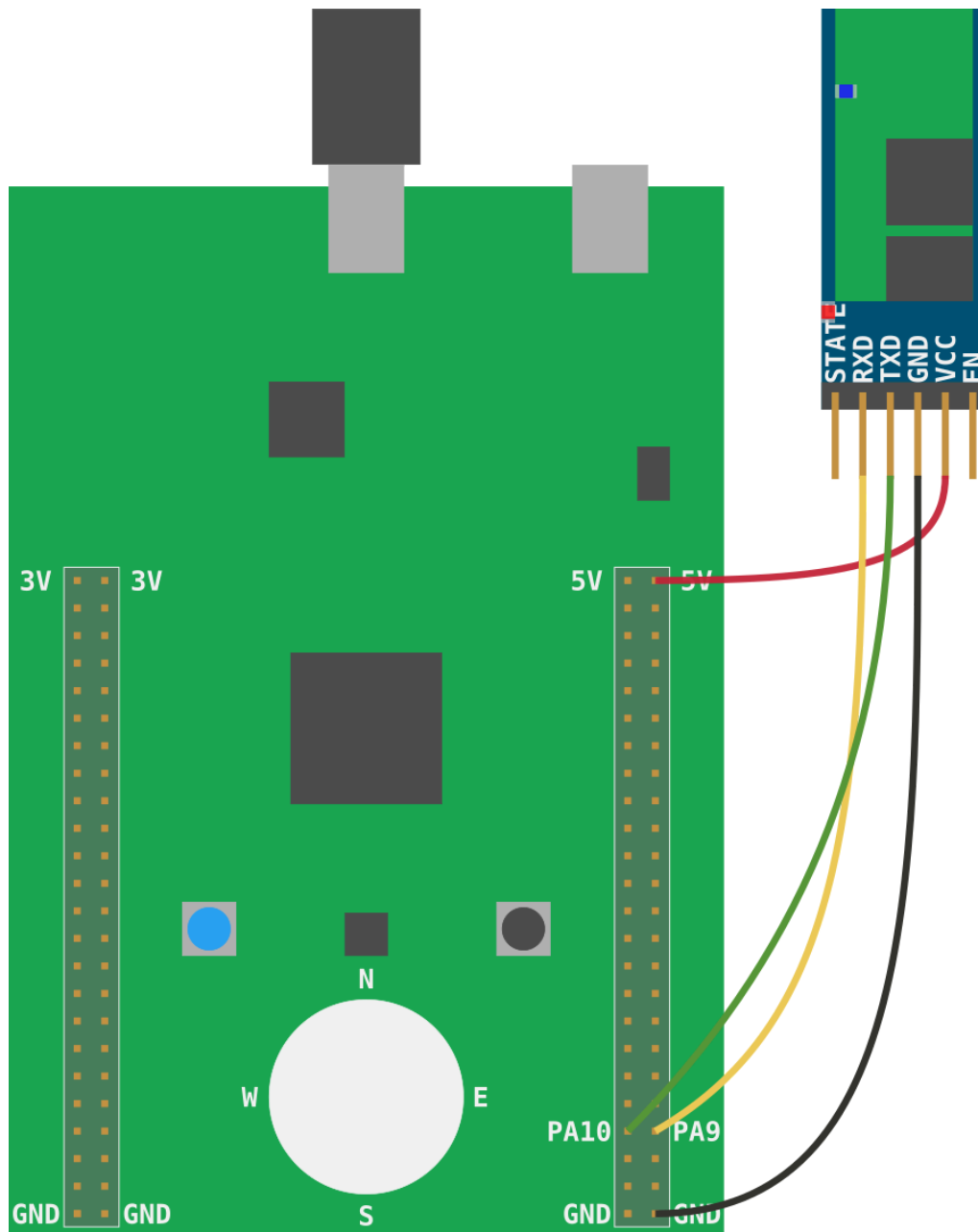
```
at+uart?
+UART:9600,0,0
OK
+UART:9600,0,0
OK
(etc ...)
```

Change the baud rate

```
$ at+uart=115200,0,0
OK
```


Serial over Bluetooth

Now that we verify that the Bluetooth module works with minicom/PuTTY, let's connect it to the microcontroller:



Recommended steps to wire this up:

- Close OpenOCD and `itmdump`.
- Disconnect the F3 from your computer.
- Connect F3's GND pin to the module's GND pin using a female to female (F/F) wire (preferably, a black one).
- Connect F3's 5V pin to the module's VCC pin using a F/F wire (preferably, a red one).
- Connect the PA9 (TX) pin on the back of the F3 to the Bluetooth's RXD pin using a F/F wire.
- Connect the PA10 (RX) pin on the back of the F3 to the Bluetooth's TXD pin using a F/F wire.
- Now connect the F3 and your computer using an USB cable.
- Re-launch OpenOCD and `itmdump`.

And that's it! You should be able to run all the programs you wrote in [section 11](#) without modification! Just make sure you open the right serial device / COM port.

NOTE If you are having trouble communicating with the bluetooth device, you may need to initialize USART1 with a lower baud rate. Lowering it from 115,200 bps to 9,600 bps might help, as described [in the code here](#)

I2C

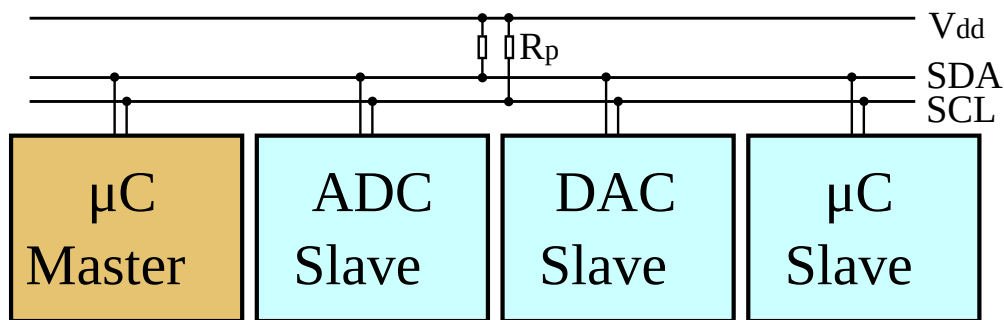
We just saw the serial communication protocol. It's a widely used protocol because it's very simple and this simplicity makes it easy to implement on top of other protocols like Bluetooth and USB.

However, it's simplicity is also a downside. More elaborated data exchanges, like reading a digital sensor, would require the sensor vendor to come up with another protocol on top of it.

(Un)luckily for us, there are *plenty* of other communication protocols in the embedded space. Some of them are widely used in digital sensors.

The F3 board we are using has three motion sensors in it: an accelerometer, a magnetometer and gyroscope. The accelerometer and magnetometer are packaged in a single component and can be accessed via an I2C bus.

I2C stands for Inter-Integrated Circuit and is a *synchronous serial* communication protocol. It uses two lines to exchange data: a data line (SDA) and a clock line (SCL). Because a clock line is used to synchronize the communication, this is a *synchronous* protocol.



This protocol uses a *master slave* model where the master is the device that *starts* and drives the communication with a slave device. Several devices, both masters and slaves, can be connected to the same bus at the same time. A master device can communicate with a specific slave device by first broadcasting its *address* to the bus. This address can be 7 bits or 10 bits long. Once a master has *started* a communication with a slave, no other device can make use of the bus until the master *stops* the communication.

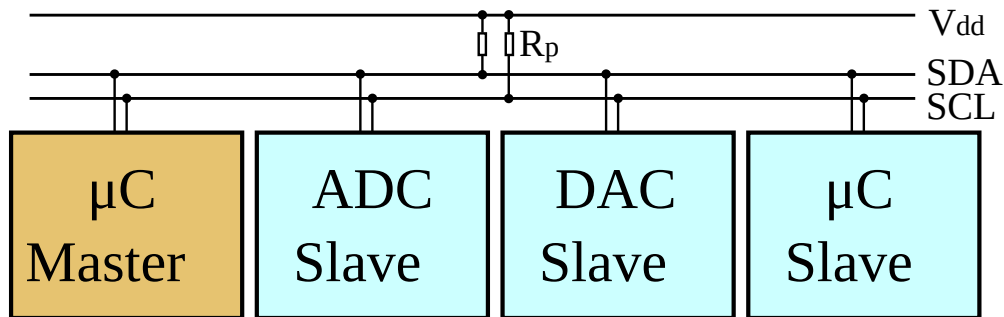
The clock line determines how fast data can be exchanged and it usually operates at a frequency of 100 KHz (standard mode) or 400 KHz (fast mode).

General protocol

The I2C protocol is more elaborate than the serial communication protocol because it has to support communication between several devices. Let's see how it works using examples:

Master -> Slave

If the master wants to send data to the slave:

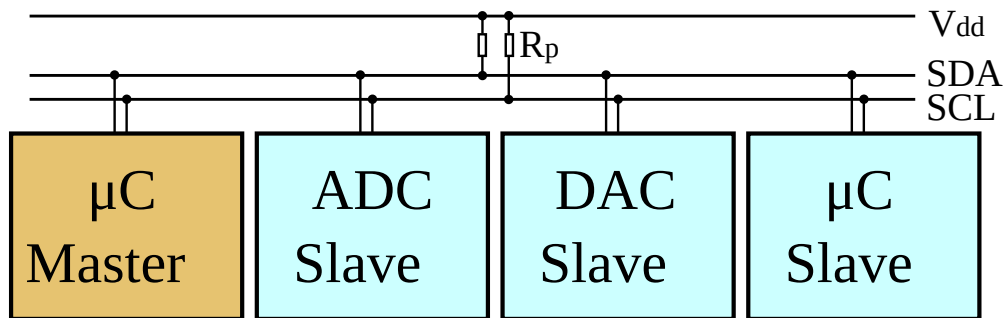


1. Master: Broadcast START
2. M: Broadcast slave address (7 bits) + the R/W (8th) bit set to WRITE
3. Slave: Responds ACK (ACKnowledgement)
4. M: Send one byte
5. S: Responds ACK
6. Repeat steps 4 and 5 zero or more times
7. M: Broadcast STOP OR (broadcast RESTART and go back to (2))

NOTE The slave address could have been 10 bits instead of 7 bits long. Nothing else would have changed.

Master <- Slave

If the master wants to read data from the slave:



1. M: Broadcast START
2. M: Broadcast slave address (7 bits) + the R/W (8th) bit set to READ
3. S: Responds with ACK
4. S: Send byte
5. M: Responds with ACK
6. Repeat steps 4 and 5 zero or more times
7. M: Broadcast STOP OR (broadcast RESTART and go back to (2))

NOTE The slave address could have been 10 bits instead of 7 bits long. Nothing else would have changed.

LSM303DLHC

* **NOTE:** Newer (from around 2020/09) Discovery boards may have an [LSM303AGR](#) rather than a [LSM303DLHC](#). Checkout the github issues like [this](#) for more details.

Two of the sensors in the F3, the magnetometer and the accelerometer, are packaged in a single component: the LSM303DLHC integrated circuit. These two sensors can be accessed via an I2C bus. Each sensor behaves like an I2C slave and has a *different* address.

Each sensor has its own memory where it stores the results of sensing its environment. Our interaction with these sensors will mainly involve reading their memory.

The memory of these sensors is modeled as byte addressable registers. These sensors can be configured too; that's done by writing to their registers. So, in a sense, these sensors are very similar to the peripherals *inside* the microcontroller. The difference is that their registers are not mapped into the microcontrollers' memory. Instead, their registers have to be accessed via the I2C bus.

The main source of information about the LSM303DLHC is its [Data Sheet](#). Read through it to see how one can read the sensors' registers. That part is in:

Section 5.1.1 I2C Operation - Page 20 - LSM303DLHC Data Sheet

The other part of the documentation relevant to this book is the description of the registers. That part is in:

Section 7 Register description - Page 25 - LSM303DLHC Data Sheet

Read a single register

Let's put all that theory into practice!

Just like with the USART peripheral, I've taken care of initializing everything before you reach `main` so you'll only have to deal with the following registers:

- `CR2` . Control register 2.
- `ISR` . Interrupt and status register.
- `TXDR` . Transmit data register.
- `RXDR` . Receive data register.

These registers are documented in the following section of the Reference Manual:

Section 28.7 I2C registers - Page 868 - Reference Manual

We'll be using the `I2C1` peripheral in conjunction with pins `PB6` (`SCL`) and `PB7` (`SDA`).

You won't have to wire anything this time because the sensor is on the board and it's already connected to the microcontroller. However, I would recommend that you disconnect the serial / Bluetooth module from the F3 to make it easier to manipulate. Later on, we'll be moving the board around quite a bit.

Your task is to write a program that reads the contents of the magnetometer's `IRA_REG_M` register. This register is read only and always contains the value `0b01001000` .

The microcontroller will be taking the role of the I2C master and the magnetometer inside the LSM303DLHC will be the I2C slave.

Here's the starter code. You'll have to implement the `TODO` s.

```

#![deny(unsafe_code)]
#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux14::{entry, iprint, iprintln, prelude::*};

// Slave address
const MAGNETOMETER: u16 = 0b0011_1100;

// Addresses of the magnetometer's registers
const OUT_X_H_M: u8 = 0x03;
const IRA_REG_M: u8 = 0x0A;

#[entry]
fn main() -> ! {
    let (i2c1, _delay, mut itm) = aux14::init();

    // Stage 1: Send the address of the register we want to read to the
    // magnetometer
    {
        // TODO Broadcast START

        // TODO Broadcast the MAGNETOMETER address with the R/W bit set to Write

        // TODO Send the address of the register that we want to read: IRA_REG_M
    }

    // Stage 2: Receive the contents of the register we asked for
    let byte = {
        // TODO Broadcast RESTART

        // TODO Broadcast the MAGNETOMETER address with the R/W bit set to Read

        // TODO Receive the contents of the register

        // TODO Broadcast STOP
        0
    };

    // Expected output: 0x0A - 0b01001000
    iprintln!(&mut itm.stim[0], "0x{:02X} - 0b{:08b}", IRA_REG_M, byte);

    loop {}
}

```

To give you some extra help, these are the exact bitfields you'll be working with:

- CR2: SADD1, RD_WRN, NBYTES, START, AUTOEND
- ISR: TXIS, RXNE, TC
- TXDR: TXDATA
- RXDR: RXDATA

The solution

```

#![deny(unsafe_code)]
#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux14::{entry, iprint, iprintln, prelude::*};

// Slave address
const MAGNETOMETER: u16 = 0b0011_1100;

// Addresses of the magnetometer's registers
const OUT_X_H_M: u8 = 0x03;
const IRA_REG_M: u8 = 0x0A;

#[entry]
fn main() -> ! {
    let (i2c1, _delay, mut itm) = aux14::init();

    // Stage 1: Send the address of the register we want to read to the
    // magnetometer
    {
        // Broadcast START
        // Broadcast the MAGNETOMETER address with the R/W bit set to Write
        i2c1.cr2.write(|w| {
            w.start().set_bit();
            w.sadd().bits(MAGNETOMETER);
            w.rd_wrn().clear_bit();
            w.nbytes().bits(1);
            w.autoend().clear_bit()
        });

        // Wait until we can send more data
        while i2c1.isr.read().txis().bit_is_clear() {}

        // Send the address of the register that we want to read: IRA_REG_M
        i2c1.txdr.write(|w| w.txdata().bits(IRA_REG_M));

        // Wait until the previous byte has been transmitted
        while i2c1.isr.read().tc().bit_is_clear() {}
    }

    // Stage 2: Receive the contents of the register we asked for
    let byte = {
        // Broadcast RESTART
        // Broadcast the MAGNETOMETER address with the R/W bit set to Read
        i2c1.cr2.modify(|_, w| {
            w.start().set_bit();
            w.nbytes().bits(1);
            w.rd_wrn().set_bit();
            w.autoend().set_bit()
        });

        // Wait until we have received the contents of the register
        while i2c1.isr.read().rxne().bit_is_clear() {}

        // Broadcast STOP (automatic because of `AUTOEND = 1`)
    }
}

```

```
        i2c1.rxdr.read().rxdata().bits()  
    };  
  
    // Expected output: 0x0A - 0b01001000  
    iprintln!(&mut itm.stim[0], "0x{:02X} - 0b{:08b}", IRA_REG_M, byte);  
  
    loop {}  
}
```


Read several registers

Reading the `IRA_REG_M` register was a good test of our understanding of the I2C protocol but that register contains uninteresting information.

This time, we'll read the registers of the magnetometer that actually expose the sensor readings. Six contiguous registers are involved and they start with `OUT_X_H_M` at address `0x03`.

We'll modify our previous program to read these six registers. Only a few modifications are needed.

We'll need to change the address we request from the magnetometer from `IRA_REG_M` to `OUT_X_H_M`.

```
// Send the address of the register that we want to read: OUT_X_H_M
i2c1.txdr.write(|w| w.txdata().bits(OUT_X_H_M));
```

We'll have to request the slave for six bytes rather than just one.

```
// Broadcast RESTART
// Broadcast the MAGNETOMETER address with the R/W bit set to Read
i2c1.cr2.modify(|_, w| {
    w.start().set_bit();
    w.nbytes().bits(6);
    w.rd_wrn().set_bit();
    w.autoend().set_bit()
});
```

And fill a buffer rather than read just one byte:

```
let mut buffer = [0u8; 6];
for byte in &mut buffer {
    // Wait until we have received the contents of the register
    while i2c1.isr.read().rxne().bit_is_clear() {}

    *byte = i2c1.rxdr.read().rxdata().bits();
}

// Broadcast STOP (automatic because of `AUTOEND = 1`)
```

Putting it all together inside a loop alongside a delay to reduce the data throughput:

```

#![deny(unsafe_code)]
#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux14::{entry, iprint, iprintln, prelude::*};

// Slave address
const MAGNETOMETER: u16 = 0b0011_1100;

// Addresses of the magnetometer's registers
const OUT_X_H_M: u8 = 0x03;
const IRA_REG_M: u8 = 0x0A;

#[entry]
fn main() -> ! {
    let (i2c1, mut delay, mut itm) = aux14::init();

    loop {
        // Broadcast START
        // Broadcast the MAGNETOMETER address with the R/W bit set to Write
        i2c1.cr2.write(|w| {
            w.start().set_bit();
            w.sadd().bits(MAGNETOMETER);
            w.rd_wrn().clear_bit();
            w.nbytes().bits(1);
            w.autoend().clear_bit()
        });

        // Wait until we can send more data
        while i2c1.isr.read().txis().bit_is_clear() {}

        // Send the address of the register that we want to read: OUT_X_H_M
        i2c1.txdr.write(|w| w.txdata().bits(OUT_X_H_M));

        // Wait until the previous byte has been transmitted
        while i2c1.isr.read().tc().bit_is_clear() {}

        // Broadcast RESTART
        // Broadcast the MAGNETOMETER address with the R/W bit set to Read
        i2c1.cr2.modify(|_, w| {
            w.start().set_bit();
            w.nbytes().bits(6);
            w.rd_wrn().set_bit();
            w.autoend().set_bit()
        });

        let mut buffer = [0u8; 6];
        for byte in &mut buffer {
            // Wait until we have received something
            while i2c1.isr.read().rxne().bit_is_clear() {}

            *byte = i2c1.rxdr.read().rxdata().bits();
        }
        // Broadcast STOP (automatic because of `AUTOEND = 1`)

        iprintln!(&mut itm.stim[0], "{:?}", buffer);

        delay.delay_ms(1_000_u16);
    }
}

```

If you run this, you should printed in the `itmdump`'s console a new array of six bytes every second. The values within the array should change if you move around the board.

```
$ # itmdump terminal
(..)
[0, 45, 255, 251, 0, 193]
[0, 44, 255, 249, 0, 193]
[0, 49, 255, 250, 0, 195]
```

But these bytes don't make much sense like that. Let's turn them into actual readings:

```
let x_h = u16::from(buffer[0]);
let x_l = u16::from(buffer[1]);
let z_h = u16::from(buffer[2]);
let z_l = u16::from(buffer[3]);
let y_h = u16::from(buffer[4]);
let y_l = u16::from(buffer[5]);

let x = ((x_h << 8) + x_l) as i16;
let y = ((y_h << 8) + y_l) as i16;
let z = ((z_h << 8) + z_l) as i16;

iprintln!(&mut itm.stim[0], "{:?}", (x, y, z));
```

Now it should look better:

```
$ # `itmdump terminal
(..)
(44, 196, -7)
(45, 195, -6)
(46, 196, -9)
```

This is the Earth's magnetic field decomposed alongside the XYZ axis of the magnetometer.

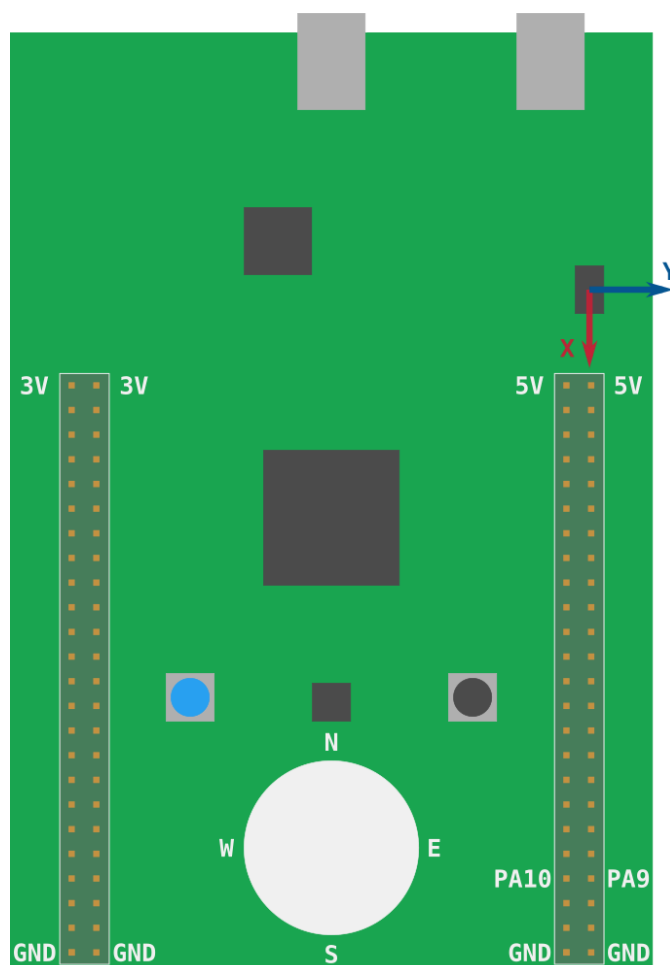
In the next section, we'll learn how to make sense of these numbers.

LED compass

In this section, we'll implement a compass using the LEDs on the F3. Like proper compasses, our LED compass must point north somehow. It will do that by turning on one of its eight LEDs; the on LED should point towards north.

Magnetic fields have both a magnitude, measured in Gauss or Teslas, and a *direction*. The magnetometer on the F3 measures both the magnitude and the direction of an external magnetic field but it reports back the *decomposition* of said field along *its axes*.

See below, the magnetometer has three axes associated to it.



Only the X and Y axes are shown above. The Z axis is pointing “out” of your screen.

Let's get familiar with the readings of the magnetometer by running the following starter code:

```

#![deny(unsafe_code)]
#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux15::{entry, iprint, iprintln, prelude::*};

#[entry]
fn main() -> ! {
    let (_leds, mut lsm303dlhc, mut delay, mut itm) = aux15::init();

    loop {
        iprintln!(&mut itm.stim[0], "{:?}", lsm303dlhc.mag().unwrap());
        delay.delay_ms(1_000_u16);
    }
}

```

This `lsm303dlhc` module provides high level API over the LSM303DLHC. Under the hood it does the same I2C routine that you implemented in the last section but it reports the X, Y and Z values in a `I16x3` struct instead of a tuple.

Locate where north is at your current location. Then rotate the board such that it's aligned "towards north": the North LED (LD3) should be pointing towards north.

Now run the starter code and observe the output. What X, Y and Z values do you see?

```

$ # itmdump terminal
(..)
I16x3 { x: 45, y: 194, z: -3 }
I16x3 { x: 46, y: 195, z: -8 }
I16x3 { x: 47, y: 197, z: -2 }

```

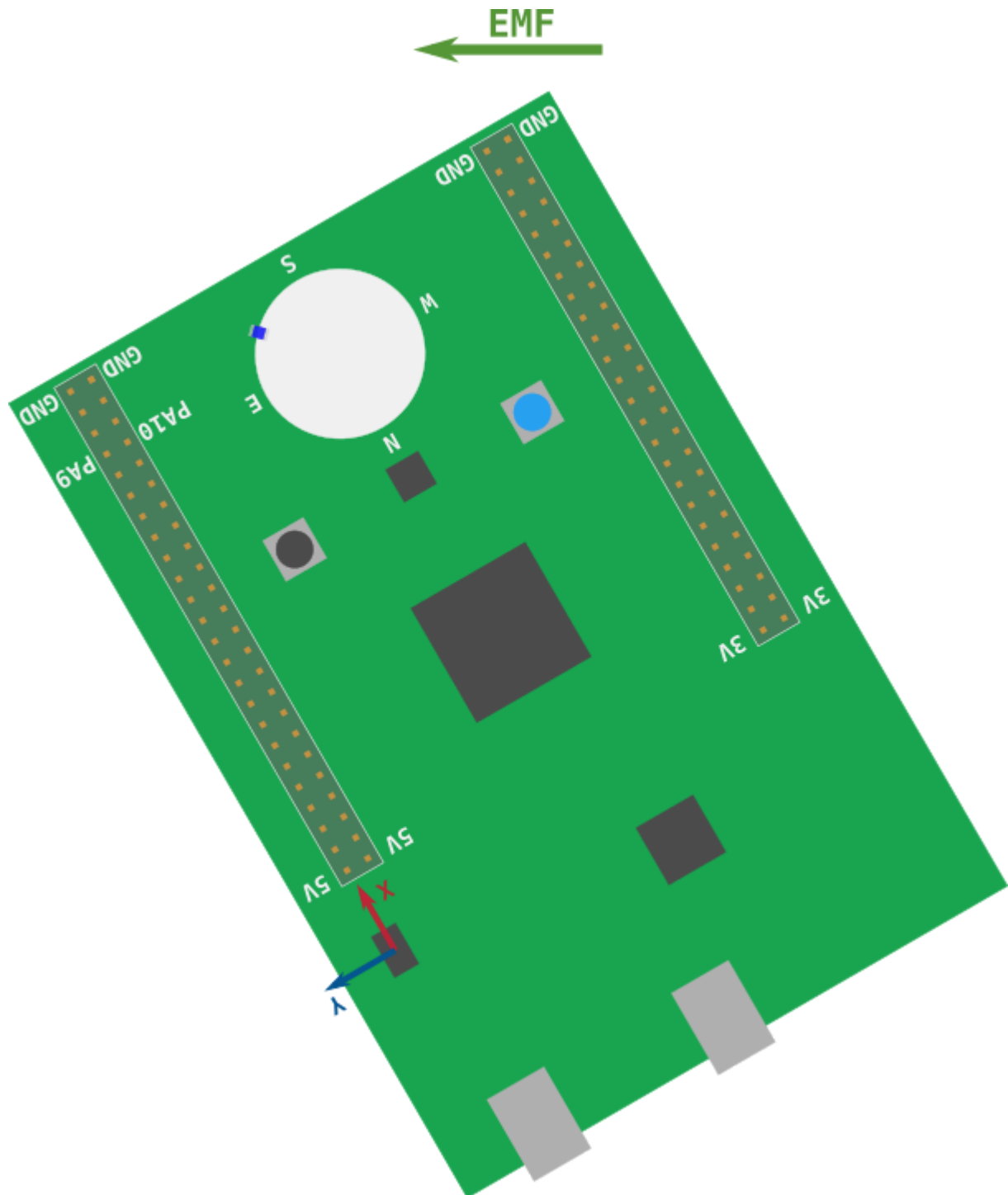
Now rotate the board 90 degrees while keeping it parallel to the ground. What X, Y and Z values do you see this time? Then rotate it 90 degrees again. What values do you see?

Take 1

What's the simplest way in which we can implement the LED compass? Even if it's not perfect.

For starters, we'd only care about the X and Y components of the magnetic field because when you look at a compass you always hold it in horizontal position thus the compass is in the XY plane.

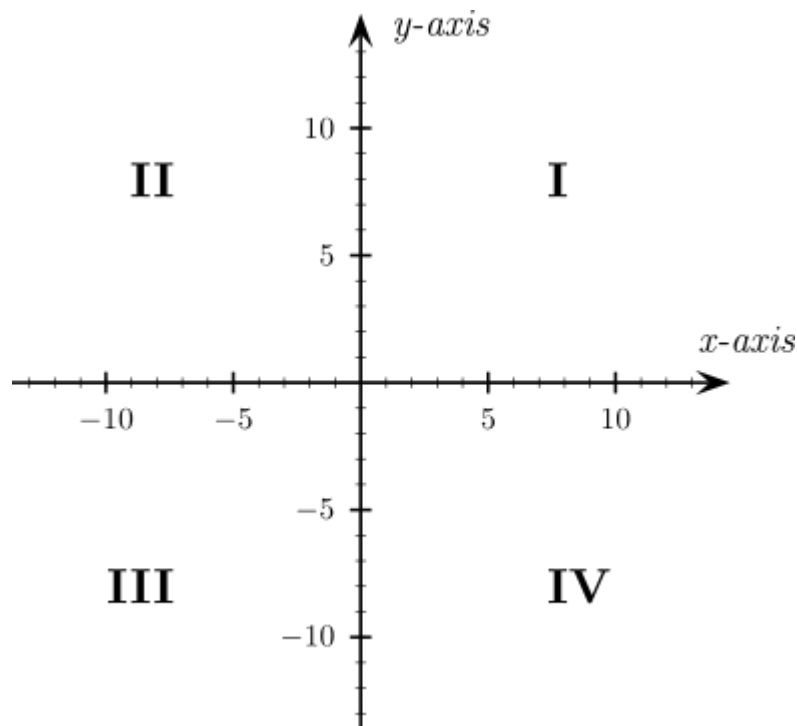
For example, what LED would you turn on in the following case. EMF stands for Earth's Magnetic Field and green arrow has the direction of the EMF (it points north).



The **Southeast** LED, right?

What *signs* do the X and Y components of the magnetic field have in that scenario? Both are positive.

If we only looked at the signs of the X and Y components we could determine to which quadrant the magnetic field belongs to.



In the previous example, the magnetic field was in the first quadrant (x and y were positive) and it made sense to turn on the **SouthEast** LED. Similarly, we could turn a different LED if the magnetic field was in a different quadrant.

Let's try that logic. Here's the starter code:

```

#![deny(unsafe_code)]
#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux15::{entry, iprint, iprintln, prelude::*, switch_hal::OutputSwitch, Direction, I16x3};

#[entry]
fn main() -> ! {
    let (leds, mut lsm303dlhc, mut delay, _itm) = aux15::init();
    let mut leds = leds.into_array();

    loop {
        let I16x3 { x, y, .. } = lsm303dlhc.mag().unwrap();

        // Look at the signs of the X and Y components to determine in which
        // quadrant the magnetic field is
        let dir = match (x > 0, y > 0) {
            // Quadrant ???
            (true, true) => Direction::Southeast,
            // Quadrant ???
            (false, true) => panic!("TODO"),
            // Quadrant ???
            (false, false) => panic!("TODO"),
            // Quadrant ???
            (true, false) => panic!("TODO"),
        };

        leds.iter_mut().for_each(|led| led.off().unwrap());
        leds[dir as usize].on().unwrap();

        delay.delay_ms(1_000_u16);
    }
}

```

There's a `Direction` enum in the `led` module that has 8 variants named after the cardinal points: `North`, `East`, `Southwest`, etc. Each of these variants represent one of the 8 LEDs in the compass. The `Leds` value can be indexed using the `Direction` enum; the result of indexing is the LED that points in that `Direction`.

Solution 1

```

#![deny(unsafe_code)]
#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux15::{entry, iprint, iprintln, prelude::*, switch_hal::OutputSwitch, Direction, I16x3};

#[entry]
fn main() -> ! {
    let (leds, mut lsm303dlhc, mut delay, _itm) = aux15::init();
    let mut leds = leds.into_array();

    loop {
        let I16x3 { x, y, .. } = lsm303dlhc.mag().unwrap();

        // Look at the signs of the X and Y components to determine in which
        // quadrant the magnetic field is
        let dir = match (x > 0, y > 0) {
            // Quadrant I
            (true, true) => Direction::Southeast,
            // Quadrant II
            (false, true) => Direction::Northeast,
            // Quadrant III
            (false, false) => Direction::Northwest,
            // Quadrant IV
            (true, false) => Direction::Southwest,
        };

        leds.iter_mut().for_each(|led| led.off().unwrap());
        leds[dir as usize].on().unwrap();

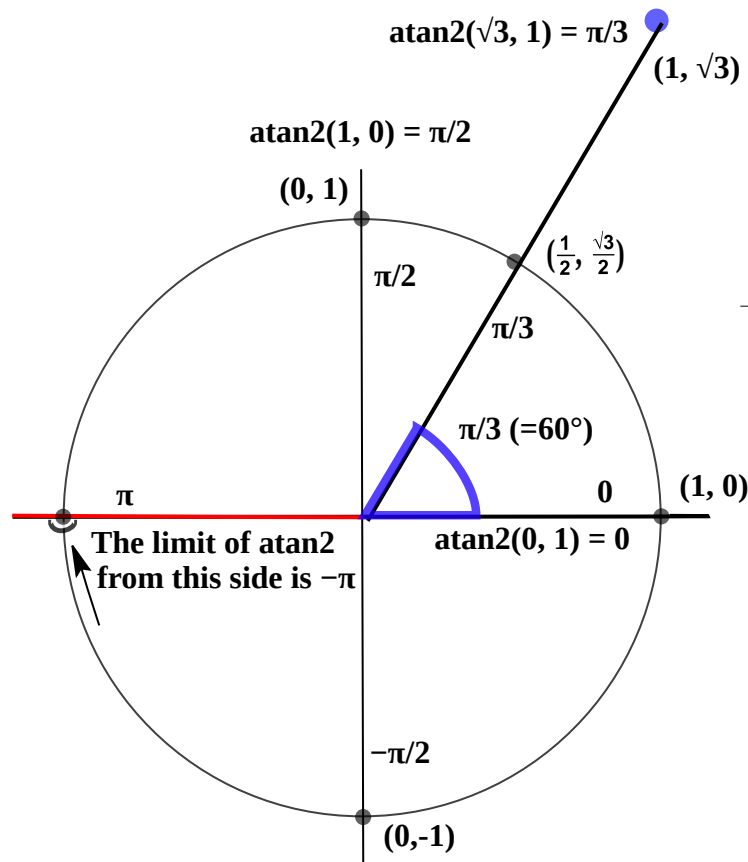
        delay.delay_ms(1_000_u16);
    }
}

```


Take 2

This time, we'll use math to get the precise angle that the magnetic field forms with the X and Y axes of the magnetometer.

We'll use the `atan2` function. This function returns an angle in the `-PI` to `PI` range. The graphic below shows how this angle is measured:



Although not explicitly shown in this graph the X axis points to the right and the Y axis points up.

Here's the starter code. `theta`, in radians, has already been computed. You need to pick which LED to turn on based on the value of `theta`.

```

#![deny(unsafe_code)]
#![no_main]
#![no_std]

// You'll find this useful ;-)
use core::f32::consts::PI;

#[allow(unused_imports)]
use aux15::{entry, iprint, iprintln, prelude::*, switch_hal::OutputSwitch, Direction, I16x3};
// this trait provides the `atan2` method
use m::Float;

#[entry]
fn main() -> ! {
    let (leds, mut lsm303dlhc, mut delay, _itm) = aux15::init();
    let mut leds = leds.into_array();

    loop {
        let I16x3 { x, y, .. } = lsm303dlhc.mag().unwrap();

        let _theta = (y as f32).atan2(x as f32); // in radians

        // FIXME pick a direction to point to based on `theta`
        let dir = Direction::Southeast;

        leds.iter_mut().for_each(|led| led.off().unwrap());
        leds[dir as usize].on().unwrap();

        delay.delay_ms(100_u8);
    }
}

```

Suggestions/tips:

- A whole circle rotation equals 360 degrees.
- `PI` radians is equivalent to 180 degrees.
- If `theta` was zero, what LED would you turn on?
- If `theta` was, instead, very close to zero, what LED would you turn on?
- If `theta` kept increasing, at what value would you turn on a different LED?

Solution 2

```

#![deny(unsafe_code)]
#![no_main]
#![no_std]

// You'll find this useful ;-)
use core::f32::consts::PI;

#[allow(unused_imports)]
use aux15::{entry, iprint, iprintln, prelude::*, switch_hal::OutputSwitch, Direction, I16x3};
use m::Float;

#[entry]
fn main() -> ! {
    let (leds, mut lsm303dlhc, mut delay, _itm) = aux15::init();
    let mut leds = leds.into_array();

    loop {
        let I16x3 { x, y, .. } = lsm303dlhc.mag().unwrap();

        let theta = (y as f32).atan2(x as f32); // in radians

        let dir = if theta < -7. * PI / 8. {
            Direction::North
        } else if theta < -5. * PI / 8. {
            Direction::Northwest
        } else if theta < -3. * PI / 8. {
            Direction::West
        } else if theta < -PI / 8. {
            Direction::Southwest
        } else if theta < PI / 8. {
            Direction::South
        } else if theta < 3. * PI / 8. {
            Direction::Southeast
        } else if theta < 5. * PI / 8. {
            Direction::East
        } else if theta < 7. * PI / 8. {
            Direction::Northeast
        } else {
            Direction::North
        };

        leds.iter_mut().for_each(|led| led.off().unwrap());
        leds[dir as usize].on().unwrap();

        delay.delay_ms(100_u8);
    }
}

```


Magnitude

We have been working with the direction of the magnetic field but what's its real magnitude? The number that the `magnetic_field` function reports are unit-less. How can we convert those values to Gauss?

The documentation will answer that question.

Section 2.1 Sensor characteristics - Page 10 - LSM303DLHC Data Sheet

The table in that page shows a *magnetic gain setting* that has different values according to the values of the GN bits. By default, those GN bits are set to `001`. That means that magnetic gain of the X and Y axes is `1100 LSB / Gauss` and the magnetic gain of the Z axis is `980 LSB / Gauss`. LSB stands for Least Significant Bits and the `1100 LSB / Gauss` number indicates that a reading of `1100` is equivalent to `1 Gauss`, a reading of `2200` is equivalent to `2 Gauss` and so on.

So, what we need to do is divide the X, Y and Z values that the sensor outputs by its corresponding *gain*. Then, we'll have the X, Y and Z components of the magnetic field in Gauss.

With some extra math we can retrieve the magnitude of the magnetic field from its X, Y and Z components:

```
let magnitude = (x * x + y * y + z * z).sqrt();
```

Putting all this together in a program:

```

#![deny(unsafe_code)]
#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux15::{entry, iprint, iprintln, prelude::*, I16x3};
use m::Float;

#[entry]
fn main() -> ! {
    const XY_GAIN: f32 = 1100.; // LSB / G
    const Z_GAIN: f32 = 980.; // LSB / G

    let (_leds, mut lsm303dlhc, mut delay, mut itm) = aux15::init();

    loop {
        let I16x3 { x, y, z } = lsm303dlhc.mag().unwrap();

        let x = f32::from(x) / XY_GAIN;
        let y = f32::from(y) / XY_GAIN;
        let z = f32::from(z) / Z_GAIN;

        let mag = (x * x + y * y + z * z).sqrt();

        iprintln!(&mut itm.stim[0], "{} mG", mag * 1_000.);

        delay.delay_ms(500_u16);
    }
}

```

This program will report the magnitude (strength) of the magnetic field in milligauss (**mG**). The magnitude of the Earth's magnetic field is in the range of **250 mG** to **650 mG** (the magnitude varies depending on your geographical location) so you should see a value in that range or close to that range – I see a magnitude of around 210 mG.

Some questions:

Without moving the board, what value do you see? Do you always see the same value?

If you rotate the board, does the magnitude change? Should it change?

Calibration

If we rotate the board, the direction of the Earth's magnetic field with respect to the magnetometer should change but its magnitude should not! Yet, the magnetometer indicates that the magnitude of the magnetic field changes as the board rotates.

Why's that the case? Turns out the magnetometer needs to be calibrated to return the correct answer.

The calibration involves quite a bit of math (matrices) so we won't cover it here but this [Application Note](#) describes the procedure if you are interested. Instead, what we'll do in this section is *visualize* how off we are.

Let's try this experiment: Let's record the readings of the magnetometer while we slowly rotate the board in different directions. We'll use the `iprintln` macro to format the readings as Tab Separated Values (TSV).

```
#![deny(unsafe_code)]
#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux15::{entry, iprint, iprintln, prelude::*, I16x3};

#[entry]
fn main() -> ! {
    let (_leds, mut lsm303dlhc, mut delay, mut itm) = aux15::init();

    loop {
        let I16x3 { x, y, z } = lsm303dlhc.mag().unwrap();

        iprintln!(&mut itm.stim[0], "{}\t{}\t{}", x, y, z);

        delay.delay_ms(100_u8);
    }
}
```

You should get an output in the console that looks like this:

```
$ # itmdump console
-76      213      -54
-76      213      -54
-76      213      -54
-76      213      -54
-73      213      -55
```

You can pipe that to a file using:

```
$ # Careful! Exit any running other `itmdump` instance that may be running
$ itmdump -F -f itm.txt > emf.txt
```

Rotate the board in many different direction while you log data for a several seconds.

Then import that TSV file into a spreadsheet program (or use the Python script shown below) and plot the first two columns as a scatter plot. 236

```
#!/usr/bin/python

import csv
import math
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
import sys

# apply plot style
sns.set()

x = []
y = []

with open(sys.argv[1], 'r') as f:
    rows = csv.reader(f, delimiter='\t')

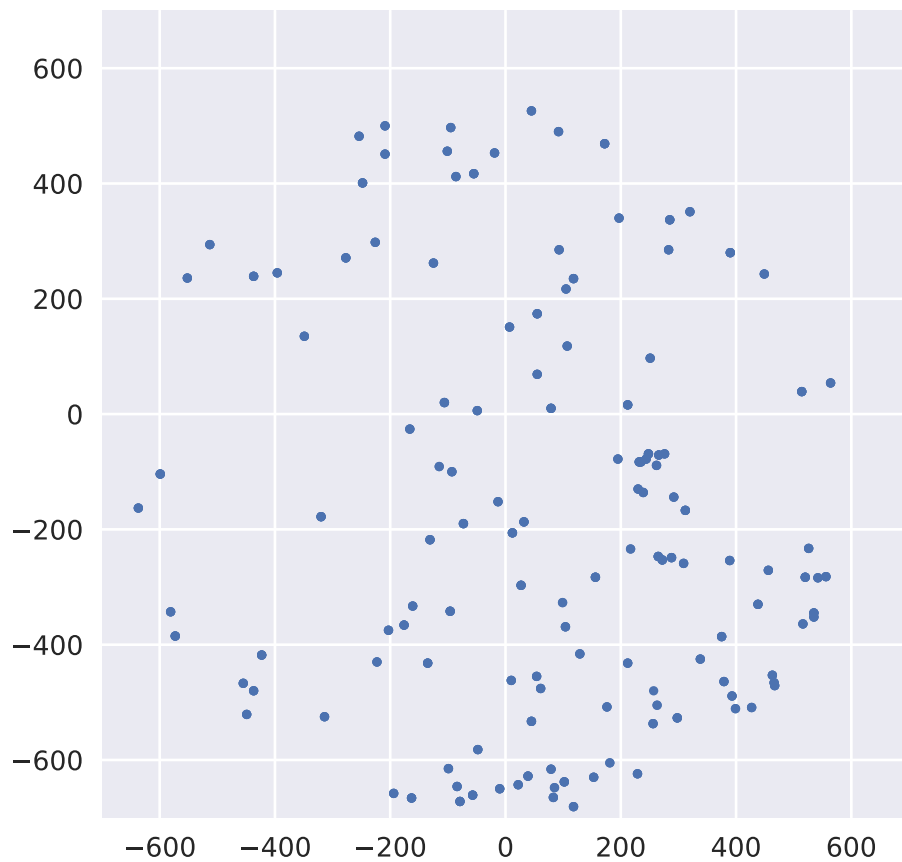
    for row in rows:
        # discard rows that are missing data
        if len(row) != 3 or not row[0] or not row[1]:
            continue

        x.append(int(row[0]))
        y.append(int(row[1]))

r = math.ceil(max(max(np.abs(x)), max(np.abs(y))) / 100) * 100

plt.plot(x, y, '.')
plt.xlim(-r, r)
plt.ylim(-r, r)
plt.gca().set_aspect(1)
plt.tight_layout()

plt.savefig('emf.svg')
plt.close
```



If you rotated the board on a flat horizontal surface, the Z component of the magnetic field should have remained relatively constant and this plot should have been a circumference (not an ellipse) centered at the origin. If you rotated the board in random directions, which was the case of the plot above, then you should have gotten a circle made of a bunch of points centered at the origin. Deviations from the circle shape indicate that the magnetometer needs to be calibrated.

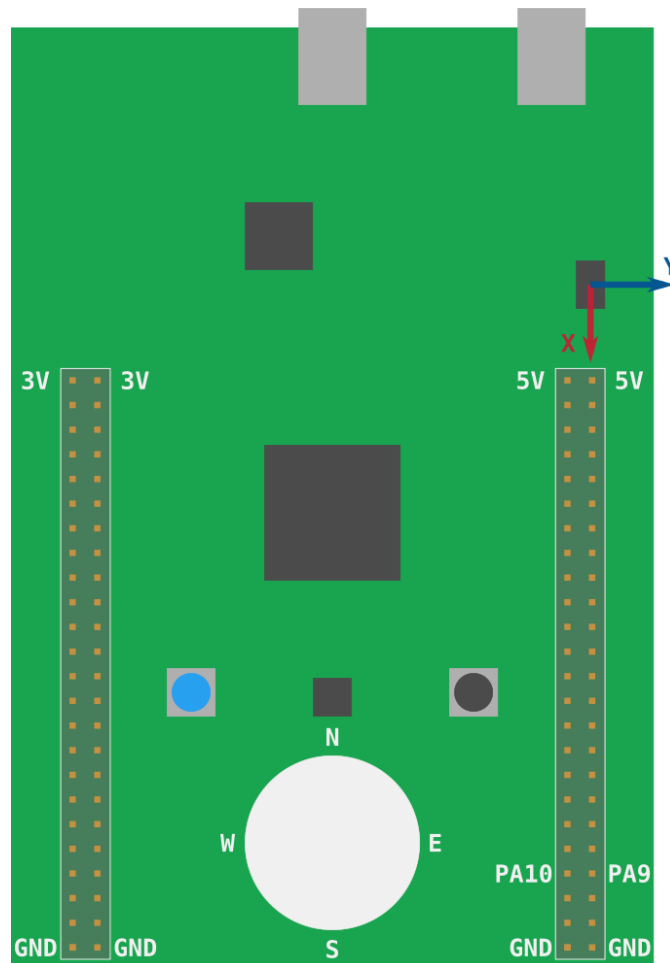
Take home message: Don't just trust the reading of a sensor. Verify it's outputting sensible values. If it's not, then calibrate it.

Punch-o-meter

In this section we'll be playing with the accelerometer that's in the board.

What are we building this time? A punch-o-meter! We'll be measuring the power of your jabs. Well, actually the maximum acceleration that you can reach because acceleration is what accelerometers measure. Strength and acceleration are proportional though so it's a good approximation.

The accelerometer is also built inside the LSM303DLHC package. And just like the magnetometer, it can also be accessed using the I2C bus. It also has the same coordinate system as the magnetometer. Here's the coordinate system again:



Just like in the previous unit, we'll be using a high level API to directly get the sensor readings in a nicely packaged `struct`.

Gravity is up?

What's the first thing we'll do?

Perform a sanity check!

The starter code prints the X, Y and Z components of the acceleration measured by the accelerometer. The values have already been “scaled” and have units of **g** s. Where **1 g** is equal to the acceleration of the gravity, about **9.8** meters per second squared.

```
#![deny(unsafe_code)]
#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux16::{entry, iprint, iprintln, prelude::*, I16x3, Sensitivity};

#[entry]
fn main() -> ! {
    let (mut lsm303dlhc, mut delay, _mono_timer, mut itm) = aux16::init();

    // extend sensing range to `[-12g, +12g]`
    lsm303dlhc.set_accel_sensitivity(Sensitivity::G12).unwrap();
    loop {
        const SENSITIVITY: f32 = 12. / (1 << 14) as f32;

        let I16x3 { x, y, z } = lsm303dlhc.accel().unwrap();

        let x = f32::from(x) * SENSITIVITY;
        let y = f32::from(y) * SENSITIVITY;
        let z = f32::from(z) * SENSITIVITY;

        iprintln!(&mut itm.stim[0], "{:?}", (x, y, z));

        delay.delay_ms(1_000_u16);
    }
}
```

The output of this program with the board sitting still will be something like:

```
$ # itmdump console
(..)
(0.0, 0.0, 1.078125)
(0.0, 0.0, 1.078125)
(0.0, 0.0, 1.171875)
(0.0, 0.0, 1.03125)
(0.0, 0.0, 1.078125)
```

Which is weird because the board is not moving yet its acceleration is non-zero. What's going on? This must be related to the gravity, right? Because the acceleration of gravity is **1 g**. But the gravity pulls objects downwards so the acceleration along the Z axis should be negative not positive ...

Did the program get the Z axis backwards? Nope, you can test rotating the board to align the gravity to the X or Y axis but the acceleration measured by the accelerometer is always pointing up.

What happens here is that the accelerometer is measuring the *proper acceleration* of the board not the acceleration *you* are observing. This proper acceleration is the acceleration of the board as seen from an observer that's in free fall. An observer that's in free fall is moving toward the center of the Earth with an acceleration of $1g$; from its point of view the board is actually moving upwards (away from the center of the Earth) with an acceleration of $1g$. And that's why the proper acceleration is pointing up. This also means that if the board was in free fall, the accelerometer would report a proper acceleration of zero. Please, don't try that at home.

Yes, physics is hard. Let's move on.

The challenge

To keep things simple, we'll measure the acceleration only in the X axis while the board remains horizontal. That way we won't have to deal with subtracting that *fictitious* `1g` we observed before which would be hard because that `1g` could have X Y Z components depending on how the board is oriented.

Here's what the punch-o-meter must do:

- By default, the app is not “observing” the acceleration of the board.
- When a significant X acceleration is detected (i.e. the acceleration goes above some threshold), the app should start a new measurement.
- During that measurement interval, the app should keep track of the maximum acceleration observed
- After the measurement interval ends, the app must report the maximum acceleration observed. You can report the value using the `iprintln` macro.

Give it a try and let me know how hard you can punch `;-)` .

My solution

```

#![deny(unsafe_code)]
#![no_main]
#![no_std]

#[allow(unused_imports)]
use aux16::{entry, iprint, iprintln, prelude::*, I16x3, Sensitivity};
use m::Float;

#[entry]
fn main() -> ! {
    const SENSITIVITY: f32 = 12. / (1 << 14) as f32;
    const THRESHOLD: f32 = 0.5;

    let (mut lsm303dlhc, mut delay, mono_timer, mut itm) = aux16::init();

    lsm303dlhc.set_accel_sensitivity(Sensitivity::G12).unwrap();

    let measurement_time = mono_timer.frequency().0; // 1 second in ticks
    let mut instant = None;
    let mut max_g = 0.;
    loop {
        let g_x = f32::from(lsm303dlhc.accel().unwrap().x).abs() * SENSITIVITY;

        match instant {
            None => {
                // If acceleration goes above a threshold, we start measuring
                if g_x > THRESHOLD {
                    iprintln!(&mut itm.stim[0], "START!");

                    max_g = g_x;
                    instant = Some(mono_timer.now());
                }
            }
            // Still measuring
            Some(ref instant) if instant.elapsed() < measurement_time => {
                if g_x > max_g {
                    max_g = g_x;
                }
            }
            _ => {
                // Report max value
                iprintln!(&mut itm.stim[0], "Max acceleration: {}g", max_g);

                // Measurement done
                instant = None;

                // Reset
                max_g = 0.;
            }
        }

        delay.delay_ms(50_u8);
    }
}

```


What's left for you to explore

We have barely scratched the surface! There's lots of stuff left for you to explore.

NOTE: If you're reading this, and you'd like to help add examples or exercises to the Discovery book for any of the items below, or any other relevant embedded topics, we'd love to have your help!

Please [open an issue](#) if you would like to help, but need assistance or mentoring for how to contribute this to the book, or open a Pull Request adding the information!

Topics about embedded software

These topics discuss strategies for writing embedded software. Although many problems can be solved in different ways, these sections talk about some strategies, and when they make sense (or don't make sense) to use.

Multitasking

All our programs executed a single task. How could we achieve multitasking in a system with no OS, and thus no threads. There are two main approaches to multitasking: preemptive multitasking and cooperative multitasking.

In preemptive multitasking a task that's currently being executed can, at any point in time, be *preempted* (interrupted) by another task. On preemption, the first task will be suspended and the processor will instead execute the second task. At some point the first task will be resumed. Microcontrollers provide hardware support for preemption in the form of *interrupts*.

In cooperative multitasking a task that's being executed will run until it reaches a *suspension point*. When the processor reaches that suspension point it will stop executing the current task and instead go and execute a different task. At some point the first task will be resumed. The main difference between these two approaches to multitasking is that in cooperative multitasking *yields* execution control at *known* suspension points instead of being forcefully preempted at any point of its execution.

Sleeping

All our programs have been continuously polling peripherals to see if there's anything that needs to be done. However, some times there's nothing to be done! At those times, the microcontroller should "sleep".

When the processor sleeps, it stops executing instructions and this saves power. It's almost always a good idea to save power so your microcontroller should be sleeping as much as possible. But, how does it know when it has to wake up to perform some action? "Interrupts" are one of the events that wake up the microcontroller but there are others and the `wfi` and `wfe` are the instructions that make the processor "sleep". 249

Topics related to microcontroller capabilities

Microcontrollers (like our STM32F3) have many different capabilities. However, many share similar capabilities that can be used to solve all sorts of different problems.

These topics discuss some of those capabilities, and how they can be used effectively in embedded development.

Direct Memory Access (DMA).

This peripheral is a kind of *asynchronous memcpy*. So far our programs have been pumping data, byte by byte, into peripherals like UART and I2C. This DMA peripheral can be used to perform bulk transfers of data. Either from RAM to RAM, from a peripheral, like a UART, to RAM or from RAM to a peripheral. You can schedule a DMA transfer, like read 256 bytes from USART1 into this buffer, leave it running in the background and then poll some register to see if it has completed so you can do other stuff while the transfer is ongoing.

Interrupts

In order to interact with the real world, it is often necessary for the microcontroller to respond *immediately* when some kind of event occurs.

Microcontrollers have the ability to be interrupted, meaning when a certain event occurs, it will stop whatever it is doing at the moment, to instead respond to that event. This can be very useful when we want to stop a motor when a button is pressed, or measure a sensor when a timer finishes counting down.

Although these interrupts can be very useful, they can also be a bit difficult to work with properly. We want to make sure that we respond to events quickly, but also allow other work to continue as well.

In Rust, we model interrupts similar to the concept of threading on desktop Rust programs. This means we also must think about the Rust concepts of `Send` and `Sync` when sharing data between our main application, and code that executes as part of handling an interrupt event.

Pulse Width Modulation (PWM)

In a nutshell, PWM is turning on something and then turning it off periodically while keeping some proportion ("duty cycle") between the "on time" and the "off time". When used on a LED with a sufficiently high frequency, this can be used to dim the LED. A low duty cycle, say 10% on time and

90% off time, will make the LED very dim whereas a high duty cycle, say 90% on time and 10% off time, will make the LED much brighter (almost as if it were fully powered).

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In general, PWM can be used to control how much *power* is given to some electric device. With proper (power) electronics between a microcontroller and an electrical motor, PWM can be used to control how much power is given to the motor thus it can be used to control its torque and speed. Then you can add an angular position sensor and you got yourself a closed loop controller that can control the position of the motor at different loads.

Digital inputs

We have used the microcontroller pins as digital outputs, to drive LEDs. But these pins can also be configured as digital inputs. As digital inputs, these pins can read the binary state of switches (on/off) or buttons (pressed/not pressed).

(*spoilers* reading the binary state of switches / buttons is not as straightforward as it sounds ;-)

Analog-to-Digital Converters (ADC)

There are a lots of digital sensors out there. You can use a protocol like I2C and SPI to read them. But analog sensors also exist! These sensors just output a voltage level that's proportional to the magnitude they are sensing.

The ADC peripheral can be use to convert that “analog” voltage level, say 1.25 Volts, into a “digital” number, say in the [0, 65535] range, that the processor can use in its calculations.

Digital-to-Analog Converters (DAC)

As you might expect a DAC is exactly the opposite of ADC. You can write some digital value into a register to produce a voltage in the [0, 3.3V] range (assuming a 3.3V power supply) on some “analog” pin. When this analog pin is connected to some appropriate electronics and the register is written to at some constant, fast rate (frequency) with the right values you can produce sounds or even music!

Real Time Clock (RTC)

This peripheral can be used to track time in “human format”. Seconds, minutes, hours, days, months and years. This peripheral handles the translation from “ticks” to these human friendly units of time. It even handles leap years and Daylight Save Time for you!

Other communication protocols

SPI, I2S, SMBUS, CAN, IrDA, Ethernet, USB, Bluetooth, etc.

Different applications use different communication protocols. User facing applications usually have an USB connector because USB is an ubiquitous protocol in PCs and smartphones. Whereas inside cars you'll find plenty of CAN "buses". Some digital sensors use SPI, others use I2C and others, SMBUS. 251

General Embedded-Relevant Topics

These topics cover items that are not specific to our device, or the hardware on it. Instead, they discuss useful techniques that could be used on embedded systems.

Gyroscopes

As part of our Punch-o-meter exercise, we used the Accelerometer to measure changes in acceleration in three dimensions. Our board also features a sensor called a Gyroscope, which allows us to measure changes in "spin" in three dimensions.

This can be very useful when trying to build certain systems, such as a robot that wants to avoid tipping over. Additionally, the data from a sensor like a gyroscope can also be combined with data from accelerometer using a technique called Sensor Fusion (see below for more information).

Servo and Stepper Motors

While some motors are used primarily just to spin in one direction or the other, for example driving a remote control car forwards or backwards, it is sometimes useful to measure more precisely how a motor rotates.

Our microcontroller can be used to drive Servo or Stepper motors, which allow for more precise control of how many turns are being made by the motor, or can even position the motor in one specific place, for example if we wanted to move the arms of a clock to a particular direction.

Sensor fusion

The STM32F3DISCOVERY contains three motion sensors: an accelerometer, a gyroscope and a magnetometer. On their own these measure: (proper) acceleration, angular speed and (the Earth's) magnetic field. But these magnitudes can be "fused" into something more useful: a "robust" measurement of the orientation of the board. Where robust means with less measurement error than a single sensor would be capable of.

This idea of deriving more reliable data from different sources is known as sensor fusion.

So where to next? There are several options:

- You could check out the examples in the [f3](#) board support crate. All those examples work for the STM32F3DISCOVERY board you have.

- You could try out [this motion sensors demo](#). Details about the implementation and source code are available in [this blog post](#).
- You could check out [Real Time for The Masses](#). A very efficient preemptive multitasking framework that supports task prioritization and dead lock free execution.
- You could try running Rust on a different development board. The easiest way to get started is to use the [cortex-m-quickstart](#) Cargo project template.
- You could check out [this blog post](#) which describes how Rust type system can prevent bugs in I/O configuration.
- You could check out my [blog](#) for miscellaneous topics about embedded development with Rust.
- You could check out the [embedded-hal](#) project which aims to build abstractions (traits) for all the embedded I/O functionality commonly found on microcontrollers.
- You could join the [Weekly driver initiative](#) and help us write generic drivers on top of the [embedded-hal](#) traits and that work for all sorts of platforms (ARM Cortex-M, AVR, MSP430, RISC-V, etc.)

General troubleshooting

OpenOCD problems

can't connect to OpenOCD - “Error: open failed”

Symptoms

Upon trying to establish a *new connection* with the device you get an error that looks like this:

```
$ openocd -f (..)
(..)
Error: open failed
in procedure 'init'
in procedure 'ocd_bouncer'
```

Cause

The device is not (properly) connected or not the correct ST-LINK interface configuration is used.

Fix

Linux:

- Check the USB connection using `lsusb`.
- You may not have enough permission to open the device. Try again with `sudo`. If that works, you can use [these instructions](#) to make OpenOCD work without root privilege.
- You might be using the wrong interface configuration for your ST-LINK. Try `interface/stlink-v2.cfg` instead of `interface/stlink-v2-1.cfg`.

Windows:

- You are probably missing the ST-LINK USB driver. Installation instructions [here](#).

can't connect to OpenOCD - “Polling again in X00ms”

Symptoms

Upon trying to establish a *new connection* with the device you get an error that looks like this:

```
$ openocd -f (..)
(..)
Error: jtag status contains invalid mode value - communication failure
Polling target stm32f3x.cpu failed, trying to reexamine
Examination failed, GDB will be halted. Polling again in 100ms
Info : Previous state query failed, trying to reconnect
Error: jtag status contains invalid mode value - communication failure
Polling target stm32f3x.cpu failed, trying to reexamine
Examination failed, GDB will be halted. Polling again in 300ms
Info : Previous state query failed, trying to reconnect
```

Cause

The microcontroller may have get stuck in some tight infinite loop or it may be continuously raising an exception, e.g. the exception handler is raising an exception.

Fix

- Close OpenOCD, if running
- Press and hold the reset (black) button
- Launch the OpenOCD command
- Now, release the reset button

OpenOCD connection lost - “Polling again in X00ms”

Symptoms

A *running* OpenOCD session suddenly errors with:

```
# openocd -f (..)
Error: jtag status contains invalid mode value - communication failure
Polling target stm32f3x.cpu failed, trying to reexamine
Examination failed, GDB will be halted. Polling again in 100ms
Info : Previous state query failed, trying to reconnect
Error: jtag status contains invalid mode value - communication failure
Polling target stm32f3x.cpu failed, trying to reexamine
Examination failed, GDB will be halted. Polling again in 300ms
Info : Previous state query failed, trying to reconnect
```

Cause

The USB connection was lost.

Fix

- Close OpenOCD
- Disconnect and re-connect the USB cable.
- Re-launch OpenOCD

Symptoms

While flashing the device, you get:

```
$ arm-none-eabi-gdb $file
Start address 0x8000194, load size 31588
Transfer rate: 22 KB/sec, 5264 bytes/write.
Ignoring packet error, continuing...
Ignoring packet error, continuing...
```

Cause

Closed `itmdump` while a program that "printed" to the ITM was running. The current GDB session will appear to work normally, just without ITM output but the next GDB session will error with the message that was shown in the previous section.

Or, `itmdump` was called **after** the `monitor tpiu` was issued thus making `itmdump` delete the file / named-pipe that OpenOCD was writing to.

Fix

- Close/kill GDB, OpenOCD and `itmdump`
- Remove the file / named-pipe that `itmdump` was using (for example, `itm.txt`).
- Launch OpenOCD
- Then, launch `itmdump`
- Then, launch the GDB session that executes the `monitor tpiu` command.

can't connect to OpenOCD - "Error: couldn't bind [telnet] to socket: Address already in use"

Symptoms

Upon trying to establish a *new connection* with the device you get an error that looks something like this:

```
$ openocd -f (..)
(..)
Error: couldn't bind telnet to socket: Address already in use
```

Cause

One or more of the ports OpenOCD requires access to, 3333, 4444, or 6666, is in use by another process. Each of these ports is used for another aspect: 3333 for gdb, 4444 for telnet, 6666 for remote procedure call (RPC) commands to TCL

You can go two routes for fixing this. A) Kill any process that's using one of those ports. B) Specify different ports you know to be free for OpenOCD to use.

Solution A

Mac:

- Get a list of processes using ports by running `sudo lsof -PiTCP -sTCP:LISTEN`
- Kill the process(es) blocking the key ports by noting their pid(s) and running `kill [pid]` for each. (Assuming you can confirm they're not running anything mission-critical on your machine!)

Solution B

All:

- Send configuration details to OpenOCD when starting it up so that it uses a different port from the default for any of the processes.
- For example, to do its telnet features on 4441 instead of the default 4444, you would run `openocd -f interface/stlink-v2-1.cfg -f target/stm32f3x.cfg -c "telnet_port 4441"`
- More details on OpenOCD's Configuration Stage can be found in their [official docs online](#).

Cargo problems

“can't find crate for core”

Symptoms

```
Compiling volatile-register v0.1.2
Compiling rlibc v1.0.0
Compiling r0 v0.1.0
error[E0463]: can't find crate for `core`

error: aborting due to previous error

error[E0463]: can't find crate for `core`

error: aborting due to previous error

error[E0463]: can't find crate for `core`

error: aborting due to previous error

Build failed, waiting for other jobs to finish...
Build failed, waiting for other jobs to finish...
error: Could not compile `r0`.

To learn more, run the command again with --verbose.
```

You are using a toolchain older than `nightly-2018-04-08` and forgot to call `rustup target add thumbv7em-none-eabihf`.

Fix

Update your nightly and install the `thumbv7em-none-eabihf` target.

```
$ rustup update nightly
$ rustup target add thumbv7em-none-eabihf
```


How to use GDB

Below are some useful GDB commands that can help us debug our programs. This assumes you have [flashed a program](#) onto your microcontroller and attached to an OpenOCD session.

General Debugging

NOTE: Many of the commands you see below can be executed using a short form. For example, `continue` can simply be used as `c`, or `break $location` can be used as `b $location`. Once you have experience with the commands below, try to see how short you can get the commands to go before GDB doesn't recognize them!

Dealing with Breakpoints

- `break $location` : Set a breakpoint at a place in your code. The value of `$location` can include:
 - `break *main` - Break on the exact address of the function `main`
 - `break *0x080012f2` - Break on the exact memory location `0x080012f2`
 - `break 123` - Break on line 123 of the currently displayed file
 - `break main.rs:123` - Break on line 123 of the file `main.rs`
- `info break` : Display current breakpoints
- `delete` : Delete all breakpoints
 - `delete $n` : Delete breakpoint `$n` (`n` being a number. For example: `delete $2`)
- `clear` : Delete breakpoint at next instruction
 - `clear main.rs:$function` : Delete breakpoint at entry of `$function` in `main.rs`
 - `clear main.rs:123` : Delete breakpoint on line 123 of `main.rs`
- `enable` : Enable all set breakpoints
 - `enable $n` : Enable breakpoint `$n`
- `disable` : Disable all set breakpoints
 - `disable $n` : Disable breakpoint `$n`

Controlling Execution

- `continue` : Begin or continue execution of your program
- `next` : Execute the next line of your program
 - `next $n` : Repeat `next` `$n` number times
- `nexti` : Same as `next` but with machine instructions instead
- `step` : Execute the next line, if the next line includes a call to another function, step into that code
 - `step $n` : Repeat `step` `$n` number times
- `stepi` : Same as `step` but with machine instructions instead
- `jump $location` : Resume execution at specified location:

- `jump 123` : Resume execution at line 123
- `jump 0x080012f2` : Resume execution at address 0x080012f2

Printing Information

- `print /$f $data` - Print the value contained by the variable `$data` . Optionally format the output with `$f` , which can include:

```
x: hexadecimal
d: signed decimal
u: unsigned decimal
o: octal
t: binary
a: address
c: character
f: floating point
```

- `print /t 0xA` : Prints the hexadecimal value `0xA` as binary (0b1010)
- `x /$n$u$u$u$u$u $address` : Examine memory at `$address` . Optionally, `$n` define the number of units to display, `$u` unit size (bytes, halfwords, words, etc), `$f` any `print` format defined above
 - `x /5i 0x080012c4` : Print 5 machine instructions starting at address `0x080012c4`
 - `x/4xb $pc` : Print 4 bytes of memory starting where `$pc` currently is pointing
- `disassemble $location`
 - `disassemble /r main` : Disassemble the function `main` , using `/r` to show the bytes that make up each instruction

Looking at the Symbol Table

- `info functions $regex` : Print the names and data types of functions matched by `$regex` , omit `$regex` to print all functions
 - `info functions main` : Print names and types of defined functions that contain the word `main`
- `info address $symbol` : Print where `$symbol` is stored in memory
 - `info address GPIOC` : Print the memory address of the variable `GPIOC`
- `info variables $regex` : Print names and types of global variables matched by `$regex` , omit `$regex` to print all global variables
- `ptype $data` : Print more detailed information about `$data`
 - `ptype cp` : Print detailed type information about the variable `cp`

Poking around the Program Stack

- `backtrace $n` : Print trace of `$n` frames, or omit `$n` to print all frames
 - `backtrace 2` : Print trace of first 2 frames
- `frame $n` : Select frame with number or address `$n` , omit `$n` to display current frame

- `up $n` : Select frame `$n` frames up
- `down $n` : Select frame `$n` frames down
- `info frame $address` : Describe frame at `$address` , omit `$address` for currently selected frame
- `info args` : Print arguments of selected frame
- `info registers $r` : Print the value of register `$r` in selected frame, omit `$r` for all registers
 - `info registers $sp` : Print the value of the stack pointer register `$sp` in the current frame

Controlling OpenOCD Remotely

- `monitor reset run` : Reset the CPU, starting execution over again
 - `monitor reset` : Same as above
- `monitor reset init` : Reset the CPU, halting execution at the start
- `monitor targets` : Display information and state of current target